ICCUB
Institute of Cosmos Sciences

EXCELENCIA MARÍA DE MAEZTU

Institute of Cosmos Sciences

REPORT OF ACTIVITIES 2015

UNIVERSITAT DE BARCELONA

FOREWORD

In 2015, the Institute of Cosmos Sciences was awarded the *Unidad de Excelencia Maria de Maeztu* distinction in the first call organized by the Ministry of Economy and Competitiveness (MINECO). This award represents not only an important recognition of the work done so far, but also a challenge for the future. In order to extract the maximum benefits of this award, the ICCUB underwent an internal reorganization intended to boost excellence and broaden our research, as recommended by our International Advisory Committee.

Currently, the additional financial support associated to the ICCUB has allowed us to reinforce some specific research groups, as well as to boost some parallel lines of research and technology, by allocating several postdoctoral researchers, PhD students, and technical personnel. Being a university institute, we lack permanent and direct support from our local or central government, and for this reason we need the support of our university, which we hope is going to materialize soon in many ways, including new allocated spaces for the technological part of ICCUB.

With this award we plan to increase our scientific achievements during the next years. In particular, we plan to pay special attention to our technological aspects of research, not only along our usual lines of space missions, particle detectors, and astronomical instrumentation, but also by opening new lines related to transferring technology to industry. We also hope to boost our internalization by enlarging our collaborations with other groups and institutes worldwide.

Lluís Garrido Beltrán
Director

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THE ICCUB

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The Institute of Cosmos Sciences of the University of Barcelona (ICCUB) is an interdisciplinary center devoted to fundamental research in the fields of cosmology, astrophysics and particle physics. In addition, the institute has a strong technology program aimed at fostering its participation in observational astronomy and particle physics international collaborations.

The Institute was created in 2006 as the instrument of the University of Barcelona for the active support of research in theoretical astrophysics and particle physics, focusing on their synergy with cosmology, in order to promote experimental physics and instrument development, to enable a significant participation of the University of Barcelona in large international collaborations, and to attract highly qualified scientific personnel.

Since its creation, the ICCUB has experienced a significant growth, becoming a consolidated research institution with 65 long term scientists, 70 postdoctoral researchers and PhD students and 19 engineers.

EXCELENCIA MARÍA DE MAEZTU

The Maria de Maeztu award

In 2015, the ICCUB was chosen as a María de Maeztu center of excellence by the Spanish Government. Selected institutes stand out for the international impact of their scientific contributions, their innovative power and for their strong relation to the social and economic environment. This award is helping to further strengthen ICCUB's scientific contribution.

Building of the Physics Faculty of the University of Barcelona, headquarters of the ICCUB

ORGANIZATION CHART 2015

Executive Board

Director: Lluís Garrido
Deputy Director: Francesca Figueras
Secretary: Bartomeu Fiol

Council of the Institute

Domènec Espriu
Bartomeu Fiol
Francesca Figueras
Lluís Garrido
Eugeni Graugés
David Mateos
Simone Migliari
Jordi Miralda
Josep Maria Paredes
Àngels Ramos
Blai Sanahuja
Joan Soto

Scientific Board

Francesca Figueras
Bartomeu Fiol
Lluís Garrido
Ricardo Graciani
Simone Migliari
Josep M. Paredes

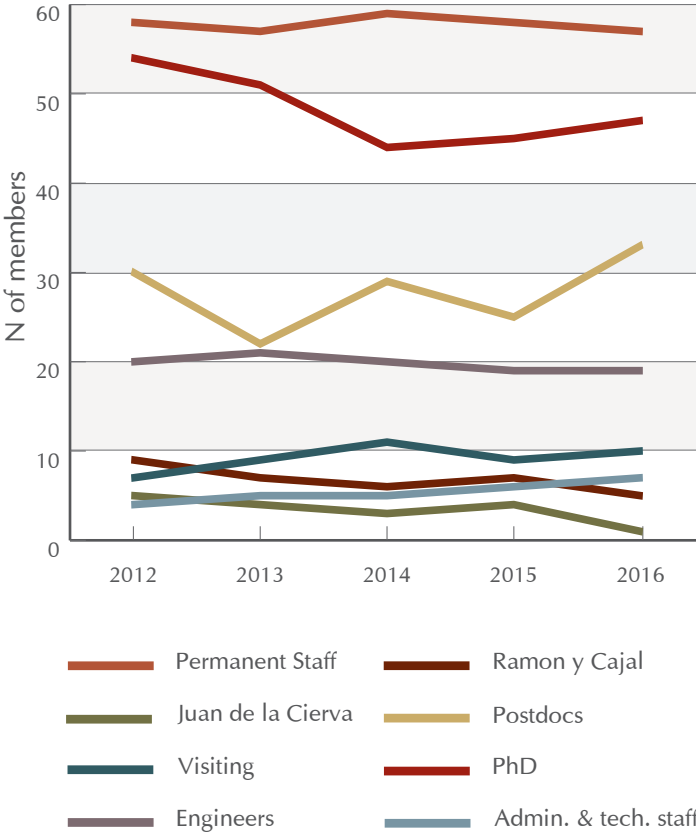
International Advisory Council

Felix Aharonian, Dublin Institute for Advanced Studies and Max Planck Institute für Kernphysik, Heidelberg (Chair)
Alan Heavens, Imperial Centre for Inference and Cosmology, Imperial College, London.
Slava Mukhanov, ASC, Physics Department, LMU, Munich.
Tatsuya Nakada, LPHE, École Polytechnique Fédérale de Lausanne, Lausanne.

2015 - THE ICCUB IN FIGURES

Staff

58 Permanent Staff
7 Ramon y Cajal Members
4 Juan de la Cierva Members
25 Postdoc Fellows
10 Visiting Scholars
45 PhD Students
19 Engineers and Technicians
6 Services and Administration Personnel



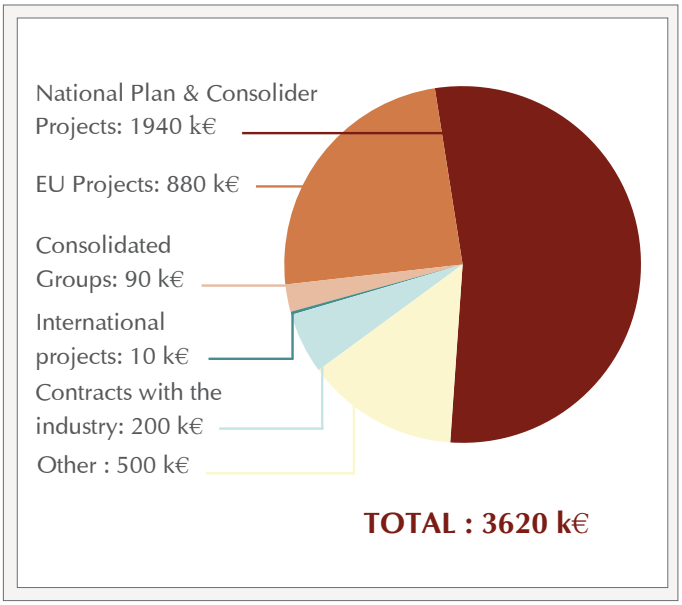
ICCUB'S STAFF EVOLUTION

Projects and Funds

2 ERC Grants
14 European Projects
3 International projects
35 National Plan & Consolider Projects
8 Consolidated Groups
9 Contracts with the industry

Publications

284 SCI Publications
86 Non SCI-Publications
83 Technical Reports



PROJECTS AND FUNDS: 2015 BUDGET

For projects with an execution period longer than one year, only the proportional amount has been considered

Theses

12 Finished PhD Theses
35 Ongoing PhD Theses
20 Finished Master Theses

Activities

7 ICCUB Colloquia
108 Group Seminars
26 Event Organization
24 Public Talks
11 Showings of itinerant exhibitions

ICCUB STAFF* 2

RESEARCHERS

Permanent Staff

Canal, Ramon (UB)
Centelles, Mario (UB)
Crusats, Joaquim (UB)
D'Enterria, David (ICREA, leave of absence)
Diéguez, Ángel (UB)
El-Hachemi, Zoubir (UB)
Emparan García, Roberto A. (ICREA)
Espriu, Domènec (UB)
Estalella, Robert (UB)
Fabricius, Claus Vilhelm (IEEC)
Fernández, José M. (UB)
Figueras, Francesca (UB)
Fiol, Bartomeu (UB)
Garrido, Lluís (UB)
Garriga, Jaume (UB)
Gómez, Gerard (UB)
Gómez, Jose M. (UB)
Gomis, Joaquim (UB)
González- García, M. Concepción (ICREA)
Graciani-Díaz, Ricardo (UB)
Graugés, Eugeni (UB)
Guasch, Jaume (UB)
Guzmán, Rafael (UB & U. Florida)
Iwasawa, Kazushi (ICREA)
Jiménez Tellado, Raúl (ICREA)
Jordi, Carme (UB)
Latorre, José I. (UB)
Llosa, Josep (UB)
López, Rosario (UB)
Luri, F. Xavier (UB)
Magas, Volodymyr (UB)
Manrique, Alberto (UB)
Mateos, David (ICREA)
Mescia, Federico (UB)
Miralda-Escudé, Jordi (ICREA)
Molina, Alfred (UB)
Núñez, Jorge C. (UB)
Padoan, Paolo (ICREA)
Paredes, Josep Maria (UB)
Parreño, Assumpta (UB)
Polls, Artur (UB)
Pons, Josep M. (UB)
Ramos, Àngels (UB)
Ribó, Marc (UB)
Ruiz, Hugo (UB, leave of absence)
Russo, G. (ICREA)
Sala, Ferran (UB)
Salvador, Eduard (UB)
Salvat, Francesc (UB)
Sanahuja, Blai (UB)
Solà, Joan (UB)
Solanes, José M. (UB)
Soto, Joan (UB)
Taron, Josep (UB)
Torra, Jordi (UB)
Verdaguer, Enric (UB)
Verde, Licia (ICREA)
Viñas, Xavier (UB)

Ramon y Cajal Members

Bosch-Ramon, Valentí
Casalderrey, Jorge
Germani, Cristiano
Iblisdir, Sofyan
Julià, Bruno
Migliari, Simone
Notari, Alessio

Juan de la Cierva Members

García, Miguel
Tarrio, Luis Javier
Tywoniuk, Konrad
Zanin, Roberta

Postdoc Fellows

Àgueda Costafreda, Neus
Aliu Ester
Aran, Àngels
Attems , Maximilian
Balaguer, Dolores
Bellini , Emilio
Bergström, Johannes
Carrasco, José M.
Cuesta, Antonio José
Fernández-Faedo, Antón
Gracia, Gonzalo
Haibo, Qiu
Hofer, Lars
Hu, Bin
Izumi, Keisuke
Masana, Eduard
Pantelidou Christiana
Portell , Jordi
Roca, Santi
Rocha, Jorge
Romero, Mercè
Simpson , Fergus Rae Goalen
Voss, Holger
Weiler, Michael
Zilhao MigueL

Visiting Scholars

Andrianov, Alexander
Casademunt, Jaume
Jorba, Ángel
Lizzi, Fedele
Ribó, Josep Maria
Ruiz-Lapuente, M. Pilar
Ruiz, Josep Xavier
Talavera, Pere
Torrelles, Txema
Yun, Joao Lin

* Staff with ICCUB affiliation between 1. January and 31. December 2015

PhD Students

Abedi, Hoda
Alsina, Daniel
Ariño Prats, Andreu
Bernal, José Luís
Casamiquela, Laia
Cervera, Alba
Cheng, Yu
Dector, Aldo
Del Ser, Daniel
Di Dato, Adriana
Farràs, Martí
Feijoo, Albert
Fernández, Isabel
Gabbanelli, Luciano
Galindo, Daniel
García, Carles
Garolera, Blai
Gómez, Adrià
Gontcho A Gontcho, Satya
Juan, Enric
Juárez, Carmen
Karimkhani, Elahe
Maneu, Jordi
Marcote, Benito
Marín, Carla
Mariño, Mauricio
Martínez, Marina
Merino, M. Teresa
Mor, Roger
Moreno, Víctor
Olikara, Zubin Philip
Oriol, Pablo
Pablos, Daniel
Pacheco, Daniel
Paita, Fabrizio
Palmer, Max
Paredes, Xavier
Pérez, Daniel
Pérez, Ignasi
Pravos, David
Renau, Albert
Rives, Vicente
Torrents, Genís
Torres, Núria
Triana, Miquel

ENGINEERS AND TECHNICIANS

Antiche, Erika
Borrachero, Raúl
Casajús, Adrià
Casas, Albert
Castañeda, Javier
Ciaglia, Dimitri
Clotet, Marcial
Garralda, Nora
Gascón, David
Gómez, Sergio
González, Juan José
Julbe, Francesc
Mauricio, Joan
Molina, Daniel
Pérez, Gabriel
Picatoste, Eduard
Roma, David
Sabater, Josep
Sanuy, Andreu

SERVICES AND ADMINISTRATION PERSONNEL

ICCUB Secretariat

Frutos, Ariadna
Moreno, Ana Belén

Group Support

Bascon, Susana
Macduff, Kayla
Olarte, Surinye

Collaborating Students

Ortiz, Elisenda

RESEARCH ACTIVITY 3

Searching for answers to fundamental questions with a multidisciplinary approach

Research at ICCUB is conducted with the aim of answering some of the most intriguing and fundamental questions in Cosmology:

What are the origin and fate of the Universe?

The Universe underwent an early phase of accelerated expansion, known as inflation, similar to the ongoing phase due to dark energy. Are these two phenomena related? Can they be derived from a fundamental theory?

Which are the ultimate constituents of the Universe?

Dark matter apparently accounts for most of the matter density of the universe. But what is the nature of dark matter and how could the Standard Model of Particle Physics be extended to accommodate it?

Why does the Universe have its present appearance?

Various galaxy properties do not conform to the galaxy formation scenario foreseen in the accelerated flat cold dark matter Universe. Is there something wrong with the models of galaxy formation? Is dark matter warm rather than cold?

These questions reveal the intimate connection between particle physics and astrophysics and therefore demand a multidisciplinary approach. Research at ICCUB intends to tackle them from the theoretical, observational and experimental viewpoints. It is organized in the following areas:

- Cosmology and Large Scale Structure
- Experimental Particle Physics
- Galaxy Structure and Evolution
- Gravitation and Cosmology
- High Energy Astrophysics
- Nuclear and Hadron Physics
- Particle Physics Phenomenology
- Star Formation
- Theoretical Physics
- Additional lines of research

Contributing to international projects to study the macro and the microcosmos

Research in particle physics and astrophysics involve the use of data collected by means of sophisticated instrumentation that cannot be afforded by individual research centers. ICCUB researchers are currently contributing to the following projects:

Space Missions:

Gaia, Solar Orbiter, Euclid, CorE.

Ground-based observatories and telescopes

MAGIC Cherenkov Telescopes, Sloan Digital Sky Survey (SDSS), Cherenkov Telescope Array (CTA), Dark Energy Survey (DES), LSST.

Accelerators and particle detectors

LHCb detector, BABAR detector.

Developing new technology for science applications

Apart from doing fundamental research, the ICCUB has a strong technology program aimed at fostering its participation in observational astronomy and particle physics international collaborations. Its expertise is organized along two transversal technological lines:

- Electronics and Instrumentation Development
- Very Large Data Processing and Analysis

Transferring technology to industry

ICCUB research generates products and technologies transferable to industry and society through licensing of patents and creation of spin-off companies and instrumental services. To date, its technology transfer program has given rise to several applications from medical imaging to space industry.

COSMOLOGY AND LARGE SCALE STRUCTURE

LINES OF RESEARCH

- Large scale structure of galaxies and the intergalactic medium
- Microwave background radiation anisotropies
- Baryonic acoustic oscillations
- Supernova cosmology
- Dark matter and dark energy
- Lyman-α emission from galaxies at high redshifts
- Reionization of the intergalactic medium

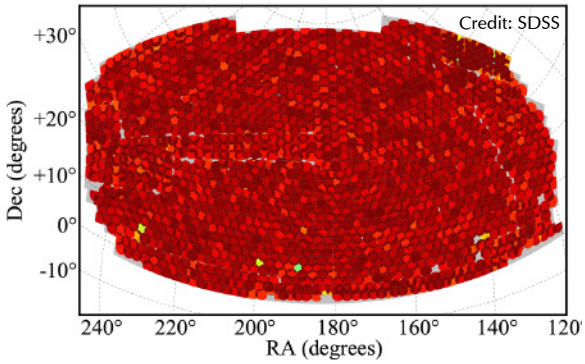
ICCUB MEMBERS

Ariño, A. • Bellini, E. • Bernal, J.L. • Canal, R. • Cuesta, A.J. • Gontcho A Gontcho, S. • Guzmán, R. • Hu, B. • Jimenez, R. • Juan, E. • Manrique, A. • Miralda-Escudé, J. • Pérez-Ràfols, I. • Porqueres, N. • Sala, F. • Salvador, E. • Simpson, F. • Verde, L.

VISITING SCHOLARS

Ruiz-Lapuente, P.

Research in Cosmology and Large Scale Structure ranges from the inflationary model, the Cosmic Microwave Background and the epoch of reionization, to the formation and evolution of galaxies and the distribution of gas in space, including statistical applications and data analysis. Research is also being carried out into the nature of dark matter and the primordial fluctuations that gave rise to galaxies and larger structures in the universe. ICCUB cosmologists participate in several cosmological projects such as SDSS, EUCLID, LSST and CorE.



BOSS SKY COVERAGE (NORTH GALACTIC CAP)

In 2015 the SDSS3 final data release 12 was made public along with the cosmological interpretation and latest BAO (baryon acoustic oscillations) and redshift-space distortions measurements. The figure shows the observations made by the BOSS project.

Activity 2015

ICCUB researchers continued their work on the metal line properties of Damped Lyman Alpha systems (DLAs) and the large-scale bias factors of MgII systems, quasars and DLAs. The cross-correlation of quasars with Lyman-α emission was also investigated. Lyman-α emission is a potential new tracer of star formation, and a first tentative detection at very low surface brightness levels has been made possible by using the BOSS fiber spectra that were designed to target low-redshift galaxies but which may contain Lyman-α light at higher redshift as well. The detection may indicate a very high escape fraction of Lyman-α photons from star-forming galaxies, or else a special Lyman-α emission process near quasars arising from the ionizing radiation or quasar jets. Furthermore, new results on the Lyman-α forest non-linear power spectrum from hydrodynamic simulations were presented, with a new practical formula for fitting observational results from BOSS and future surveys.

Apart from this, the implications of the latest cosmological observations for neutrino properties were elucidated, producing stringent and robust constraints on its mass scale and developing the methodology to further strengthen the indirect detection of the cosmic neutrino background.

ICCUB members also presented a new way to analyze cosmic shear surveys and a new way to analyze cosmological survey data, which yield insights on the nature of dark energy.

Finally, the first lower limit on possible values of the cosmological constant was reported. The observed value of the cosmological constant is a great puzzle. Upper limits on this value are known: a value too high would not enable gravity to create enough galaxies, stars and planets to support life. But following this argument any infinitely small (or even negative) value would be acceptable. Why then is the observed value small yet non zero and positive? It was shown that a too low value is also bad because of the effect of cosmic explosions on the survival of complex life on Earth-like planets.

EXPERIMENTAL PARTICLE PHYSICS

LINES OF RESEARCH

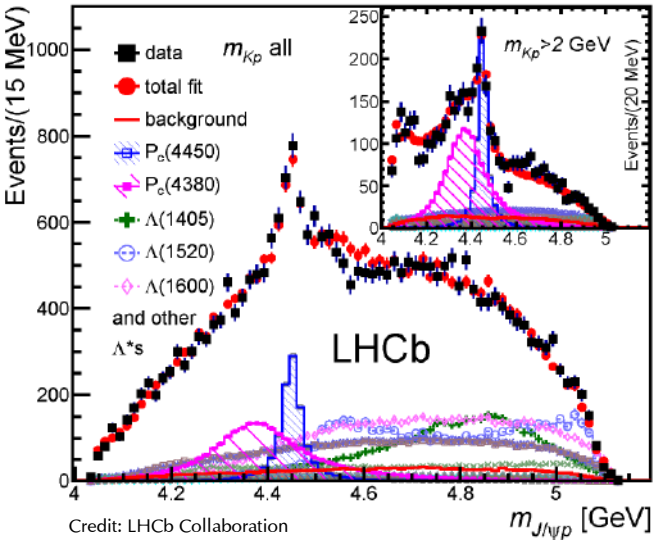
- Physics of beauty and charm mesons
- Charge-Parity symmetry violation
- Search for deviations from the Standard
- Model in rare B and charm meson decays
- Development of distributed calculation methods using grid and cloud computing
- Simulation and study of the radiation hardness of avalanche photodetectors
- Design, construction, and operation of instrumentation for astrophysics and high energy physics experiments as well as medical imaging techniques
- Design of Geiger mode avalanche photodiodes for tracking detectors of future accelerators
- Development of time of flight readout electronics for photo-sensors

ICCUB MEMBERS

Casajús, A. • Ciaglia, D. • Garrido, L. • Gascón, D. • Gómez, S. • Graciani, R. • Graugés, E. • Marín, C. • Mauricio, J. • Picatoste, E. • Rives, V. • Ruiz, H. • Sanuy, A.

ICCUB's experimental particle physicists are specialized in the study of flavor physics. Specifically, in measuring charge-parity violation effects and rare decays of particles containing b or c quarks. Currently the group is fully involved in LHCb experiment data analysis and on its upgrade project.

The LHCb detector, one of the four detectors of the Large Hadron Collider (LHC) in CERN, is designed to study this asymmetry through the b and anti-b particle pairs produced in proton collisions. The ICCUB, aside from its participation at a scientific level, undertook the design, production and installation of the electronics of the SPD (Scintillator Pad Detector) part of the calorimeter and participated in the development of the Worldwide LHC Computing Grid (WLCG) computer network and the DIRAC software. An updated LHCb detector is currently being designed and scheduled for mid 2018 to start operation. ICCUB researchers participate in the design of the readout electronics of both the calorimeter and the new central tracker, which will be based on scintillating fibers. An important program of technology transfer was consolidated in 2015, with partnership from industry, where applications to medical imaging and time of flight measurements of photo-sensor electronics are exploited.



Credit: LHCb Collaboration

OBSERVATION OF PENTA-QUARKS IN LHCb

In July 2014 the LHCb collaboration reported the discovery of a new class of particle, the "pentaquark", a particle formed by five quarks. The figure shows a fit to the J/ψp invariant mass spectrum for the $\Lambda^b_0 \rightarrow J/\psi p K^-$ decay, with each fit component shown individually. The contribution of the pentaquarks is shown by hatched histograms.

Activity 2015

In 2015 the research focused on the study of the radiative B meson decays. This kind of decays offer a unique opportunity to look for new physics beyond the Standard Model by precisely measuring the photon polarization of such decays, analyzing either the B meson decay time or the angular distribution of its decay products.

Apart from this, the ICCUB has contributed to the design and development of the readout electronics for the calorimeter and the Scintillating Fiber Tracking of the LHCb experiment upgrade. The cross-application of the LHCb-generated knowledge in photo-sensor read out electronics has generated several service contracts with semiconductor companies and several medical applications (PET devices).

GALAXY STRUCTURE AND EVOLUTION

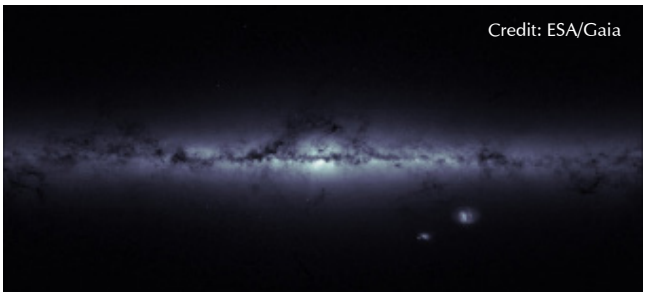
LINES OF RESEARCH

- The stellar constituents of the galactic disk and halo
- The stellar luminosity calibration
- Modeling of galaxy aggregations
- Formation and evolution of galaxies

ICCUB MEMBERS

Abedi, H. • Balaguer, D. • Carrasco, J.M. • Casamiquela , L. • Castañeda, J. • Fabricius, C.V. • Figueras, F. • García, C. • Gracia, G. • Jordi, C. • Luri, F.X. • Masana, E. • Miralda, J. • Mor, R. • Palmer, M. • Reina, M. • Roca, S. • Romero, M. • Solanes, J.M. • Torra, J. • Voss, H. • Weiler, M.

At present, the research in Galactic Astronomy at the ICCUB is highly influenced by the preparation of the scientific exploitation of the ESA's Gaia mission, in which ICCUB researchers are deeply involved (see *Very Large Data Processing and Analysis*). Beyond this, research is also being carried out in the modeling of galaxies and their aggregations, the study of stellar constituents and the calibration of stellar luminosity. Some members are also devoting their efforts to investigating the very complex process of galaxy formation and to explore the physical mechanisms driving their evolution. ICCUB members coordinate the Red Española de Explotación Científica de Gaia and are active members of the international networks created for the scientific exploitation of Gaia.



GAIA STELLAR DENSITY MAP

The outline of our Galaxy, the Milky Way, and of its neighbouring Magellanic Clouds, in an image based on housekeeping data from ESAs Gaia satellite, indicating the total number of stars detected every second in each of the satellite's fields of view.

Activity 2015

In 2015 new tools were explored for the scientific exploitation of Gaia —innovative and far from traditional. ICCUB members took part of the Gaia-ESO survey, the Open Clusters OCCASO survey and the WEAVE@WHT consortium to complement Gaia data with high resolution spectroscopy from ground. They also worked in the definition of synergies with large present and future surveys.

The work developed in galaxy modeling included the characterization of the central bar, the development of novel methods to unveil the nature of the spiral structure and the update of the kinematic model to describe different morphologies of the galactic warp, such as lopsidedness. High-resolution cosmological N-body with hydrodynamics simulations for Milky Way-like galaxies allowed new insights in the disk large scale structures and baryonic content to be provided.

Concerning the characterization of the Milky Way constituents, ICCUB researchers made the astrometric and photometric analysis of open clusters and they determined their radial velocities. The availability of the Gaia Object Generator (GOG) was used in this occasion for scientific analysis and work is in progress to improve the Besançon-Barcelona Population Synthesis Galaxy Model. The Besançon Galaxy Model was used as a background to evaluate the Gaia capabilities to detect ultra-faint dwarf galaxies. Besides, by means of controlled collisionless simulations of the previrialization stage of galaxy groups, it was possible to demonstrate that the fraction of the merger time in which strong dual AGN activity can be detected is independent of the orbital configuration and typical of the order of a few percent.

Apart from this, controlled simulations of forming galaxy groups were used to demonstrate that the multiple mergers that take place during the hierarchical build-up of these systems are able to create fully realistic first-ranked galaxies without the additional consideration of a dissipative component, thus making a strong case for considering hierarchical dissipationless merging a viable route for the formation of the largest galaxies of the universe.

GRAVITATION AND COSMOLOGY

LINES OF RESEARCH

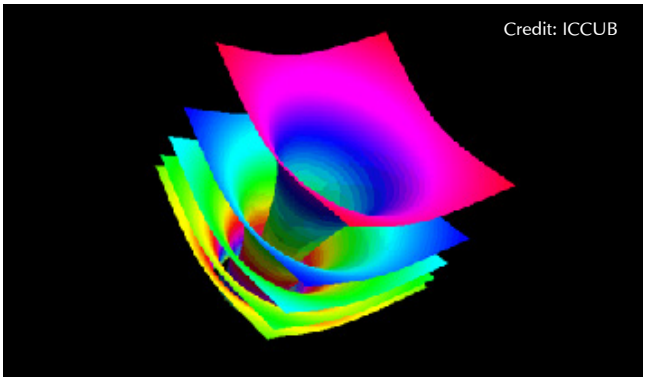
- Dark matter and dark energy in cosmology and in particle physics
- Quantum and semiclassical gravity
- AdS/CFT correspondence and holography
- Black holes

ICCUB MEMBERS

Di Dato, A. • Emparan, R.A. • Fernández, A. • Fernández, I. • Fiol, B. • Garolera, B. • Garriga, J. • Germani, C. • Guasch, J. • Izumi, K. • Karimkhani, E. • Llosa, J. • Martínez, M. • Molina, A. • Mateos, D. • Notari, A. • Pantelidou, C. • Pravos, D. • Rocha, J. • Solà, J. • Tarrio, L.J. • Torrents, G. • Triana , M. • Verdaguer, E. • Zilhao, M.

Research made by this ICCUB group focuses on cosmology, gravity and holography. Its members also collaborate with researchers in the Theoretical Physics group to make incursions into particle physics and the gauge/gravity correspondence.

In the area of cosmology, research is focused on models of dark energy, inflationary models, the study of the Cosmic Microwave Background and multiverse scenario, whereas in the area of gravity, research is focused on the study of black holes in string theory and in higher-dimensional spacetimes, on the one hand, and on quantum gravity in de Sitter spaces, on the other hand. In the area of gauge/gravity correspondence, research is devoted to the study of the quark-gluon plasma and the interplay between the gauge/gravity correspondence and supersymmetric localization.



CURVED SPACETIME: FROM BLACK HOLES TO COSMOLOGY

Einstein's theory of gravity tells us that the geometry of space and time gets dramatically distorted in the vicinity of black holes and at the beginning of the universe. This challenges the foundations of Einstein's theory itself and calls for the incorporation of the effects of quantum mechanics. A surprising spin-off of this research is the possibility of describing a ball of quark-gluon plasma as a black hole in a higher-dimensional space.

Activity 2015

Concerning cosmology, in 2015 ICCUB researchers of this group extended their research about anomalies in the Cosmic Microwave Background. Furthermore, they considered possible observational imprints of black holes and topological defects in multiverse scenarios, and they discussed a mechanism to stabilize the electroweak vacuum via non-minimal couplings to gravity. Finally, they developed and studied cosmological models with a dynamical vacuum energy.

In regard to gravity, ICCUB researchers continued the development of the inverse-dimensional expansion for black holes and they applied it to study several of their properties. Additionally, they studied the breakdown of the semi-classical description of black holes.

Lastly, concerning holography ICCUB researchers analytically constructed gravity solutions dual to three-dimensional gauge theories with flavor, both with zero and non-zero quark density. These solutions display quasi-conformal dynamics (walking) in a range of parameters. ICCUB researchers also proposed a new way to compute the energy loss by radiation of heavy probes coupled to conformal field theories.

HADRONIC, NUCLEAR AND ATOMIC PHYSICS

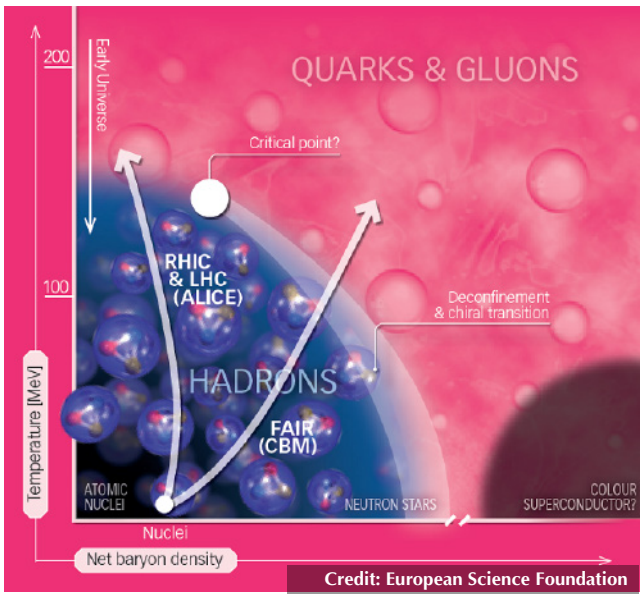
LINES OF RESEARCH

- Nuclear structure. Nuclear symmetry energy.
- Dense and hot nuclear matter and applications in nuclear astrophysics.
- Hadronic physics. Strangeness and charm in the nuclear medium.
- Relativistic heavy ion collisions.
- Lattice QCD calculations of light nuclear systems.
- Radiation transport and interactions of radiation with matter.
- Ultra-cold atomic gases. Bose-Einstein condensates.

ICCUB MEMBERS

Centelles, M. • Feijoo, A. • Fernández, J.M. • Haibo, Q. • Julià, B. • Magas, V. • Maneu, J. • Parreño, A. • Polls, A. • Ramos, A. • Salvat, F. • Viñas, X.

The group has conducted research on the theoretical description of hadron systems with strangeness and/or charm in the vacuum, in nuclear matter, and in the hot medium generated in relativistic heavy ion collisions. In the atomic sector, models for the numerical study of the interaction of electrons, photons and ions with matter have been improved, while many efforts have also been devoted to the study of ultra-cold atomic gases in various geometrical configurations. These multidisciplinary activities have a direct connection with experiments at world leading physics laboratories, as BNL, CERN, ELSA-Bonn, Fz-Jülich, GSI, GANIL, JLAB, JPARC, MAMI-Mainz, RIA or RIKEN.



THE PHASE DIAGRAM OF NUCLEAR MATTER

The phases of nuclear matter over different values of temperatures and pressure (taken from the NuPECC Long Range Plan 2010 Booklet "Nuclear physics: science and applications")..

Activity 2015

A unified equation of state for neutron stars has been obtained from the outer crust to the core of the star based on modern microscopic calculations. In addition, new constraints on the neutron skin thickness of nuclei and the nuclear symmetry energy have been derived from the recent experimental determination of the electric dipole polarizability in ^{68}Ni, ^{120}Sn, and ^{208}Pb.

Using Lattice QCD calculations, the magnetic moments and magnetic polarizabilities of nuclear systems with A≤4 have been determined, along with the cross section for the radiative capture process np → dγ from extrapolations to physical quark masses.

In the exotic sector, a coupled channel chiral model up to next-to-leading of the $\bar{K}$N interaction has been developed and applied to give predictions for the Λb→ $J/\Psi K\Xi$ and Λb→ $J/\Psi\eta\Lambda$ processes, the latter permitting to analyze the possible existence of a S=-1 partner of the LHCb pentaquark.

Research performed in atomic physics included a theoretical description of the quantum synchronization between coupled ultracold atomic gases, the mesoscopic superpositions of Tonks-Girardeau states and the Bose-Fermi mapping in one-dimensional systems. Simultaneously, code development and applications

HIGH ENERGY ASTROPHYSICS

LINES OF RESEARCH

- High-Energy and Very-High-Energy gamma-ray sources in the Galaxy
- Multi-wavelength observations and theoretical modeling
- Microquasars
- Gamma-ray binaries
- Pulsar wind nebulae
- Active galactic nuclei
- MAGIC and Cherenkov Telescope Array

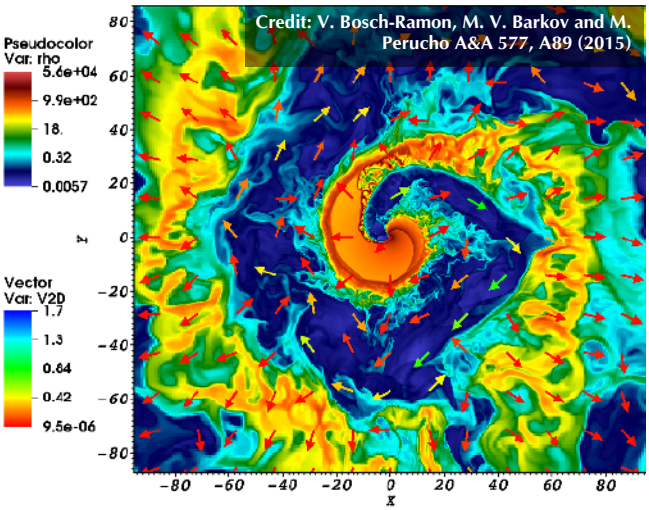
ICCUB MEMBERS

Aliu, E. • Bosch-Ramon, V. • Galindo, D. • Iwasawa, K. • Marcote, B. • Migliari, S. • Moreno , V. • Paredes, X. • Paredes, J.M. • Ribó, M. • Torres, N. • Zanin, R.

A general aim of ICCUB researchers working on this field is to achieve a better understanding of the high energy galactic and extragalactic sources, gathering data over a large wavelength range (from radio to TeV energies) as well as modeling emission processes in different scenarios (pulsar winds, jets, shocks, interaction with the interstellar medium, etc.).

In particular, ICCUB researchers are interested in the observational and theoretical study of microquasars, gamma-ray binaries and active galactic nuclei, as well as in their interactions, as objects that exhibit a non-thermal behavior in the emission from radio to gamma rays related to powerful relativistic outflows generated in a compact object.

ICCUB high energy astrophysicists have been members of the MAGIC Collaboration since February 2006 and they are currently participating, together with experimental physicists and engineers from the ICCUB, in the Cherenkov Telescope Array (CTA) project, an initiative to build the next generation of ground-based gamma-ray instrument (see *Electronic and Instrumentation Development*).



THE ORBITAL EVOLUTION OF COLLIDING STAR AND PULSAR WINDS

A relativistic and a non-relativistic wind interact in a high-mass binary hosting a non-accreting pulsar. The orbital motion induces non-ballistic unstable structures that are potential sites of high-energy radiation, in one of the most powerful gamma-ray emitting sources in the Galaxy.

Activity 2015

In 2015 research in High Energy Astrophysics was focused on observations, theoretical modeling and numerical simulations to understand the physics of relativistic astrophysical outflows. Semi-analytical and numerical calculations of the hydrodynamics and the radiation processes taking place in high energy sources were developed in order to enable predictions of observable features in the different scenarios under investigation (small and large scales, galactic and extragalactic sources of outflows), which can be tested observationally.

On the observational side, ICCUB researchers conducted simultaneous Chandra/VLA observations of MWC 656, the first Be/black hole system, which revealed an extremely quiescent state. They also detected the gamma-ray binary LSI+61303 at 150 MHz with both GMRT and LOFAR for the first time, and they studied its low-frequency variability. In addition, ICCUB members continued the monitoring of gamma-ray binaries with the TFRM and they started dedicated campaigns with the TJO. Finally, they conducted the first interferometric observations with VLTI to resolve the discs of Be stars in gamma-ray binaries.

PARTICLE PHYSICS PHENOMENOLOGY

LINES OF RESEARCH

- Standard Model and beyond at the LHC
- Physics of neutrinos, with an emphasis on astrophysics and cosmology
- B-physics, with an emphasis on the analysis and physical reach of the LHCb detector
- QCD in extreme conditions: heavy ion experiments at the LHC, FAIR and other accelerators
- Heavy quark effective theory and other effective theories of QCD
- Perturbative QCD: parton distribution functions. Jet physics
- Axions and other dark matter candidates
- Physics of future colliders

ICCUB MEMBERS

Attems, M. • Bergström, J. • Casalderrey, J. • D'Enterria, D. • Espriu, D. • García-Echevarria, M. • Gómez, A. • González, M.C. • Guasch, J. • Hofer, L. • Latorre, J.I. • Mescia, F. • Pablos, D. • Renau, A. • Solà, J. • Soto, J. • Taron, J.M. • Tywoniuk, K.

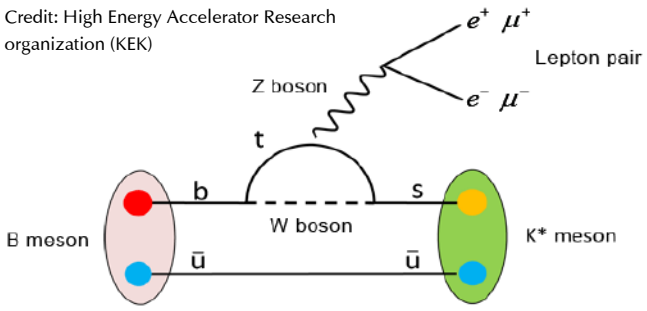
VISITING SCHOLARS

Andrianov, A.

The phenomenological aspects of Particle Physics are central to ICCUB research, covering many aspects of the areas reported in the hep-ph, hep-th and hep-lat archives. The composition of the group reveals this variety of interests, extending to many of the forefront areas of research in Particle Physics.

Several members share their activity in phenomenological aspects with their work in more formal parts of theoretical physics and gravitation. Furthermore, they have an ever-growing interest in the cosmological and astrophysical implications of particle physics phenomenology. There is also a considerable overlap of interests with the Experimental Particle Physics group members from the LHCb experiment in the area of b-physics.

Credit: High Energy Accelerator Research organization (KEK)



SCHEMATIC VIEW OF A B DECAY

Weak semileptonic B decays, such as the one in the figure, show some desviations from the Standard Model predictions which are being actively investigated by ICCUB researchers. This research will allow to address issues relevant for the phenomenology of the Standard Model and its possible extensions within Lattice Quantum Chromodynamics (QCD).

Activity 2015

The activity was influenced to a large extent by the first LHC results. Studies in this area focused on effective theories from the symmetry breaking sector of the Standard Model, some aspects of supersymmetric theories, string phenomenology, flavor physics (particularly b-physics) and physics beyond the Standard Model that the LHC will continue exploring in the years to come. ICCUB members also sharpened their theoretical tools to take adequate stock of run II of the LHC.

In the area of b-physics, ICCUB researchers continued studying possible new observables that might reveal the presence of new physics, and participated actively in the European Twisted Mass lattice Collaboration (ETMC).

Apart from this, effective theories of QCD, especially in the heavy quark sector, were intensively studied. Several features of heavy ion collisions, particularly in the domain of hard probes and the study of properties of QCD under extreme conditions, also received considerable attention. Furthermore, QCD-related research included work on lattice field theory, especially in connection with b-physics, parton distribution functions using neural networks, and jet physics using resummation techniques and effective theories.

Finally, very relevant contributions were made in the field of neutrino physics, mostly in the form of global fits to the data. Axion physics and other dark matter candidates as well as dark energy also merited attention from ICCUB members of this group, who worked in collaboration with other groups at ICCUB.

STAR FORMATION

LINES OF RESEARCH

- High-angular-resolution observations of the first stages of stellar evolution
- Outflows, jets, and accretion disks in young stellar objects, and jets in planetary nebulae
- Computational models of star-forming regions, from large-scale SN-driven turbulence to individual stars

ICCUB MEMBERS

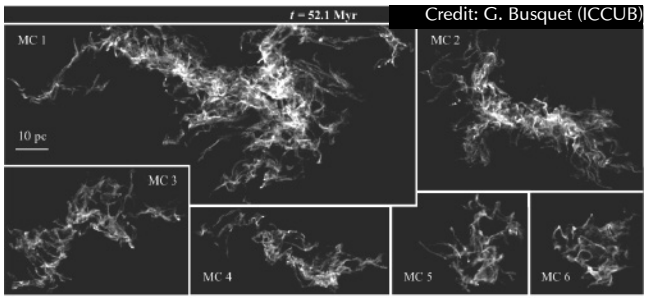
Estalella, R. • Juárez, C. • López, R. • Padoan, P.

VISITING SCHOLARS

Torrelles, J.M. • Yun, J.

ICCUB research in this field focuses on the investigation of the dynamics of star-forming regions in our galaxy and on the study of the first stages of stellar evolution. ICCUB researchers intend to acquire a perspective as wide as possible by pursuing both an observational approach, ranging from the optical to the radio domain, and a theoretical approach, based on state-of-the-art supercomputer simulations of the evolution of star-forming regions. Observations and simulations are compared through the generation of synthetic observations of the simulations.

Specific areas of research include: the characterization of the role of the magnetic field in the star formation process and in the launching and collimation of the astrophysical jets associated with young stellar objects and planetary nebulae; the investigation of the early stages of the formation of massive stars; the search for signatures of planet formation within the protoplanetary disks; the study of the transition from hot molecular cores to bright HII regions; the numerical modeling of the turbulent fragmentation process to understand the origin of the stellar initial mass function and the star formation rate in molecular clouds; and the numerical modeling of the ISM on very large scale to study the role of supernova explosions in the driving of the ISM turbulence and in the formation and disruption of giant molecular clouds.



SYNTHETIC MOLECULAR CLOUDS

Examples of molecular clouds selected from a snapshot of a simulation of SN-driven turbulence in a volume of 250 pc size.

Activity 2015

The observational study of fragmentation in high-mass star-forming regions continued with high angular resolution observations of the 1.3 mm continuum emission carried out with the Submillimeter Array (SMA) toward two hubs, G14.2-hub-N and G14.2-hub-S, in the Infrared Dark Cloud G14.225-0.506 together with observations of the dust emission at 870 and 350 μm obtained with APEX and the CSO single-dish telescopes. The density structure of the two hubs was studied by means of a simultaneous fit of the radial intensity profile at 870 and 350 μm and the spectral energy distribution. ICCUB researchers also studied the interplay between magnetic field, turbulence, gravity and UV radiation feedback.

The computational effort was focused on the study of supernova (SN) feedback in the star-forming gas. ICCUB researchers carried out the highest-resolution simulation to date of SN-driven turbulence, on a scale of 250pc, to study the origin and dynamics of molecular clouds (MCs). Thanks to the large dynamic range of the simulation, not only individual MCs and SN remnants were well resolved, but they also formed self-consistently as the result of larger-scale dynamics, with realistic initial and boundary conditions. The simulation provided a large sample of clouds and thus constrained also the probability distribution of cloud properties. It was found that clouds selected from the simulations (see figure above) have mass and size distributions, and velocity-size and mass-size relations consistent with the observations. Using tracer particles, ICCUB members also studied their evolution and found that they form and disperse in approximately four dynamical times. They concluded that SN-driven turbulence alone can explain the formation and dynamics of star-forming clouds.

THEORETICAL PHYSICS

LINES OF RESEARCH

- String and superstring theory
- Exact results in supersymmetric field theories
- Applications of the gauge/string duality to strongly coupled systems
- Properties of many-body quantum systems. Quantum phase transitions
- Quantum information
- Quantum simulations

ICCUB MEMBERS

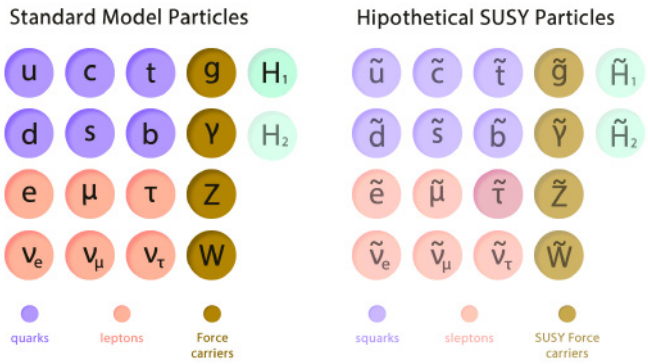
Alsina, D. • Attems, M. • Casalderrey, J. • Cervera, A. • Dector, A. • Emparan, R.A.• Espriu, D. • Fernández-Faedo, A. • Fiol, B. • Gabbanelli, L. • Garolera, B. • Gomis, J. • Iblisdir, S. • Latorre, J.I. • Mariño, M. • Mateos, D. • Pablos, D. • Pantelidou, C. • Pons, J.M. • Rocha, J. • Russo, J.G. • Solà, J. • Tarrio, L.J. • Tywoniuk, K. • Zilhao, M.

VISITING SCHOLARS

Lizzi, F. · Talavera, P.

ICCUB activities in theoretical physics cover a wide spectrum of areas reported in the hep-th and quant-ph archives.

Many ICCUB researchers are interested in string theory, a field which has recently been inspired by enormous activity in the gauge/string duality conjecture that allows a treatment of several types of strongly coupled theories in terms of gravity duals. Supersymmetric field theories are studied seeking in an effort to understand the ultraviolet behavior of theories with extended supersymmetries and deriving exact results. Research in quantum information is also active too focusing on several topics such as entanglement entropy, tensor networks, quantum error correction, many-body quantum systems, topological order, ultra-cold gases and quantum simulation. The quantum information researchers at the ICCUB are in close collaboration with some of the groups at ICFO.



Credit: ICCUB

THE STANDARD MODEL AND BEYOND

Supersymmetry (SUSY) is one of the possible extentions of the Standard Model in which ICCUB members are working. It predicts a partner for each particle of the Standard Model, and it could solve a major problem of this model: fixing the mass of the Higgs boson. For SUSY theories to work, a second Higgs particle must be also introduced in the Minimal Standard Model

Activity 2015

In 2015, a very active research line searched for exact results in nonabelian gauge theories. Using supersymmetric localization, matrix integrals and resurgence techniques, various aspects of supersymmetric gauge theories in three and four dimensions were studied, including quantum phase transitions and vacuum expectation values of Wilson operators. In addition, ICCUB researchers used the AdS/CFT duality to study various Yang-Mills theories at finite heavy-quark density and their renormalization group flows. Strongly coupled condensed matter systems and other theories involving large couplings (such as some extended EWSB sectors) were also investigated and ICCUB researchers plan to continue this investigation in the future. The group also worked in the more theoretical aspects of the holographic treatment to heavy ion collisions. Concerning quantum information theory, research focused on maximally entangled states, quantum simulations, quantum statistical systems and other related issues. Compression algorithms were also studied. Finally, there was also some activity in the areas of non-commutative geometry, emergent gravity and black hole physics.

ADDITIONAL AREAS OF RESEARCH

ASTRODYNAMICS AND CELESTIAL MECHANICS

ICCUB researchers on Astrodynamics are devoting their efforts to address some fundamental issues concerning the problems related to orbit and attitude control in formation flying of swarms of spacecraft; the development of some dynamical indicators to determine regions and structures that separate different dynamic regimes in autonomous and non-autonomous dynamical systems; the optimal transfer to polar orbits around the Moon; the analysis of the phase space in the vicinity of an irregular asteroid; and the study of mass transport mechanisms in the Solar System.

LINES OF RESEARCH

- Develop tools to explain in a natural way different astronomical and astrodynamical patterns

ICCUB MEMBERS

Cheng, Y. • Gómez, G. • Olikara, Z.P. • Paita, F. • Pérez, D.

VISITING SCHOLARS

Jorba, À.

Activity 2015

During 2015 ICCUB researchers worked on the computation of Lagrangian Coherent Structures for the determination of invariant manifolds and long-term stability regions in Celestial Mechanics and Astrodynamics.

Moreover, they finished their activities related to the end of life disposal of spacecraft at the libration point regions. The completed their studies on the LEO-GEO transfers visiting the libration point regions of the Sun-Earth system.

Lastly, ICCUB researchers continued their research in spacecraft formation flight considering both the orbit control using Lorentz forces and attitude control by means of a behavioral approach.

ASTRONOMICAL IMAGE PROCESSING AND HIGH ANGULAR RESOLUTION TECHNIQUES

The group on Image Processing at ICCUB is focused on the use of the wavelet and curvelet transforms to improve the ability of image sensors to detect faint stars and moving objects, as well as the development of applications to astronomy and remote sensing. Particularly, the group is studying the effects of the curvelet transform over interferometric images and the effect of deconvolution (using wavelets and curvelets based maximum likelihood estimator) in adaptive optics observations. Work is also under way to obtaining super-resolution using additive-substitutive wavelets techniques on remotely sensed images. The group also studies observational techniques and software development for the automatic detection of Space debris using the TFRM telescope under the SSA/SST programs of the ESA and the European Union.

LINES OF RESEARCH

- Image deconvolution by means of multiresolution analysis (wavelet and curvelet transform)
- Image fusion and super-resolution by means of multi-resolution analysis (wavelet transforms)
- Application of Image deconvolution to Space Debris observation
- Observational techniques and software for space debris detection

ICCUB MEMBERS

Merino, M.T • Núñez,J.C. • del Ser, D.

Activity 2015

During the year 2015 the main activities were focused on the application of deconvolution to increase the limiting magnitude of images obtained for space debris detection. Apart from this, a considerable effort was devoted to obtaining standalone software that could detect and extract the position of space debris in astronomical images. In addition, the group continued studying the deconvolution by multiresolution of images obtained using adaptive optics and the comparison of classical, myopic and blind algorithms. Studies of image fusion and superresolution also continued.

CHIRALITY AND PREBIOTIC CHEMISTRY

The experimental expertise in this area is the study of the stereo and enantioselective effect of hydrodynamic flows in the formation of supramolecular systems by self-assembly as well as the phase transitions from achiral building blocks to chiral supramolecules. The group also works in the definition of thermodynamic scenarios in applied chemistry in which such a transition to chirality is possible. The general objective of these works is to understand chiral polarizations in the spontaneous emergence of chirality during the chemical evolution.

LINES OF RESEARCH

- Effect of mechanical forces (flows with gradient of shear rates) on the emergence of chirality in soft matter
- Mirror symmetry breaking in autocatalytic systems: Crystallizations and aggregations showing critical phenomena

ICCUB MEMBERS

Crusats, J. • El-Hachemi, Z.

VISITING SCHOLARS

Ribó, J.M.

Activity 2015

In 2015, the effect of mechanical forces on racemic mixtures on chiral nanotubes/self-assembly of achiral amphiphilic porphyrins were studied. Additionally, new substituted amphiphilic porphyrins were synthesized in order to elucidate which were the structural patterns leading either to chiral or to achiral aggregates and, in the case of formation of racemic conglomerate mixtures, which were the effects and the experimental conditions necessary to lead to a bias from the racemic composition. Finally, results will soon be published of research into the theoretical studies on how enantioselective autocatalysis couple to reactions implying the deletion of chiral substance leads to parity breaking.

HELIOSPHERIC PHYSICS AND SPACE WEATHER

LINES OF RESEARCH

- Solar energetic particle (SEP) events, interplanetary shocks and related solar activity
- Modeling gradual proton events: magnetohydrodynamic shock simulations plus particle transport simulations and applications
- Modeling near-relativistic electron events: inversion methods and applications
- Space weather: Engineering models for prediction of peak flux and fluences of solar energetic particle events

ICCUB MEMBERS

Àgueda, N. • Aran, A. • Pacheco, D. • Sanahuja, B.

The Heliospheric Physics and Space Weather group of the ICCUB mainly focuses on solar energetic particle (SEP) events triggered by solar activity and by interplanetary disturbances. Solar flares and coronal mass ejections, the main agents of SEP acceleration, together with proxies of solar activity, the solar wind plasma and the interplanetary magnetic field, are the background components of the SEP scenario. In this context, ICCUB researchers are performing both data analysis and modeling of SEP events including multi-spacecraft events.

ICCUB members in this area also provide scientific support to other Spanish groups involved in the development of instrumentation for the ESA's Solar Orbiter mission (see *Electronic and Instrumentation Development*).

Activity 2015

During 2015 ICCUB researchers worked in HESPERIA (High Energy Solar Particle Events foRecastIng and Analysis), a project funded by the EU under the H2020 program. Their role in this project is to lead the development of the first inversion method to infer the sources of relativistic SEPs (> 500 MeV), based on the observations provided by the worldwide network of ground-based neutron monitors. Relativistic SEPs provide key observations for understanding the most energetic acceleration processes occurring during solar eruptive events.

The team also continued to study the variation of the proton intensity–time profiles in gradual SEP events in relation to the relative observer's position in the interplanetary space to the direction of propagation of the particles sources.

Finally, ICCUB members worked in an ESA-sponsored project, SOL2UP, to update the physics-based code SOLPENCO2 (SOLar Particle ENgineering COde 2). This tool provides the heliocentric radial distance scaling (from 0.2 AU to 1.6 AU) of the maximum intensity and fluence during SEP events for 5-200 MeV protons.

MICROGRAVITY AND TWO-PHASE FLOWS

LINES OF RESEARCH

- Dynamics of turbulent bubble flows in microgravity.
- Controlled boiling in microgravity.

VISITING SCHOLARS

Casademunt, J. • Ruiz, J.X.

In recent years, ICCUB's visiting scholars have been consistently studying the formation and management of small bubbles under microgravity conditions, in particular in the context of turbulent flows, an area of fundamental interest in multiphase flows and with important applications in space technology, from life support systems to thermal control of space vehicles.

Experiments are conducted in the European Space Agency (ESA) Drop Tower facility at ZARM (Bremen, Germany). The use of the tower is supported by ESA and the research is financially supported by the US Air Force Office of Scientific Research (AFOSR) through the European Office of Aerospace Research and Development. The main objective is to elucidate physical mechanisms that control bubble formation, bubble-flow interactions, and heat exchange in the absence of gravity, in view of improving current designs and searching for new strategies for efficient thermal control in microgravity environments.

Activity 2015

In 2015 the group finished the optimization and characterization of an innovative design for capillary boiling through controlled localized nucleation. The prototype system was proven to work robustly and independently of the gravity environment. The different regimes of bubble formation were identified and quantitatively characterized. The device generates regular slug flows that can be used as input for heat-exchange or other devices. Some fundamental questions concerning scaling of nucleation were also raised and solved by means of a simplified simulation model. Possible extensions of this work are currently being explored.

ELECTRONIC AND INSTRUMENTATION DEVELOPMENT

ACTIVITIES

- Cameras
- Radiation detectors and photo detectors
- Microelectronics
- Radiation hard design and qualification
- Hardware for Instrumentation and Communications
- Embedded Software and Firmware
- Signal processing

ICCUB MEMBERS

Casas, A. • Ciaglia, D. • Diéguez, A. • Garrido, L. • Gascón, D. • Gómez, J.M. • Graciani, R. • Graugés, E. • Mauricio, J. • Núñez, J.C. • Oriol, P. • Picatoste, E. • Ribó, M. • Roma, D. • Ruiz, H. • Sabater, J. • Sanuy, A.

ICCUB members develop instrumentation, front-end electronics, read-out and real-time processing systems, application-specific integrated circuits (ASICs) and signal processing algorithms/tools for astrophysics, space projects, high energy physics and medical imaging among other fields. At the moment, the ICCUB is participating in the following projects:

CTA: The CTA (Cherenkov Telescope Array) project is an initiative to build the next generation ground-based very high energy gamma-ray instrument, which will serve as an open observatory to a wide astrophysics community and which will provide a deep insight into the non-thermal high-energy universe. In addition to physics case studies, the ICCUB is involved in the design and prototyping of the CTA telescope cameras. This involves designing ASICs for preamplification, signal processing and triggering.

LHCb: LHCb is one of the four general purpose detectors of the Large Hadron Collider (LHC) at CERN, designed to study CP violation through the analysis of b and anti-b particle pairs produced in proton collisions. The ICCUB undertook the design, production and installation of the electronics of the SPD (Scintillator Pad Detector) part of the calorimeter. An updated LHCb detector is currently being designed and scheduled for late 2018 to start operation. The ICCUB participates in the design of the readout electronics of both the calorimeter and the new central tracker.

MIRADAS: The ICCUB is part of the MIRADAS project of the Gran Telescopio de Canarias (GTC). The basic MIRADAS concept is a near-infrared multi-object echelle spectrograph operating at spectral resolution R=20,000 over the 1-2.5μm bandpass. MIRADAS selects targets using ~12 deployable probe arms with pickoff mirror optics, each feeding a 4.0x1.2-arcsec field of view to the spectrograph. The ICCUB is responsible of the software that controls the probe arms. It determines the targets of each arm, and the sequence followed to reach it.

Solar Orbiter: The ICCUB is part of the Polarimetric and Helioseismic Imager instrument for the Solar Orbiter mission (SO/ PHI). The ICCUB responsibility is the development and implementation of an Image Stabilization System (ISS) that includes a camera, a controller for a piezo-electric based Tip-Tilt mirror, and the control firmware for the FPGA that controls the whole system. The ISS has been optimized minimizing the power consumption while reaching the required performance.

Telescope Fabra-ROA: ICCUB researchers are working since more than ten years ago in the Telescope Fabra-ROA Montsec (TFRM), a remote operated Baker-Nunn Camera located at the Observatori Astronomic del Montsec (OAdM). Since the beginning of science TFRM operations, ICCUB researchers have been conducting two specific and distinct surveys: super-Earths transiting around M-type dwarfs stars and geostationary debris in the context of Space Situational Awareness / Space Surveillance and Tracking (SSA/SST) programs. In parallel other programs as the search for near earth objects (NEO) and the observations of high energy sources are carried out regularly.

PET Imaging: The experience gained in the development of scientific instrumentation, particularly in the design of ASICs for high speed photodetectors like photomultiplier tubes (PMTs) or Silicon photomultipliers (SiPM, MPPC, GAPDs, etc.) is now being applied to the medical imaging field. The ICCUB has developed several ASICs for Positron Emission Tomography (PET) with Time of Flight (ToF) capabilities.

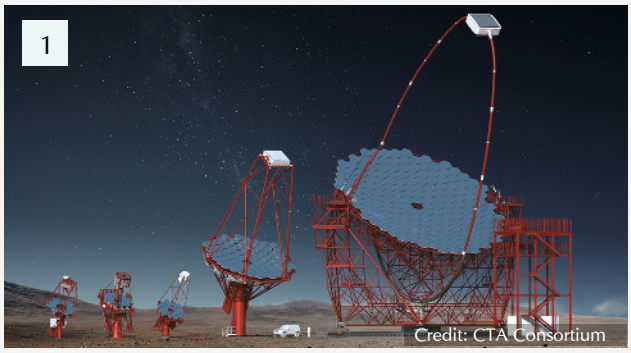
Activity 2015

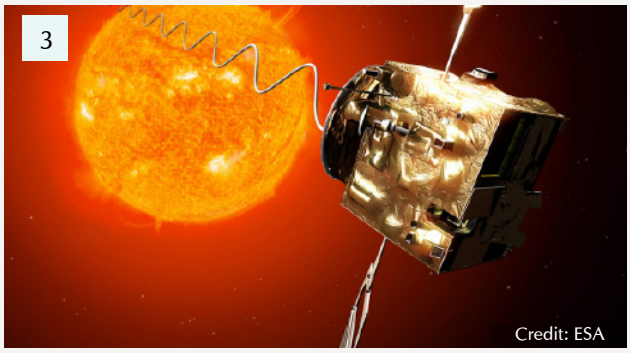
Concerning CTA, three ASICs developed and successfully tested in 2013 and 2014, were prepared for production: PACTAv1.4 (a wideband 16 bit dynamic range current mode PreAmplifier), ACTAf 2Ch F (a wideband pulse amplifier for NECTAr chip) and TL0R1 (a versatile ASIC for L0 triggering in Cherenkov Telescopes). PACTA and TL0R will presumably be used in the Large Size Telescopes (LST) and in the Middle Size Telescopes cameras of CTA, whereas ACTA was designed only to be used in MST. Also, another ASIC, the Multiple Use SiPM Integrated Circuit (MUSIC), was designed to alleviate SiPM limitations due to high capacitance. It is an 8 channel ASIC for SiPM anode readout based on a novel low input impedance current conveyor that can deal with sensors from different manufacturers, covering most of the available SiPM devices in the market. MUSIC offers three main features: (1) differential output of the sum of (selectable) individual input channels for 15 bit accuracy; (2) 8 individual single ended analog outputs and; (3) 8 individual binary outputs.

In regard to the upgrade of the LHCb calorimeter, the final chip (ICECALv3) of the analog signal processing channel was designed, produced, tested and validated for irradiated environment. Apart from this, a third prototype of the PACIFIC ASIC with the input stage and shaper was designed for SiPM readout in the new planned Scintillating Fiber Tracker.

With respect to PET, a multi-purpose 4-channel time to digital converter (TDC) was designed based on a patented resistive clock interpolation mesh circuit, a two-dimensional regular structure with outstanding performance in terms of power consumption, area and low process variability. New PET modules based on ICCUB and CIEMAT ASIC FlexToT were also developed for monolithic crystals.

Finally, in the context of space debris, the TFRM collaborated in the ISON network, and participated in the contract Test-Bed for the Remote Control of an Automated Follow-Up Telescope and in the Spanish Space Surveillance and Tracking (S3T) coordinated by the ESA. The total number of observations reported was 197731, corresponding to 29807 objects (tracks) with a mean of 263.8 objects per night. A mean of 24.6 objects per night without TLE (not in the Space-Track catalog) were observed. Regarding the exoplanet survey, ICCUB researchers continued observing 48 selected fields detecting hundreds of new variable stars and studying the possibility of new exoplanets in them using newly implemented algorithms. Also, the optical counterparts of four high-energy sources were systematically observed to establish the variations of their light curves and the models to explain them.





1 CTA prototype designs From Left: Three different SST, the MST and the LST-
2. Members of the LHCb collaboration posing in front of the detector.
3. Artist view of Solar Orbiter.
4. Telescope Fabra-ROA at Montec Astronomical Observatory.

VERY LARGE DATA PROCESSING AND ANALYSIS

LINES OF RESEARCH

- Star tracker algorithms
- Onboard real-time data processing
- High-performance data compression
- Massive data processing pipelines
- Clustering and cross-matching algorithms
- Data mining techniques
- Grid Computing

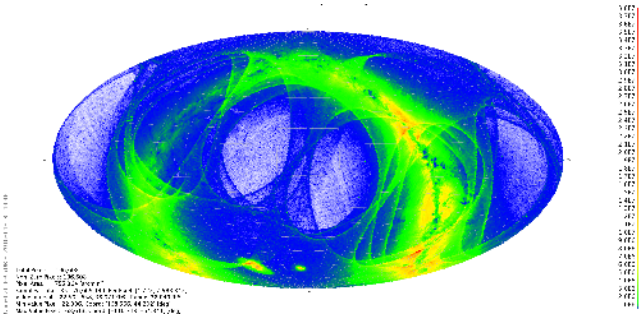
ICCUB MEMBERS

Antiche, E. • Balaguer, D. • Borrachero, R. • Carrasco, J. M. • Casajús, A. • Castañeda, J. • Clotet, M. • Fabricius, C. • Farràs, M. • Figueras, F. • Garralda, N. • González, J.J. • Graciani, R. • Jordi, C. • Julbe, F. • Luri, F.X. • Masana, E. • Molina, D. • Portell, J. • Torra, J. • Voss, H. • Weiler, M.

The ICCUB provides massive data processing algorithms and software which are essential for the success of large international projects. Expertise of ICCUB members is applicable to ground and space The institute is currently participating in the following projects:

Gaia DPAC: ICCUB researchers have a leading role at the highest technological, scientific and management levels of the Gaia Data Processing & Analysis Consortium (DPAC). They have responsibilities in the Simulations, Core Processing and Photometry units, and they manage the Data Processing Center of Barcelona. The ICCUB leads the Gaia Archive unit, funded through an EU project (GENIUS), The knowledge gained on data compression has allowed ICCUB to patent a SW/HW solution and to create a spin-off, DAPCOM.

DIRAC: The DIRAC (Distributed Infrastructure with Remote Agent Control) project is a complete grid solution for a community of users developed by CERN, CNRS and ICCUB. It was created to handle the distributed computing of the LHCb experiment, and now other communities, such as CTA, have begun to use it. The ICCUB is now responsible of the continuous updates. During 2015, the Consortium decided to allow access to new partners, and by the end of 2015 these new partners included the Institute of High of High Energy, IHEP at Beijing, China, and the High Energy Accelerator Research Organization, KEK at Sukuba, Japan.



DENSITY OF TRANSITS PROCESSED BY THE GAIA IDT SYSTEM IN EQUATORIAL COORDINATES

The data was taken during the first year and a half of nominal operations. The Galactic Plane can be seen, as well as the effect of the scanning law. More than 35 billion transits (over 350 billion images) have been processed. .

Activity 2015

Concerning Gaia DPAC, during 2015 ICCUB members continued analyzing the data received from the satellite to precisely evaluate its performance. Instrumental and photometric models were improved. The daily data processing system (IDT), led by the ICCUB, reached a routine operation state. The first execution of the IDU Cross-Matching was done at BSC/MareNostrum, which determines the contents of the first Gaia Data Release envisaged for 2016. Large simulations have been generated to test systems such as data mining and archive, which are becoming ready for the first data release.

Concering DIRAC, the Consortium developed new partnerships with the Institute of High of High Energy, IHEP at Beijing, China, and the High Energy Accelerator Research Organization, KEK at Sukuba, Japan. The extension of the DIRAC user based beyond high energy physics consolidated thanks to the DIRAC4EGI instance operated in collaboration with CYFRONET at Krakow and the UAB. The ESFRI projects EISCAT-3D and ENVRI+ also started to use this DIRAC service as a possible baseline for the future distributed data processing environment. In regard to the updates of DIRAC ICCUB during 2015 a campaign started to raise private funds for the development of the new version in collaboration with a major ICT company.

KNOWLEDGE TRANSFER AND INNOVATION

SERVICES

- DAPCOM Data Services.
- Ideas Service (SiUB).

ICCUB MEMBERS

Casajús, A. • Castañeda, J. • Clotet, M. • Garrido, L. • Gascón, D. • Gaciani-Díaz, R. • Graugés, E. • Guzmán, R. • Julbe, F. • Luri, X. • Mauricio, J. • Picatoste, E. • Portell, J. • Salvador, E. • Sanuy, A. • Torra, J.

ICCUB research generates products and technologies transferable to industry and society through the licensing of patents and creation of services. To date, its technology transfer program has given rise to several applications from medical imaging to space industry. Also, one spin-off, DAPCOM, and one instrumental service, SiUB, have been created.

DAPCOM Data Services (ESA BIC Barcelona)

DAPCOM Data Services S.L. is a spin-off company participated by the University of Barcelona (UB) and the Technical University of Catalonia (UPC) which is specialized on the handling and processing of large amounts of data. It provides software engineering solutions and high-performance data compression strategies including proprietary implementations.

DAPCOM commercializes FAPEC (Fully Adaptive Prediction Error Coder), a patented lossless data compression algorithm originally created for satellite payloads. Besides being resilient to outliers in the data, it offers an optimum compromise between resources consumption and compression ratio. It can be applied to scientific research projects, supercomputing or companies dealing with Big Data scenarios.

http://www.dapcom.es

Ideas Service (SiUB)

The IDEAS Service (*Servei per la Innovació del Disseny Electrònic Avançat de Sistemes a la UB*), or SiUB, is an instrumentation service of the Physics Faculty of the UB which, by the one hand, provides a service on electronics and microelectronics instrumentation design, development and test to research groups of the UB and other research institutions, and by the other hand, enhances the industry technology transfer. SiUB staff and associate members hold more than ten years of experience developing instrumentation at different levels: design and test of application-specific integrated circuits (ASICs), design and test of cards and PCBs, and development of equipment and systems.

http://siub.ub.edu

Activity 2015

In 2014 DAPCOM became one of the first companies selected for the new ESA Business Incubation Centre (BIC) of Barcelona due to the potential of FAPEC for being transferred to terrestrial applications. The two-years incubation program at the UPC Campus of Castelldefels provides the necessary resources to boost the R&D activities, including the IEEC-DAPCOM PICFAPEC project (Parallel and Imaging Capabilities for FAPEC) which was concluded during 2015. As a result, the first public release (FAPEC Core 2015.0) was issued in September 2015, offering high-performance and reliable data compression for science data, images, spectral imaging, raw video or text data, either lossless or lossy, for space or ground systems.

Concerning SiUB, several ASIC designs were produced for the LHCb and CTA projects, using different technologies (IBM 130nm, TSMC 130nm and AMS 350nm SiGe BICMOS). The PACTA family of ASICs (wideband 16 bit dynamic range current mode preamplifiers) showed good characteristics for a variety of applications related to photomultiplier readout, which allowed using it in two other projects for ICFO and CNRS.

PROJECTS AND FUNDS 4

EUROPEAN PROJECTS AND FUNDS

ERC Projects

Cosmological physics with future large scale structure surveys (PHYSS.LSS)
Reference: 240117 (FP7-IDEAS-ERC)
PI: Licia Verde
Agency: European Research Council (ERC)
Period: 01/11/2009 - 30/11/2015

Holography for the LHC era (HoloLHC)
Reference: 306605 (FP7-IDEAS-ERC)
PI: David Mateos
Agency: European Community (EC)
Period: 01/10/2012 - 30/09/2017

Other European Projects

Advanced European Infrastructures for Detectors at Accelerators (AIDA)
Reference: 262025 (FP7-INFRASTRUCTURES)
PI: Laurent Serin (CERN) (ICCUB: Ángel Diéguez)
Agency: European Community (EC)
Period: 01/02/2011 - 31/01/2015

European Particle physics Latin American NETwork (EPLANET)
Reference: PIRSES-GA-2009-246806 (FP7-PEOPLE)
PI: Luciano Maiani (Univ. degli Studi di Roma, "La Sapienza") (ICCUB: Domènec Espriu)
Agency: European Community (EC)
Period: 01/02/2011 - 2016

Fast Thermalization of the Quark-Gluon Plasma (FastTh)
Reference: 658574 (HORIZON 2020-PILLAR 1 -MSCA)
PI: David Mateos
Agency: European Community (EC)
Period: 01/09/2015 - 31/08/2017

Gaia European Network for Improved User Services (GENIUS)
Reference: 606740 - GENIUS (FP7- SPACE)
PI: Xavier Luri
Agency: European Community (EC)
Period: 01/10/2013 - 31/03/2017

Gaia Research for European Astronomy Training (GREAT)
Reference: 08-RNP-118 (Research Networking Program)
PI: Nick Walton (ICCUB: Carme Jordi)
Agency: European Science Foundation (ESF)
Period: 18/02/2010 - 17/02/2015

Gaia Research for European Astronomy Training (GREAT-ITN)
Reference: PITN-GA-2010-264895 (FP7-PEOPLE)
PI: Nick Walton (ICCUB: Francesca Figueras)
Agency: European Community (EC)
Period: 01/03/2011 - 28/02/2015

High Energy Solar Particle Events foRecastIng and Analysis (HESPERIA)
Reference: 637324 (HORIZON 2020. PILLAR 2-LEIT-Space.)
PI: Olga Malandraki (National Observatory Athens) (ICCUB: Neus Àgueda)
Agency: European Community (EC)
Period: 01/05/2015 - 30/04/2017

INVISIBLES
Reference: PITN-GA-2011-289442 (FP7-PEOPLE)
PI: B. Gavela (UAM) (ICCUB: Mª Concepción González-García)
Agency: European Community (EC)
Period: 01/04/2012 - 31/03/2016

Probing strongly coupled deconfined matter at the LHC (DECOLHC)
Reference: PCIG12-GA-2012-333786 (FP7-PEOPLE)
PI: Joan Soto, Jorge Casalderrey-Solana
Agency: European Community (EC)
Period: 01/03/2013 - 31/10/2016

Rotational effects on strongly gravitating systems with matter (REGMat)
Reference: 656882 (HORIZON 2020-PILLAR 1 -MSCA)
PI: Roberto Emparan
Agency: European Community (EC)
Period: 01/10/2015 - 30/09/2017

Studying at high energies the dynamical and non-thermal processes in astrophysical outflows (ASTFLOW)
Reference: PCIG11-GA-2012-321520 (FP7-PEOPLE)
PI: Josep Mª Paredes, Valentí Bosch-Ramon
Agency: European Community (EC)
Period: 01/03/2013 - 28/02/2017

The Astrodynamics Network (ASTRONET-II)
Reference: PITN-GA-2011-289240 (FP7-PEOPLE (RTN))
PI: Gerard Gómez
Agency: European Community (EC)
Period: 01/01/2012 - 31/12/2015

The String Theory Universe
Reference: MP1210 (Cost Action)
PI: Silvia Penati (Univ. di Milano-Bicocca)
(ICCUB: Roberto Emparan)
Agency: COST Action (European Cooperation in Science and Technology)
Period: 04/03/2013 - 03/03/2017

Updating SOLPENCO2 and New Analysis on Downstream Fluence
Reference: Contr. 4000114116/15/NL/H
PI: Àngels Aran
Agency: Agencia Espacial Europea (ESA)
Period: 01/06/2015 - 01/02/2016

INTERNATIONAL PROJECTS

Contract for the final design of the MIRADAS Instrument for the Gran Telescopio Canarias
Reference: MIRADAS III
PI: Jordi Torra
Agency: University of Florida
Period: 01/01/2015 - 31/12/2015

Injection of nucleate-boiling slug flows into a heat exchange chamber
Reference: FA8655-12-1-2060
PI: Jaume Casademunt
Agency: Air Force Office of Scientific Research (USA)
Period: 20/03/2012 - 19/03/2015

Preconditioning of the interplanetary medium as responsible for large intense SEP events: Radial and longitudinal effects
Reference: NNX11AO83G
PI: David Lario (Johns Hopkins University/APL)
(ICCUB: Neus Águeda)
Agency: NASA
Period: 01/08/2011 - 01/07/2015

NATIONAL PLAN PROJECTS

Contribución al desarrollo científico y tecnológico de la misión Gaia III
Reference: ESP2013-48318-C2-1-R
PI: Jordi Torra
Agency: MINECO
Period: 01/01/2014 - 30/09/2015

Contribución al desarrollo científico y tecnológico de la misión Gaia IV
Reference: ESP2014-55996-C2-1-R
PI: Xavier Luri
Agency: MINECO
Period: 01/01/2015 - 31/12/2016

Cosmología Física en la época de grandes cartografiados
Reference: AYA2014-58747-P
PI: Licia Verde
Agency: MINECO
Period: 01/01/2015 - 31/12/2018

Desafíos presentes y futuros del experimento LHCb del CERN
Reference: FPA2014-57896-C4-2-R
PI: Eugeni Graugés
Agency: MINECO
Period: 01/01/2015 - 31/12/2017

Desarrollo de nuevos detectores y estudios de física para futuros colisionadores lineales
Reference: FPA2013-48387-C6-4-P
PI: Ángel Diéguez
Agency: MINECO
Period: 01/01/2014 - 31/12/2015

Estructura a Gran Escala, Cuásares y las Primeras Estrellas con los Espectros de Absorción de Cuásares de BOSS
Reference: AYA2012-33938
PI: Jordi Miralda-Escudé
Agency: MICINN
Period: 01/01/2013 - 31/12/2016

Estudio de la Violación de CP con el detector LHCb
Reference: FPA2011-30163-C02-01
PI: Eugeni Graugés
Agency: MICINN
Period: 01/01/2012 - 31/12/2015

Eyecciones astrofísicas en procesos de alta energía no térmicos. Teoría y observaciones multi-longitud de onda
Reference: AYA2013-47447-C3-1-P
PI: Josep Mª Paredes
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Fabricación e integración de SOIPHI (Polarimetric and Helioseismic Imager for Solar Orbiter)
Reference: ESP2013-47349-C6-3-R
PI: Josep Mª Gómez-Cama
Agency: MINECO
Period: 01/01/2014 - 31/12/2015

Física Hadrónica
Reference: FIS2014-57026-REDT
PI: Assumpta Parreño
Agency: MINECO
Period: 01/12/2014 - 30/11/2016

Formación Estelar Multi-Escala: de la Fuente Galáctica a Estrellas Individuales
Reference: AYA2014-57134-P
PI: Paolo Padoan
Agency: MINECO
Period: 01/01/2015 - 31/12/2017

Hadrones, núcleos y átomos - Nuevos retos para problemas de muchos cuerpos
Reference: FIS2014-54672-P
PI: Artur Polls
Agency: MINECO
Period: 01/01/2015 - 31/12/2017

Información Cuántica: entrelazamiento, frustración, gases fríos y orden topológico
Reference: FIS2013-41757-P
PI: José Ignacio Latorre
Agency: MINECO
Period: 01/01/2014 - 31/12/2015

Institut de Ciències del Cosmos (ICC)
Reference: MDM-2014-0369
PI: Josep M. Paredes
Agency: MINECO
Period: 01/07/2015 - 30/06/2019

Interstellar medium at high-angular resolution: preparing for the ALMA era
Reference: AYA2011-30228-C03-03
PI: Robert Estalella
Agency: MICINN
Period: 01/01/2012 - 30/09/2015

Las Componentes del Universo
Reference: AYA2012-36353
PI: M. Pilar Ruiz Lapuente
Agency: MINECO
Period: 01/01/2013 - 31/12/2015

Materia blanda forzada, activa y viva
Reference: FIS2013-41144-P
PI: Jaume Casademunt
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Métodos constructivos en sistemas dinámicos y aplicaciones
Reference: MTM2012-32541
PI: Ángel Jorba
Agency: MINECO
Period: 01/01/2013 - 31/12/2015

Modelado de la reionización del universo y de las galaxias que la causan
Reference: AYA2012-39168-C03-02
PI: Eduard Salvador-Solé
Agency: MINECO
Period: 01/01/2013 - 31/12/2015

Non-thermal high-energy processes in astrophysical outflows. Theory and multi-wavelength observations
Reference: AYA2013-47447-C3-2-P
PI: Kazushi Iwasawa
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Núcleos de gas molecular, discos y jets: el efecto del campo magnético
Reference: AYA2014-57369-C3-2-P
PI: Rosario López
Agency: MINECO
Period: 01/01/2015 - 31/12/2017

Participación española en el diseño y prototipado del Cherenkov Telescope Array: contribución del ICC-UB
Reference: FPA2014-55819-C4-4-P
PI: Marc Ribó
Agency: MINECO
Period: 01/01/2015 - 31/12/2015

Participación española en el diseño y prototipado del Cherenkov Telesope Array: perspectivas de física, prototipado de ASICs y explotación de DIRAC
Reference: FPA2013-48381-C6-6-P
PI: Marc Ribó
Agency: MINECO
Period: 01/01/2014 - 31/12/2015

Red de infraestructuras de astronomía
Reference: AYA2014-53365-REDT
PI: Jordi Torra
Agency: MINECO
Period: 01/12/2014 - 30/11/2016

Red Española de Explotación Científica de Gaia
Reference: AYA2015-71820-REDT
PI: Xavier Luri
Agency: MINECO
Period: 15/10/2015 - 14/10/2017

Simulación Monte Carlo del transporte de radiación. Emisión de electrones secundarios
Reference: FPA2013-44549-P
PI: Francesc Salvat
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Simulaciones de interacciones y fusiones de galaxias durante la formación de grupos
Reference: AYA2013-40609-P
PI: Josep Mª Solanes
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Sistemas de Fermi fuertemente correlacionados: átomos, núcleos y hadrones
Reference: FIS2011-24154
PI: Xavier Viñas
Agency: MICINN
Period: 01/01/2012 - 31/12/2015

Spanish Participation in the LHCb experiment at CERN: Physics exploitation and Upgrade
Reference: FPA2013-48020-C3-3-P
PI: Ricardo Graciani Díaz
Agency: MINECO
Period: 01/01/2014 - 31/12/2015

Sucesos solares de partículas energéticas: análisis y modelos. Aplicaciones para Solar Orbiter y herramientas para el tiempo espacial
Reference: AYA2013-42614-P
PI: Blai Sanahuja
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Teoría y fenomenología de las interacciones fundamentales: física de partículas y unificación de las fuerzas
Reference: FPA2013-46570-C2-1-P
PI: Domènec Espriu
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Teoría y fenomenología de las interacciones fundamentales: gravitación y cosmología
Reference: FPA2013-46570-C2-2-P
PI: Roberto Emparan
Agency: MINECO
Period: 01/01/2014 - 31/12/2016

Teorías efectivas de las interacciones fuertes: aplicaciones a quarkonium pesado y a QCD bajo condiciones externas
Reference: FPA2010-16963
PI: Joan Soto
Agency: MICINN
Period: 01/01/2011 - 30/06/2015

CONSOLIDER INGENIO PROJECTS & CONSOLIDER NETWORKS

Canfranc Underground Physics
Reference: CSD2008-00037
PI: Mª Concepción González-García
Agency: MEC
Period: 15/12/2008 - 14/12/2015

Centro nacional de física de partículas, astropartículas y nuclear
Reference: CSD2007-00042
PI: Antonio Pich, IFIC (ICCUB: Lluís Garrido)
Agency: MEC
Period: 01/10/2007 – 09/06/2015

Red Consolider Centro Nacional de Física de Partículas, Astroparticulas y Nuclear, (CPAN)
Reference: FPA2015-69037-REDC
PI: Lluís Garrido
Agency: MINECO
Period: 15/10/2015 - 14/10/2017

CONSOLIDATED GROUPS

Astronomia i Astrofísica
Reference: 2014SGR86
PI: Josep Mª Paredes
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Física nuclear teòrica i de moltes partícules en interacció
Reference: 2014SGR401
PI: Ángels Ramos
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Gravitation, Strings, and Cosmology
Reference: 2014SGR1474
PI: Jaume Garriga
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Grup de Física experimental d'altes energies
Reference: 2014SGR769
PI: Lluís Garrido
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Grup de Física Teòrica d'Altes Energies (FISALTEN)
Reference: 2014SGR104
PI: Jorge Russo
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Grup de Sistemes Dinàmics
Reference: 2014SGR1145
PI: Ángel Jorba
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Grup d'informació i simulació quàntiques (UB)
Reference: 2014SGR727
PI: José Ignacio Latorre
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

Physical Cosmology, PhysCos
Reference: 2014SGR921
PI: Licia Verde
Agency: AGAUR
Period: 01/01/2014 - 31/12/2016

CONTRACTS WITH THE INDUSTRY

Assessment on radiation detectors
Reference: FBG 201502759
PI: Andreu Sanuy
Agency: Arktis Radiation Detectors Ltd
Period: 24/08/2015-31/12/2015

Assessorament en física de l'estat sòlid
Reference: FBG 308561
PI: Francesc Salvat
Agency: Rovalma S.A
Period: 25/11/2015 - 30/01/2016

Assessorament i investigació aplicada en el camp de la física
Reference: FBG 307886
PI: J. M. Fernández Varea
Agency: Fundació Bosch i Gimpera
Period: 01/01/2015 - 31/12/2015

Assessorament, transferència, col•laboració i investigació aplicada en el camp de la instrumentació electrònica avançada
Reference: FBG 307594
PI: Ricardo Graciani Díaz
Agency: Fundació Bosch i Gimpera
Period: 01/01/2015 - 31/12/2015

Contract for intellectual services relative to Penelope course 2015
Reference: FBG 308375
PI: Francesc Salvat
Agency: OECD. Organisation for Economic Co-operation and Development
Period: 29/06/2015 - 03/07/2015

Design and production of preamplifier for synchrotron facility
Reference: FBG 201503042
PI: David Gascón
Agency: Scientifica Internacional, S.L
Period: 2015

Further development of the modification and/or customization of PENELOPE
Reference: FBG 307269
PI: Francesc Salvat
Agency: Hamamatsu Photonics K.K.
Period: 01/04/2013 - 31/03/2015

Further development of the modification and/or customization of PENELOPE
Reference: FBG 308407
PI: Francesc Salvat
Agency: Hamamatsu Photonics K.K.
Period: 01/04/2015 - 31/03/2016

Upgraded electronics for Hamamatsu detectors I II and III
Reference: FBG 308324
PI: Ricardo Graciani Díaz
Agency: Hamamatsu Photonics France S.A.R.L.
Period: 03/05/2015 - 15/12/2015

PUBLICATIONS 5

SCI PUBLICATIONS

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Differential branching fraction and angular analysis of $\Lambda_b^0 \to \Lambda \mu^+ \mu^-$ *decays*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 115 (2015)
10.1007/JHEP06(2015)115

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Determination of the quark coupling strength $|Vub|$ *using baryonic decays*. Article.
Nature Physics, Vol. 11, Num. 9, p. 743 - 747 (2015)
10.1038/nphys3415

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the time-dependent CP asymmetries in $B_s^0 \to J/\psi\ K_s^0$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 131 (2015)
10.1007/JHEP06(2015)131

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Observation of the $B_s^0 \to \eta'\eta'$ *Decay*. Article.
Physical Review Letters, Vol. 115, Num. 51801 (2015)
10.1103/PhysRevLett.115.051801

Aaij, R.; et al (LHCb collaboration. ICCUB: Mohr, R. Casanova; **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Benito, C. Marin**; **Olloqui, E. Picatoste**; **Molina, V. Rives**; **Ruiz, H.**)
First observation and amplitude analysis of the $B^- \to D^+K^-\pi^-$ *decay*. Article.
Physical Review D, Vol. 91, Num. 92002 (2015)
10.1103/PhysRevD.91.092002

Aaij, R.; et al (LHCb collaboration. ICCUB: Mohr, R. Casanova; **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Olloqui, E. Picatoste**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the forward Z boson production cross-section in pp collisions at $\sqrt{s} = 7\ TeV$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 39 (2015)
10.1007/JHEP08(2015)039

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of CP Violation in $B^0 \to J/\psi\ K_S^0$ *Decays*. Article.
Physical Review Letters, Vol. 115, Num. 31601 (2015)
10.1103/PhysRevLett.115.031601

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Graciani Diaz, R.** ; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the $\eta_c(1S)$ *production cross-section in proton–proton collisions via the decay* $\eta_c(1S) \to p\bar{p}$. Article.
European Physical Journal C, Vol. 75, Num. 311, p. 1 - 12 (2015)
10.1140/epjc/s10052-015-3502-x

Aaij, R.; et al (LHCb collaboration. ICCUB: Camboni, A.; **Garrido, L.**; **Diaz, R. Graciani**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Model-independent confirmation of the $Z(4430)^-$ *state*. Article.
Physical Review D, Vol. 92, Num. 112009 (2015)
10.1103/PhysRevD.92.112009

Aaij, R.; et al (LHCb collaboration. ICCUB: Casanova Mohr, R. C. M.; **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Angular analysis of the $B^0 \to K^{*0} e^+ e^-$ decay in the low- q^2 region. Article.
Journal of High Energy Physics, Vol. 2015, Num. 64 (2015)
10.1007/JHEP04(2015)064

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of indirect CP asymmetries in $D^0 \to K^- K^+$ and $D^0 \to \pi^- \pi^+$ decays using semileptonic B decays. Article.
Journal of High Energy Physics, Vol. 2015, Num. 43 (2015)
10.1007/JHEP04(2015)043

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Mann Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the exclusive Y production cross-section in pp collisions at $\sqrt{s}$ = 7 TeV and 8 TeV. Article.
Journal of High Energy Physics, Vol. 2015, Num. 84 (2015)
10.1007/JHEP09(2015)084

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Mann Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Precise measurements of the properties of the $B_1(5721)^{0,+}$ and $B_2^*(5747)^{0,+}$ states and observation of $B^{+,0}\pi^{-,+}$ mass structures. Article.
Journal of High Energy Physics, Vol. 2015, Num. 24 (2015)
10.1007/JHEP04(2015)024

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Olloqui, E. Picatoste**; **Molina, V. Rives**; **Ruiz, H.**)
Quantum numbers of the $X(3872)$ state and orbital angular momentum in its $\rho^0 J/\psi$ decay. Article.
Physical Review D, Vol. 92, Num. 11102 (2015)
10.1103/PhysRevD.92.011102

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Olloqui, E. Picatoste**; **Rives Molina, V.**; **Ruiz, H.**)
Search for the decay $B_s^0 \to \bar{D}^0 f_0(980)$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 5 (2015)
10.1007/JHEP08(2015)005

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Molina, V. Rives**; **Ruiz, H.**)
Dalitz plot analysis of $B^0 \to \bar{D}^0 \pi^+ \pi^-$ decays. Article.
Physical Review D, Vol. 92, Num. 32002 (2015)
10.1103/PhysRevD.92.032002

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Amplitude analysis of $B^0 \to \bar{D}^0 K^+ \pi^-$ decays. Article.
Physical Review D, Vol. 92, Num. 12012 (2015)
10.1103/PhysRevD.92.012012

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
First observation and measurement of the branching fraction for the decay $B_s^0 \to D_s^{*m} K^\pm$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 130 (2015)
10.1007/JHEP06(2015)130

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Study of $B^- \to D K^-\pi^+\pi^-$ and $B^- \to D \pi^-\pi^+\pi^-$ decays and determination of the CKM angle γ. Article.
Physical Review D, Vol. 92, Num. 112005 (2015)
10.1103/PhysRevD.92.112005

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**; Valls, P.R.)
A study of CP violation in $B^m \to Dh^m$ ($h = K, \pi$) with the modes $D \to K^m\pi^\pm\pi^0$, $D \to \pi^+ \pi^- \pi^0$ and $D \to K^+ K^- \pi^0$. Article.
Physical Review D, Vol. 91, Num. 112014 (2015)
10.1103/PhysRevD.91.112014

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Picatoste Olloqui, E.**; **Molina, V. Rives**; **Ruiz, H.**)
Observation of the $B^0 \to \rho^0\rho^0$ decay from an amplitude analysis of $B^0 \to (\pi^+\pi^-)(\pi^+\pi^-)$ decays. Article.
Physics Letters B, Vol. 747, p. 468 - 478 (2015)
10.1016/j.physletb.2015.06.027

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Diaz, R. Graciani**; **Grauges, E.**; **Benito, C. Marin**; **Olloqui, E. Picatoste**; **Molina, V. Rives**; **Ruiz, H.**)
Study of W boson production in association with beauty and charm. Article.
Physical Review D, Vol. 92, Num. 52001 (2015)
10.1103/PhysRevD.92.052001

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Mann Benito, C.**; **Olloqui, E. Picatoste**; **Rives Molina, V.**)
First measurement of the differential branching fraction and CP asymmetry of the $B^\pm \to \pi^\pm\mu^+\mu^-$ decay. Article.
Journal of High Energy Physics, Vol. 2015, Num. 34 (2015)
10.1007/JHEP10(2015)034

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Olloqui, E. Picatoste**; **Rives Molina, V.**)
Observation of $J/\psi p$ Resonances Consistent with Pentaquark States in $\Lambda_b^0 \to J/\psi K^- p$ Decays. Article.
Physical Review Letters, Vol. 115, Num. 72001 (2015)
10.1103/PhysRevLett.115.072001

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Evidence for the Strangeness-Changing Weak Decay $\Xi_b^- \to \Lambda_b^0 \pi^-$. Article.
Physical Review Letters, Vol. 115, Num. 241801 (2015)
10.1103/PhysRevLett.115.241801

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Forward production of Y mesons in pp collisions at $\sqrt{s}$ = 7 and 8 TeV. Article.
Journal of High Energy Physics, Vol. 2015, Num. 103, p. 1 - 34 (2015)
10.1007/JHEP11(2015)103

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Measurement of CP violation parameters and polarisation fractions in $B_s^0 \to J/\psi \bar{K}^{*0}$ decays. Article.
Journal of High Energy Physics, Vol. 2015, Num. 82 (2015)
10.1007/JHEP11(2015)082

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Measurement of the time-integrated CP asymmetry in $D^0 \to K_S^0 K_S^0$ decays. Article.
Journal of High Energy Physics, Vol. 2015, Num. 55 (2015)
10.1007/JHEP10(2015)055

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Search for Hidden-Sector Bosons in $B^0 \to K^{*0}\mu^+\mu^-$ Decays. Article.
Physical Review Letters, Vol. 115, Num. 161802 (2015)
10.1103/PhysRevLett.115.161802

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Search for the rare decays $B^0 \to J/\psi\gamma$ and $B_s^0 \to J/\psi\gamma$. Article.
Physical Review D, Vol. 92, Num. 112002 (2015)
10.1103/PhysRevD.92.112002

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
B flavour tagging using charm decays at the LHCb experiment. Article.
Journal Of Instrumentation, Vol. 10, Num. P10005 (2015)
10.1088/1748-0221/10/10/P10005

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
First Observation of Top Quark Production in the Forward Region. Article.
Physical Review Letters, Vol. 115, Num. 112001 (2015)
10.1103/PhysRevLett.115.112001

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the $B_s^0 \to \phi\phi$ *branching fraction and search for the decay* $B^0 \to \phi\phi$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 53 (2015)
10.1007/JHEP10(2015)053

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the branching fraction ratio $B(B_c^+ \to \psi(2S)\pi^+) / B(B_c^+ \to J/\psi\pi^+)$. Article.
Physical Review D, Vol. 92, Num. 72007 (2015)
10.1103/PhysRevD.92.072007

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the Ratio of Branching Fractions $B(\bar{B}^0 \to D^{*+}\tau^-\bar{\nu}_\tau) / B(\bar{B}^0 \to D^{*+}\mu^-\bar{\nu}_\mu)$. Article.
Physical Review Letters, Vol. 115, Num. 111803 (2015)
10.1103/PhysRevLett.115.111803

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graugés, E.**; **Graciani Díaz, R.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Measurement of forward J/ψ *production cross-sections in pp collisions at* $\sqrt{s}$ *=13 TeV*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 172, p. 1 - 29 (2015)
10.1007/JHEP10(2015)172

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of the track reconstruction efficiency at LHCb. Article.
Journal Of Instrumentation, Vol. 10 (2015)
10.1088/1748-0221/10/02/P02007

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Determination of the branching fractions of $B_s^0 \to D_s^m K^m$ *and* $B^0 \to D_s^- K^+$. Article.
Journal of High Energy Physics, Vol. 5, p. 19 (2015)
10.1007/Jhep05(2015)019

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Identification of beauty and charm quark jets at LHCb. Article.
Journal Of Instrumentation, Vol. 10, Num. P06013 (2015)
10.1088/1748-0221/10/06/P06013

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Search for long-lived heavy charged particles using a ring imaging Cherenkov technique at LHCb. Article.
European Physical Journal C, Vol. 75, Num. 595, p. 1 - 14 (2015)
10.1140/epjc/s10052-015-3809-7

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Search for the $\Lambda_b^0 \to \Lambda\eta'$ *and* $\Lambda_b^0 \to \Lambda\eta$ *decays with the LHCb detector*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 6, p. 1 - 21 (2015)
10.1007/JHEP09(2015)006

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Measurement of the forward-backward asymmetry in $Z/\gamma^* \to \mu^+\mu^-$ *decays and determination of the effective weak mixing angle*. Article.
Journal of High Energy Physics, Vol. 2015 (2015)
10.1007/JHEP11(2015)190

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Grauges, E.**; **Graciani Diaz, R.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Ruiz, H.**; Valls, PR.)
Measurement of forward $Z \to e^+e^-$ *production at root* $\sqrt{s}$ *= 8 TeV*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 109, (2015)
10.1007/JHEP05(2015)109

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin**

Benito, C.; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Measurement of CP asymmetries and polarisation fractions in $\rightarrow K^{*0}\ \bar{K}^{0}$ *decays*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 166 (2015)
10.1007/JHEP07(2015)166

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Observation of the decay $\bar{B}_s^0 \rightarrow \psi(2S)\, K^+\pi^-$. Article.
Physics Letters B, Vol. 747, p. 484 - 494 (2015)
10.1016/j.physletb.2015.06.038

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Grauges, E.**; **Benito, C. Marin**; **Olloqui, E. Picatoste**; **Molina, V. Rives**; **Ruiz, H.**)
Angular analysis and differential branching fraction of the decay $B_s^0 \rightarrow \phi\mu^+\mu^-$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 179 (2015)
10.1007/JHEP09(2015)179

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
LHCb detector performance. Article.
International Journal of Modern Physics A, Vol. 30, Num. 1530022 (2015)
10.1142/S0217751X15300227

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of B_c^+ *production in proton-proton collisions at* $\sqrt{s}$ *= 8TeV*. Article.
Physical Review Letters, Vol. 114, Num. 132001 (2015)
10.1103/PhysRevLett.114.132001

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of the CP-violating phase β *in* $B^0 \rightarrow J/\Psi\ \pi^+\pi^-$ *decays and limits on penguin effects*. Article.
Physics Letters B, Vol. 742, p. 38 - 49 (2015)
10.1016/j.physletb.2015.01.008

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of the inelastic pp cross-section at a centre-of-mass energy of $\sqrt{s}$ *= 7 TeV*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 129 (2015)
10.1007/JHEP02(2015)129

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of the semileptonic CP asymmetry in B^0 - $\bar{B}^0$ *mixing*. Article.
Physical Review Letters, Vol. 114, Num. 41601 (2015)
10.1103/PhysRevLett.114.041601

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of the Z+b-jet cross-section in pp collisions at $\sqrt{s}$ *= 7 TeV in the forward region*. Article.
Journal of High Energy Physics, Vol. 2015, Num. 64 (2015)
10.1007/JHEP01(2015)064

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Observation of two new Ξ_b^- *Baryon resonances*. Article.
Physical Review Letters, Vol. 114, Num. 62004 (2015)
10.1103/PhysRevLett.114.062004

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Precision measurement of CP violation in $B_s^0 \rightarrow J/\psi\ K^+K^-$ *decays*. Article.
Physical Review Letters, Vol. 114, Num. 41801 (2015)
10.1103/PhysRevLett.114.041801

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Search for CP violation in $D^0 \rightarrow \pi^-\pi^+\pi^0$ *decays with the energy test*. Article.
Physics Letters B, Vol. 740, p. 158 - 167 (2015)
10.1016/j.physletb.2014.11.043

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Search for long-lived particles decaying to jet pairs. Article.

European Physical Journal C, Vol. 75, p. 1 - 12 (2015)
10.1140/epjc/s10052-015-3344-6

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Study of the rare B_s^0 and B^0 decays into the $\pi^+\pi^-\mu^+\mu^-$ final state. Article.
Physics Letters B, Vol. 743, p. 46 - 55 (2015)
10.1016/j.physletb.2015.02.010

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Study of $\eta - \eta'$ mixing from measurement of $B^0_{(s)} \to J/\psi\, \eta^{(')}$ decay rates. Article.
Journal of High Energy Physics, Vol. 2015, Num. 24, p. 1 - 24 (2015)
10.1007/JHEP01(2015)024

+

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani**; **R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Measurement of the lifetime of the B_c^+ meson using the $B_c^+ \to J/\psi\, \pi^+$ decay mode. Article.
Physics Letters B, Vol. 742, p. 29 - 37 (2015)
10.1016/j.physletb.2015.01.010

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Graciani, R.**; **Graugés, E.**; **Marin, C.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Determination of γ and -2βs from charmless two-body decays of beauty mesons. Article.
Physics Letters B, Vol. 741, p. 1 - 11 (2015)
10.1016/j.physletb.2014.12.015

Aaij, R.; et al (LHCb collaboration. ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani, R.**; **Graugés, E.**; **Picatoste, E.**; **Rives, V.**; **Ruiz, H.**)
Search for the lepton flavour violating decay $\tau^- \to \mu^-\mu^+\mu^-$. Article.
Journal of High Energy Physics, Vol. 2015, Num. 121 (2015)
10.1007/JHEP02(2015)121

Abdesselam, A.; et al (BaBar Collaboration and Belle Collaboration; ICCUB: **Graugés, E.**)
First Observation of CP Violation in $\overline{B}^0 \to D^{(*)}_{CP}\, h^0$ Decays by a Combined Time-Dependent Analysis of BABAR and Belle Data. Article.
Physical Review Letters, Vol. 115, Num. 121604 (2015)
10.1103/PhysRevLett.115.121604

Abedi, H.; **Figueras, F.**; Aguilar, L.; Mateu, C.; **Romero-Gomez, M.**; Lopez-Corredoira, M.; Lopez, F.G.
On the characterization of the galactic warp in the gaia era. Book in Series.
EAS Publications Series, Vol. 67-68, p. 237 - 240 (2015)
10.1051/eas/1567042

Afonin, S.S.; **Andrianov, A.A.**; **Espriu, D.**
The masses of vector mesons in holographic QCD at finite chiral chemical potential. Article.
Physics Letters B, Vol. 745, p. 52 - 55 (2015)
10.1016/j.physletb.2015.04.027

Ahnen, M.L.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Very High Energy γ-Rays from the Universe's Middle Age: Detection of the z = 0.940 Blazar PKS 1441+25 with MAGIC. Article.
Astrophysical Journal, Vol. 815, Num. 2, p. L23 - L31 (2015)
10.1088/2041-8205/815/2/L23

Alam, S.; et al (ICCUB: **Cuesta, A.J.**; **Gontcho, S.G.A.**; **Miralda-Escude, J.**; **Perez-Rafols, I.**; **Verde, L.**
The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III. Article.
Astrophysical Journal Supplement Series, Vol. 219, Num. 1, p. 12 - 27 (2015)
10.1088/0067-0049/219/1/12

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; **Munar-Adrover, P.**; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
MAGIC observations of MWC 656, the only known Be/BH system. Article.
Astronomy and Astrophysics, Vol. 576, Num. A36, p. 1 - 5 (2015)
10.1051/0004-6361/201424879

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Very high-energy gamma-ray observations of novae and dwarf novae with the MAGIC telescopes. Article.
Astronomy and Astrophysics, Vol. 582, p. A67 - A74 (2015)
10.1051/0004-6361/201526478

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)

Unprecedented study of the broadband emission of Mrk 421 during flaring activity in March 2010. Article.
Astronomy and Astrophysics, Vol. 578, Num. A22, p. 1 - 26 (2015)
10.1051/0004-6361/201424811

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Measurement of the Crab Nebula spectrum over three decades in energy with the MAGIC telescopes. Article.
Journal of High Energy Astrophysics, Vol. 5-6, p. 30 - 38 (2015)
10.1016/j.jheap.2015.01.002

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
MAGIC detection of short-term variability of the high-peaked BL Lac object 1ES 0806+524. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 451, Num. 1, p. 5258 - 5269 (2015)
10.1093/mnras/stv895

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
The 2009 multiwavelength campaign on Mrk 421: Variability and correlation studies. Article.
Astronomy and Astrophysics, Vol. 576, Num. A126, p. 1 - 18 (2015)
10.1051/0004-6361/201424216

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Probing the very high energy γ-ray spectral curvature in the blazar PG 1553+113 with the MAGIC telescopes. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 450, Num. 4, p. 4399 - 4410 (2015)
10.1093/mnras/stv811

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Discovery of very high energy gamma-ray emission from the blazar 1ES 0033+595 by the MAGIC telescopes. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 446, Num. Issue 1, p. 217 - 225 (2015)
10.1093/mnras/stu2024

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Marcote, B.**; Munar-Adrover, P.; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
Multiwavelength observations of Mrk 501 in 2008. Article.
Astronomy and Astrophysics, Vol. 573, Num. A50 (2015)
10.1051/0004-6361/201322906

Alert, R.; **Casademunt, J.**; Brugués, J.; Sens, P.
Model for probing membrane-cortex adhesion by micropipette aspirations and fluctuation spectroscopy. Article.
Biophysical Journal, p. 1878 - 1886 (2015)
10.1016/j.bpj.2015.02.027

Aliu, E.; et al (ICCUB: **Aliu, E.**)
A search for pulsations from geminga above 100 GeV with veritas. Article.
Astrophysical Journal, Vol. 800, Num. 61 (2015)
10.1088/0004-637X/800/1/61

Altavilla, G.; Marinoni, S.; Pancino, E.; Galleti, S.; Ragaini, S.; Bellazzini, M.; Cocozza, G.; Bragaglia, A.; **Carrasco, J.M.**; Castro, A.; Di Fabrizio, L.; Federici, L.; **Figueras, F.**; Gebran, M.; **Jordi, C.**; **Masana, E.**; Schuster, W.; Valentini, G.; **Voss, H.**
The Gaia spectrophotometric standard stars survey -- II. Instrumental effects of six ground-based observing campaigns. Article.
Astronomische Nachrichten, Vol. 336, Num. 6, p. 515 - 529 (2015)
10.1002/asna.201512176

Anderson, L.; **Russo, J.G.**
ABJM Theory with mass and FI deformations and quantum phase transitions. Article.
Journal of High Energy Physics, Vol. 5, p. 64 (2015)
10.1007/JHEP05(2015)064

Andrianov, A.A.; **Espriu, D.**; Kolevatov, S.S.
Search for parity violation in cosmic rays. Article.
Theoretical and Mathematical Physics, Vol. 184, Num. 3, p. 1234 - 1238 (2015)
10.1007/s11232-015-0330-y

Aniceto, I; **Russo, J.G.**; Schiappa, R.
Resurgent analysis of Localizable Observables in Supersymmetric Gauge theories. Article.
Journal of High Energy Physics, Vol. 3, p. 172 (2015)
10.1007/JHEP03(2015)172

Antiche, E.; Masana, E.; Julbe, F.; Borrachero, R.
The gaia object generator (gog). Book in Series.
EAS Publications Series, Vol. 67-68, p. 355 - 355 (2015)
10.1051/eas/1567064

Antoja, T.; Mateu, C.; **Aguilar, L.; Figueras, F.**; Antiche, E.; Hernández-Pérez, F.; Brown, A. G. A.; Valenzuela, O.; Aparicio, A.; Hidalgo, S.; Velázquez, H.
Detection of satellite remnants in the Galactic halo with Gaia- III. Detection limits for ultrafaint dwarf galaxies.
Article.
Monthly Notices of the Royal Astronomical Society, Vol. 453, p. 541 – 560 (2015)
10.1093/mnras/stv1622

Areán, D.; **Tarrío, J.**
Bifundamental superfluids from holography. Article.
Journal of High Energy Physics, Vol. 2015, Num. 83 (2015)
10.1007/JHEP04(2015)083

Arinyo-i-Prats, A.; **Miralda-Escudé, J.**; Viel, M.; Cen, R.
The non-linear power spectrum of the Lyman alpha forest.
Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 17 (2015)
10.1088/1475-7516/2015/12/017

Arrabito, L.; Bregeon, J.; Haupt, A.; **Graciani Diaz, R.**; Stagni, F.; Tsaregorodtsev, A.
Prototype of a production system for cherenkov telescope array with DIRAC. Conference Paper.
Journal of Physics: Conference Series, Vol. 664, Num. 32001 (2015)
10.1088/1742-6596/664/3/032001

Arteaga, O.; Canillas, A.; **El-Hachemi, Z.**; **Crusats, J.**; **Ribó, J.M.**
Structure vs. excitonic transitions in self-assembled porphyrin nanotubes and their effect on light absorption and scattering. Article.
Nanoscale, Vol. 7, p. 20435 - 20441 (2015)
10.1039/c5nr05633k

Attems, M.
Chromo-Weibel instabilities in classical Yang-Mills evolution. Proceeding.
Journal of Physics: Conference Series, Vol. 612, Num. 1 (2015)
10.1088/1742-6596/612/1/012015

Audren, B.; **Bellini, E.**; **Cuesta, A.J.**; **Gontcho A Gontcho, S.**; Lesgourgues, J.; Niro, V.; Pellejero-Ibanez, M.; **Perez-Rafols, I.**; Poulin, V.; Tram, T.; Tramonte, D.; **Verde, L.**
Robustness of cosmic neutrino background detection in the cosmic microwave background. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 3, Num. 36 (2015)
10.1088/1475-7516/2015/03/036

Baldi, M.; **Simpson, F.**
Simulating momentum exchange in the dark sector.
Article.
Monthly Notices of the Royal Astronomical Society, Vol. 449, p. 2239 - 2249 (2015)
10.1093/mnras/stv405

Ball, R.D.; Bertone, V.; Carrazza, S.; Deans, C. S.; Del Debbio, L.; Forte, S.; Guffanti, A.; Hartland, N. P.; **Latorre, J. I.**; Rojo, J.; Ubiali, M.
Parton distributions for the LHC run II. Article.
Journal of High Energy Physics, Vol. 2015, Num. 40 (2015)
10.1007/JHEP04(2015)040

Barberan, N.; Dagnino. D.; Garcia-March, M.A.; Trombettoni, A.; **Taron, J.**; Lewenstein, M
Quantum simulation of conductivity plateaux and fractional quantum Hall effect using ultracold atoms.
Article.
New Journal of Physics, Vol. 17, Num. 125009, p. 125009-1 - 125009-11 (2015)
10.1088/1367-2630/17/12/125009

Barros, S.F.; Vanin, V.R.; Maidana, N.L.; **Fernández-Varea, J.M.**
Ionization cross sections of the L subshells of Au by 50 to 100 keV electron impact. Article.
Journal of Physics B: Atomic, Molecular and Optical Physics, Vol. 48, p. 175201-1 - 175201-12 (2015)
10.1088/0953-4075/48/17/175201

Basilakos, S.; **Solà, J.**
Growth index of matter perturbations in Running Vacuum models. Article.
Physical Review D, Vol. D92, Num. 12, p. 123501 (2015)
10.1103/PhysRevD.92.123501

Bautista, J. E.; Bailey, S.; Font-Ribera, A.; Pieri, M. M.; Busca, N. G.; **Miralda-Escudé, J.**; Palanque-Delabrouille, N.; Rich, J.; Dawson, K.; Feng, Y.; Ge, J.; **Gontcho A Gontcho, S.**; Ho, S.; Le Goff, J.M.; Noterdaeme, P.; Pâris, I.; Rossi, G.; Schlegel, D.

Mock Quasar-Lyman-α forest data-sets for the SDSS-III Baryon Oscillation Spectroscopic Survey. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 60 (2015)
10.1088/1475-7516/2015/05/060

Beane, S.R. Chang, E.; Detmold, W.; Orgino, K.; **Parreno, A.**; Savage, M.J.; Tiburzi, B.C.
Ab initio Calculation of the np → dγ Radiative Capture Process. Article.
Physical Review Letters, Vol. 115, Num. 13, p. 132001 (2015)
10.1103/PhysRevLett.115.132001

Beane, S.R.; Chang, E.; Cohen, S.D.; Detmold, W.; Lin, H.W.; Orginos, K.; **Parreno, A.**; Savage, M.J.
Quarkonium-Nucleus Bound States from Lattice QCD. Article.
Physical Review D, Vol. 91, Num. 11, p. 114503 (2015)
10.1103/PhysRevD.91.114503

Beane, S.R.; Chang, E.; Detmold, W.; Orginos, K.; **Parreño, A.**; Savage, M.J.; Tiburzi, B.C.
The magnetic structure of light nuclei from Lattice QCD. Article.
Physical Review D, Vol. 92, Num. 11, p. 114502 (2015)
10.1103/PhysRevD.92.114502

Behera, B.; **Viñas, X.**; Routray, T.R., **Centelles, M.**
Study of spin polarized nuclear matter and finite nuclei with finite range simple effective interaction. Article.
Journal of Physics G: Nuclear and Particle Physics, Vol. 42, Num. 4, p. 045103-1 - 045103-27 (2015)
10.1088/0954-3899/42/4/045103

Bellini, E.; **Jimenez, R.**; **Verde, L.**
Signatures of Horndeski gravity on the dark matter bispectrum. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 57, (2015)
10.1088/1475-7516/2015/05/057

Bellini, E.; Zumalacárregui, M.
Nonlinear evolution of the baryon acoustic oscillation scale in alternative theories of gravity. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 92, Num. 63522 (2015)
10.1103/PhysRevD.92.063522

Benaglia, P.; **Marcote, B.**; **Moldón, J.**; Nelan, E.; De Becker, M.; Dougherty, S. M.; Koribalski, B. S.
A radio map of the colliding winds in the very massive binary system HD 93129A. Article.
Astronomy and Astrophysics, Vol. 579, Num. A99 (2015)
10.1051/0004-6361/201425595

Bergström, J.; **Gonzalez-Garcia, M. C.**; Maltoni, M.; Schwetz, T.
Bayesian global analysis of neutrino oscillation data. Article.
Journal of High Energy Physics, (2015)
10.1007/JHEP09(2015)200

Bitlloch, P.; Ruiz, X.; Ramírez-Piscina, L.; **Casademunt, J.**
Turbulent Bubble Jets in Microgravity. Spatial Dispersion and Velocity Fluctuations. Article.
Microgravity Science and Technology, Vol. 27, Num. 3, p. 207 - 220 (2015)
10.1007/s12217-015-9436-y

Blanco, C.; **Ribó, J.M.**; Hochberg, D.
Modelling spontaneous chiral symmetry breaking and deracemization phenomena: Discrete versus continuum approaches. Article.
Physical Review E, Vol. 91, Num. 2, p. 022801 (2015)
10.1103/PhysRevE.91.022801

Blomqvist, M.; Kirkby, D.; Bautista, J. E.; **Arinyo-i-Prats, A.**; Busca, N. G.; **Miralda-Escudé, J.**; Slosar, A.; Font-Ribera, A.; Margala, D.; Schneider, D.P.; Vazquez, J.A.
Broadband distortion modeling in Lyman-α forest BAO fitting. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 34 (2015)
10.1088/1475-7516/2015/11/034

Boada, O.; Celi, A.; Rodríguez-Laguna, J.; **Latorre, J.I.**; Lewenstein, M.
Quantum simulation of non-trivial topology. Article.
New Journal of Physics, Vol. 17, Num. 45007 (2015)
10.1088/1367-2630/17/4/045007

Bosch-Ramon, V.
Non-thermal emission from standing relativistic shocks: an application to red giant winds interacting with AGN jets. Article.
Astronomy and Astrophysics, Vol. 575, p. 109 - 114 (2015)
10.1051/0004-6361/201425208

Bosch-Ramon, V.; Barkov, M.V.; Perucho, M.
Orbital evolution of colliding star and pulsar winds in 2D

and 3D; effects of: dimensionality, EoS, resolution, and grid size. Article.
Astronomy and Astrophysics, Vol. 577, Num. A89 (2015)
10.1051/0004-6361/201425228

BOSS Collaboration (UB: **Cuesta, A.J.**; **Gontcho A Gontcho, S.**; **Miralda-Escude, J.**)
Cosmological implications of baryon acoustic oscillation measurements. Article.
Physical Review D, Vol. 92, Num. 12, p. 123516 (2015)
10.1103/PhysRevD.92.123516

Cabrer, A.; **Ribó, J.M.**; **El-Hachemi, Z.**; **Crusats, J.**
5,10,15,20-Tetrasulfonatophenylporphyrin regioisomers: How the location of the sulfonato groups determines the formation of their supramolecular aggregates. Article.
Journal of Porphyrins and Phthalocyanines, Vol. 19, p. 852 - 857 (2015)
10.1142/S1088424615500625

Caceres, E.; **Kundu, A.**; Pedraza, J.F.; Yang, D.-L.
Weak field collapse in AdS: introducing a charge density. Article.
Journal of High Energy Physics, Vol. 2015, Num. 111 (2015)
10.1007/JHEP06(2015)111

Camanho, X.O.; Dadhich, N.; **Molina, A.**
Pure Lovelock Kasner metrics. Article.
Classical and Quantum Gravity, Vol. 32, Num. 175016; 2015)
10.1088/0264-9381/32/17/175016

Carrasco, J.M.; **Jordi, C.**; **Fabricius, C.**; **Voss, H.**; **Weiler, M.**
Characterisation of the gaia photometry. Book in Series.
EAS Publications Series, Vol. 67-68, p. 359 - 359 (2015)
10.1051/eas/1567066

Carrasco-González, C.; **Torrelles, J.M.**; Cantó, J.; Curiel, S.; Surcis, G.; Vlemmings, W. H. T.; van Langevelde, H. J.; Goddi, C.; Anglada, G.; Kim, S.-W.; Kim, J.-S.; Gómez, J. F.
Observing the onset of outflow collimation in a massive protostar. Article.
Science, Vol. 348, Num. 6230, p. 114 - 117 (2015)
10.1126/science.aaa7216

Carrazza, S.; Forte, S.; Kassabov, Z.; **Latorre, J. I.**; Rojo, J.
An unbiased Hessian representation for Monte Carlo PDFs. Article.
European Physical Journal C, Vol. 75, Num. 369 (2015)
10.1140/epjc/s10052-015-3590-7

Carrazza, S.; **Latorre, J. I.**; Rojo, J.; Watt, G.
A compression algorithm for the combination of PDF sets. Revisió.
European Physical Journal C, Vol. 75, Num. 474 (2015)
10.1140/epjc/s10052-015-3703-3

Carrera, R.; **Casamiquela, L.**; Ospina, N.; **Balaguer-Núñez, L.**; **Jordi, C.**; Monteagudo, L.
Radial velocities and metallicities from infrared Ca II triplet spectroscopy of open clusters II. Berkeley 23, King 1, NGC 559, NGC 6603 and NGC 7245. Article.
Astronomy and Astrophysics, Vol. 578, Num. A27, p. 27 - 36 (2015)
10.1051/0004-6361/201425531

Casalbuoni, R.; **Gomis, J.**
Addendum to 'Conformal symmetry for relativistic point particles'. Article.
Physical Review D, Vol. 91, Num. 47901 (2015)
10.1103/PhysRevD.91.047901

Casalderrey-Solana, J.; Gulhan, D.C.; Milhano, J.G.; **Pablos, D.**; Rajagopal, K.
A hybrid strong/weak coupling approach to jet quenching. Erratum.
Journal of High Energy Physics, Vol. 2015, Num. 175 (2015)
10.1007/JHEP09(2015)175

Corbett, T.; Eboli, O.J.P.; **Gonzalez-Garcia, M.C.**
Unitarity Constraints on Dimension-Six Operators. Article.
Physical Review D, Vol. 91, Num. 35014 (2015)
10.1103/PhysRevD.91.035014

Crosby, N.; Heynderickx, D.; Jiggens, P; **Aran, A.**; **Sanahuja, B.**; Truscott. P.; Lei, F.; Jacobs, C.; Poedts S.; Gabriel, S.; Sandberg, I.; Glover, A.; Hilgers, A.
SEPEM: A tool for statistical modeling the solar energetic particle environment. Article.
Space Weather-The International Journal Of Research And Applications, Vol. 13, Num. 7, p. 406 - 426 (2015)
10.1002/2013SW001008

Cuesta, A.J.; **Verde, L.**; Riess, A.; **Jimenez, R.**
Calibrating the Cosmic distance scale ladder: the role of the sound horizon scale and the local expansion rate as distance anchors. Article.
Monthly Notices of the Royal Astronomical Society,

Vol. 448, Num. 4, p. 3463 - 3471 (2015)
10.1093/mnras/stv261

Curran, P.A.; et al (ICCUB: **Migliari, S.**;)
Radio polarimetry as a probe of unresolved jets: The 2013 outburst of XTE J1908+094. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 451, p. 3975 - 3985 (2015)
10.1093/mnras/stv1252

Dadhich, N.; **Pons, J.M.**
On static black holes solutions in Einstein and Einstein-Gauss-Bonnet gravity with topology. Article.
European Physical Journal C, Vol. 6, p. 208 (2015)
10.1140/epjc/s10052-015-3481-y

Dadhich, N.; **Pons, J.M.**
Static pure Lovelock black hole solutions with horizon topology $S^{(n)} \times S^{(n)}$. Article.
Journal of High Energy Physics, Num. 67 (2015)
10.1007/JHEP05(2015)067

Dector, A.
Magnetic phenomena in holographic superconductivity with Lifshitz scaling. Article.
Nuclear Physics B, Vol. 898, p. 132 - 156 (2015)
10.1016/j.nuclphysb.2015.06.018

Del Palacio, S.; **Bosch-Ramon, V.**; Romero, G. E.
Gamma-ray binaries beyond one-zone models: an application to LS 5039. Article.
Astronomy and Astrophysics, Vol. 575, Num. A112 (2015)
10.1051/0004-6361/201424713

Delubac, T.; Bautista, J. E.; Busca, N. G.; Rich, J.; Kirkby, D.; Bailey, S.; Font-Ribera, A.; Slosar, A.; Lee, K-G.; Pieri, M. M.; Hamilton, J-C.; Aubourg, E.; Blomqvist, M.; Bovy, J.; Brinkmann, J.; Carithers, W.; Dawson, K.S.; Eisenstein, D. J.; **Gontcho A Gontcho, S.**; Kneib, J-P.; Le Goff, J-M.; Margala, D.; **Miralda-Escudé, J.**; Myers, A. D.; Nichol, R. C.; Noterdaeme, P.; O'Connell, R.
Baryon acoustic oscillations in the Lyα forest of BOSS DR11 quasars. Article.
Astronomy and Astrophysics, Vol. 574, Num. A59 (2015)
10.1051/0004-6361/201423969

Devastato, A.; **Lizzi, F.**; Flores, C.V.; Vassilevich, D.
Unification of coupling constants; dimension 6 operators and the spectral action. Article.
International Journal of Modern Physics A, Vol. 30, Num. 1550033 (2015)
10.1142/S0217751X15500335

Di Dato, A.; **Fröb, M.B.**
Mapping AdS to dS spaces and back. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 91, Num. 64028 (2015)
10.1103/PhysRevD.91.064028

Díaz-Londoño, G.; García-Pareja, S.; **Salvat, F.**; Lallena A.M.
Monte Carlo calculation of specific absorbed fractions: variance reduction techniques. Article.
Physics in Medicine and Biology, Vol. 60, Num. 7 (2015)
10.1088/0031-9155/60/7/2625

Di Dato, A.; Gath, J.; Vigand Perdersen, A.
Probing the hydrodynamic limit of (super)gravity. Article.
Journal of High Energy Physics, Vol. 171 (2015)
10.1007/JHEP04(2015)171

Donos, A.; Gauntlett, J.P.; **Pantelidou, C.**
Conformal field theories in d=4 with a helical twist. Article.
Physical Review D, Vol. 91, Num. 066003 (2015)
10.1103/PhysRevD.91.066003

Emparan, R.; Shiromizu, T.; Suzuki, R.; Tanabe, K.; Tanaka, T.
Effective theory of black holes in the 1/D expansion. Article.
Journal of High Energy Physics, Vol. 1506, p. 159 (2015)
10.1007/JHEP06(2015)159

Emparan, R.; Suzuki, R.; Tanabe, K.
Evolution and End Point of the Black String Instability: Large D Solution. Article.
Physical Review Letters, Vol. 115, p. 91102 (2015)
10.1103/PhysRevLett.115.091102

Emparan, R.; Suzuki, R.; Tanabe, K.
Quasinormal modes of (anti-)de Sitter black holes in the 1/D expansion. Article.
Journal of High Energy Physics, Vol. 1504, p. 85 (2015)
10.1007/JHEP04(2015)085

Espriu, D.
Searching for P- and CP-odd effects in HIC. Article.
Journal of Physics: Conference Series, Vol. 623, Num. 12012 (2015)
10.1088/1742-6596/623/1/012012

Espriu, D.; **Mescia, F.**; **Renau, A.**
Axion-Higgs interplay in the two-Higgs-doublet model. Article.
Physical Review D, Vol. 92, Num. 95013 (2015)
10.1103/PhysRevD.92.095013

Espriu, D.; **Renau, A.**
Photons in a cold axion background and strong magnetic fields: Polarimetric consequences. Article.
International Journal of Modern Physics A, Vol. 30, Num. 1550099 (2015)
10.1142/S0217751X15500992

Fabricius, C.
Gaia Validation Tasks. Article.
EAS Publications Series, Vol. 67-68, p. 57 - 60 (2015)
10.1051/eas/1567008

Faedo, A.F.; Fraser, B.; Kumar, S.P.
Emergent Lifshitz scaling from N = 4 SYM with supersymmetric heavy-quark density. Article.
Journal of High Energy Physics, Vol. 2015, Num. 84 (2015)
10.1007/JHEP05(2015)084

Faedo, A.F.; **Kundu, A.**; **Mateos, D.**; **Tarrio, J.**
(Super)Yang-Mills at Finite Heavy-Quark Density. Article.
Journal of High Energy Physics (2015)
10.1007/JHEP02(2015)010

Faedo, A.F.; **Mateos, D.**; **Tarrío, J.**
Three-dimensional super Yang-Mills with unquenched flavor. Article.
Journal of High Energy Physics, Vol. 2015, Num. 56 (2015)
10.1007/JHEP07(2015)056

Falzone, N.; **Fernández-Varea, J.M.**; Flux, G.; Vallis, K.A.
Monte Carlo evaluation of Auger electron-emitting theranostic radionuclides. Article.
Journal of Nuclear Medicine, Vol. 56, p. 1441 - 1446 (2015)
10.2967/jnumed.114.153502

Farrés, A.; **Jorba, À.**; Mondelo, J.-M.
Orbital dynamics for a non-perfectly reflecting solar sail close to an asteroid. Conference Paper.
Advances in the Astronautical Sciences, Vol. 153, p. 1229 - 1247 (2015)

Feijoo, A.; **Magas, V.K.**; **Ramos, A.**
The $\bar{K}N \to K\Xi$ reaction in coupled-channels chiral models up to next-to-leading order. Article.
Physical Review C, Vol. 92, p. 015206-1 - 015206-14 (2015)
10.1103/PhysRevC.92.015206

Feijoo, A.; **Magas, V.K.**; **Ramos, A.**; Oset, E.
$\Lambda_b \to J/\psi K \Xi$ decay a nd the higher order chiral terms of the meson baryon interaction. Article.
Physical Review D, Vol. 92, p. 076015-1 - 076015-10 (2015)
10.1103/PhysRevD.92.076015

Fernández, J. E.; Scot, V.; Di Giulio, E.; **Salvat, F.**
Bremsstrahlung contribution to the X-ray spectrum in coupled photon-electron transport. Article.
X-Ray Spectrometry, Vol. 44, p. 248 - 254 (2015)
10.1002/xrs.2615

Fernández, J. E.; Scot, V.; Di Giulio, E.; **Salvat, F.**
Evaluation of bremsstrahlung contribution to photon transport in coupled photon-electron problems. Article.
Radiation Physics and Chemistry, Vol. 116, p. 203 - 207 (2015)
10.1016/j.radphyschem.2015.05.026

Fiol, B.; Guijosa, A.; Pedraza, J.
Branes from Light: Embeddings and Energetics for symmetric k-quarks in N=4 SYM. Article.
Journal of High Energy Physics, Vol. 2015, Num. 149 (2015)
10.1007/JHEP01(2015)149

Fritzsch, H.; **Solà, J.**
Fundamental Constants and Cosmic Vacuum: the micro and macro connection. Article.
Modern Physics Letters A, Vol. A30, Num. 22, p. 1540034 (2015)
10.1142/S0217732315400349

Furniss, A.; et al (NuSTAR Team; MAGIC Collaboration; VERITAS Collaboration; F-Gamma Consortium. ICCUB: **Galindo, D.**; **Marcote, B.**; **Paredes, J.M.**; **Paredes-Fortuny, X.**; **Ribó, M.**; **Zanin, R.**)
First NuSTAR Observations of Mrk 501 within a Radio to TeV Multi-Instrument Campaign. Article.
Astrophysical Journal, Vol. 812, Num. 65 (2015)
10.1088/0004-637X/812/1/65

Gallemí, A.; Guilleumas, M.; Martorell, J.; Mayol, R.; **Polls, A.**; **Juliá-Díaz, B.**
Fragmented condensation in Bose-Hubbard trimers with tunable tunneling. Article.
New Journal of Physics, Vol. 17, p. 073014-1 - 073014-

16 (2015)
10.1088/1367-2630/17/7/073014

Garcia-Alvarez, J.A.; Maidana, N.L.; Silva, T.F.; **Fernández-Varea, J.M.**; Martins, M.N.; Vanin, V.R.
Preliminary measurements of the Bremsstrahlung doubly differential cross section for electrons between 20 and 100 keV in Au. Conference Paper.
Journal of Physics: Conference Series, Vol. 635, Num. 52084 (2015)
10.1088/1742-6596/635/5/052084

Garcia-March, M.A.; Carr, L.D.
Vortex macroscopic superpositions in ultracold bosons in a double-well potential. Article.
Physical Review A - Atomic; Molecular; and Optical Physics, Vol. 91, Num. 33626 (2015)
10.1103/PhysRevA.91.033626

Garcia-March, M.A.; Yuste, A.; **Julia-Diaz, B.**; **Polls, A.**
Mesoscopic superpositions of Tonks-Girardeau states and the Bose-Fermi mapping. Article.
Physical Review A, Vol. 92, p. 33621 (2015)
10.1103/PhysRevA.92.033621

Garriga, J.; Skenderis, K.; Urakawa, Y.
Multi-field inflation from holography. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 1501, Num. 28, p. 028-1 - 028-24 (2015)
10.1088/1475-7516/2015/01/028

Gascon, D.; Chanal, H.; Comerma, A.; **Gomez, S.**; Han, X..; Mazorra, J.; **Mauricio, J.**; Pillet, N.; Yengui, F.; Vandaele, R.
PACIFIC: A 64-channel ASIC for scintillating fiber tracking in LHCb upgrade. Article.
Journal Of Instrumentation, Vol. 10, Num. 4, p. C04030 (2015)
10.1088/1748-0221/10/04/C04030

Germani, C.; Sarkar D.
Firewalls as artefacts of inconsistent truncations of quantum geometries. Article.
Fortschritte der Physik-Progress of Physics, Vol. 64, Num. 1, p. 131 - 143 (2015)
10.1002/prop.201500057

Gil-Marín, H.; Noreña, J.; **Verde, L.**; Percival, W.J.; Wagner, C.; Manera, M.; Schneider, D.P.
The power spectrum and bispectrum of SDSS DR11 BOSS galaxies - I. Bias and gravity. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 451, p. 539 - 580 (2015)
10.1093/mnras/stv961

Gil-Marín, H.; **Verde, L.**; Noreña, J.; **Cuesta, A.J.**; Samushia, L.; Percival, W.J.; Wagner, C.; Manera, M.; Schneider, D.P.
The power spectrum and bispectrum of SDSS DR11 BOSS galaxies II: cosmological interpretation. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 452, Num. 2, p. 1914 - 1921 (2015)
10.1093/mnras/stv1359

Gómez, J. F.; Rizzo, J. R.; Suárez, O.; Palau, A.; Miranda, L. F.; Guerrero, M. A.; Ramos-Larios, G.; **Torrelles, J.M.**
A search for water maser emission toward obscured post-AGB star and planetary nebula candidates. Article.
Astronomy and Astrophysics, Vol. 578, Num. A119 (2015)
10.1051/0004-6361/201526009

Gómez-Herrero, R.; Dresing, N.; Klassen, A.; Heber, B.; Lario, D.; **Agueda, N.**; Malandraki, O.E.; Blanco, J.J.; Rodríguez-Pacheco, J.; Banjac, S.
Circumsolar energetic particle distribution on 2011 November 3. Article.
Astrophysical Journal, Vol. 799, Num. 55 (2015)
10.1088/0004-637X/799/1/55

Gómez-Valent, A.; **Karimkhani, E.**; **Solà, J.**
Background history and cosmic perturbations for a general system of self-conserved dynamical dark energy and matter. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 12, p. 1 - 48 (2015)
10.1088/1475-7516/2015/12/048

Gómez-Valent, A.; **Solà J.**
Vacuum models with a linear and a quadratic term in H: structure formation and number counts analysis. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 448, Num. 3, p. 2810 - 2821 (2015)
10.1093/mnras/stv209

Gómez-Valent, A.; **Solà, J.**; Basilakos, S.
Dynamical vacuum energy in the expanding Universe confronted with observations: a dedicated study. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, p. 4 - 62 (2015)
10.1088/1475-7516/2015/01/004

Gonzalez-Vidal, J.J.; Perez-Pueyo, R.; Soneira, M.J.; Ruiz-Moreno, S.
Independent component analysis-based algorithm for automatic identification of raman spectra applied to artistic pigments and pigment mixtures. Journal. Applied Spectroscopy, Vol. 69, p. 314 - 322 (2015)
10.1366/14-07502

Goyeneche, D.; **Alsina, D.**; **Latorre, J.I.**; Riera, A.; Zyczkowski, K.
Absolutely maximally entangled states, combinatorial designs, and multiunitary matrices. Article.
Physical Review A, Vol. 92, Num. 32316 (2015)
10.1103/PhysRevA.92.032316

Gridnev, K.A.; Tarasov, V.N.; Gridnev, D.K., Greiner, W.; **Viñas, J.**
Resonance capture of multineutrons by the 88Sr and 27Al nuclei. Article.
Jetp Letters, Vol. 102, Num. 6, p. 321 - 323 (2015)
10.1134/S002136401518006X

Grudzinska, M.; Belczynski, K.; Casares, J.; de Mink, S. E.; Ziolkowski, J.; Negueruela, I.; **Ribó, M.**; Ribas, I.; **Paredes, J.M.**; Herrero, A.; Benacquista, M.
On the formation and evolution of the first Be star in a black hole binary MWC 656. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 452, Num. 3, p. 2773 - 2787 (2015)
10.1093/mnras/stv1419

Gürsoy, U.; **Tarrío, J.**
Horizon universality and anomalous conductivities. Article.
Journal of High Energy Physics, Vol. 2015, Num. 58 (2015)
10.1007/JHEP10(2015)058

Herrero, E.; Ribas, I.; **Jordi, C.**
Correcting EChO data for stellar activity by direct s caling of activity signals. Article.
Experimental Astronomy, Vol. 40, Num. 2, p. 695 - 710 (2015)
10.1007/s10686-014-9387-0

Iudica, R.; Artigues, G.; **Portell, J.**; García-Berro, E.
Image data compression with hierarchical pixel averaging and fully adaptive prediction error coder. Article.
Journal of Applied Remote Sensing, Vol. 9, Num. 97493 (2015)
10.1117/1.JRS.9.097493

Iwasawa, K.; Vignali, C.; Comastri, A.; Gilli, R.; Vito, F.; Brandt, W. N.; Carrera, F. J.; Lanzuisi, G.; Falocco, S.; Vagnetti, F.
The XMM deep survey in the CDF-S. VIII. X-ray properties of the two brightest sources. Article.
Astronomy and Astrophysics, Vol. 574, Num. A144 (2015)
10.1051/0004-6361/201425086

Jaén, X.; **Molina, A.**
On the meaning of Painlevé-Gullstrand synchronization. Article.
General Relativity and Gravitation, Vol. 47, Num. 12, p. 152 - 168 (2015)
10.1007/s10714-015-1994-1

Jorba, A.; Rabassa, P.; Tatjer, J. C.
A renormalization operator for 1D maps under quasiperiodic perturbations. Article.
Nonlinearity, Vol. 28, p. 1017 - 1042 (2015)
10.1088/0951-7715/28/4/1017

Khachatryan, V; et al (CMS and LHCb Collaborations. (ICCUB: **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**; **Ruiz, H.**)
Observation of the rare $B_s^0 \rightarrow \mu^+\mu^-$ *decay from the combined analysis of CMS and LHCb data*. Article.
Nature, Vol. 522, p. 68 - 72 (2015)
10.1038/nature14474

Knight, J.C.; Topping, C.; Mosley, M.; Kersemans, V.; Falzone, N.; **Fernández-Varea, J.M.**; Cornelissen, B.
PET imaging of DNA damage using ^{89}*Zr-labelled anti-*γ H_2AX*-TAT immunoconjugates*. Article.
European Journal of Nuclear Medicine and Molecular Imaging, Vol. 42, p. 1707 - 1717 (2015)
10.1007/s00259-015-3092-8

Kurkov, M.A.; **Lizzi, F.**; Sakellariadou, M.; Watcharangkool, A.
Spectral action with zeta function regularization. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 91, Num. 65013 (2015)
10.1103/PhysRevD.91.065013

Labraña, P.
Emergent universe scenario and the low CMB multipoles. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 91, Num. 83534 (2015)
10.1103/PhysRevD.91.083534

Lanzuisi, G.; et al. (ICCUB: **Iwasawa, K.**)
Compton thick AGN in the XMM-COSMOS survey. Article.
Astronomy and Astrophysics, Vol. 573, Num. A137 (2015)
10.1051/0004-6361/201424924

Latorre, J.I.; Sierra, G.
There is entanglement in the primes. Article.
Quantum Information & Computation, Vol. 15, p. 622 - 659 (2015)

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Collins asymmetries in inclusive charged KK and Kπ pairs produced in e^+e^- annihilation. Article.
Physical Review D, Vol. 92, Num. 111101 (2015)
10.1103/PhysRevD.92.111101

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Dalitz plot analyses of $B^0 \to D^-D^0K^+$ and $B^+ \to \bar{D}^0 D^0K^+$ decays. Article.
Physical Review D, Vol. 91, Num. 52002 (2015)
10.1103/PhysRevD.91.052002

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Measurement of initial-state-final-state radiation interference in the processes $e^+e^- \to \mu^+\mu^-\gamma$ and $e^+e^- \to \pi^+\pi^-\gamma$. Article.
Physical Review D, Vol. 92, Num. 72015 (2015)
10.1103/PhysRevD.92.072015

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Measurement of the branching fractions of the radiative leptonic τ decays $\tau \to e\gamma\ \nu\ \bar{\nu}$ and $\tau \to \mu\gamma\nu\ \bar{\nu}$ at BABAR. Article.
Physical Review D Vol. 91, Num. 51103 (2015)
10.1103/PhysRevD.91.051103

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Measurement of the $D^0 \to \pi^-e^+\ \nu_e$ differential decay branching fraction as a function of q^2 and study of form factor parametrizations. Article.
Physical Review D, Vol. 91, Num. 52022 (2015)
10.1103/PhysRevD.91.052022

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Observation of the baryonic decay $\bar{B}^0 \to \Lambda_c^+\ \bar{p}\ K^-K^+$. Article.
Physical Review D, Vol. 91, Num. 31102 (2015)
10.1103/PhysRevD.91.031102

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Search for a light Higgs resonance in radiative decays of the Y(1S) with a charm tag. Article.
Physical Review D, Vol. 91, Num. 71102 (2015)
10.1103/PhysRevD.91.071102

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Search for Long-Lived Particles in e^+e^- Collisions. Article.
Physical Review Letters, Vol. 114, Num. 171801 (2015)
10.1103/PhysRevLett.114.171801

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Study of $B^{\pm,0} \to J/\psi K^+K^-K^{\pm,0}$ and search for $B^0 \to J/\psi\varphi$ at BaBar. Article.
Physical Review D, Vol. 91, Num. 12003 (2015)
10.1103/PhysRevD.91.012003

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Study of CP Asymmetry in B^0 - $\bar{B}^0$ Mixing with Inclusive Dilepton Events. Article.
Physical Review Letters, Vol. 114, Num. 81801 (2015)
10.1103/PhysRevLett.114.081801

Lees, J.P.; et al (BaBar collaboration. ICCUB: **Graugés, E.**)
Study of the $e^+e^- \to K^+K^-$ reaction in the energy range from 2.6 to 8.0 GeV. Article.
Physical Review D, Vol. 92, Num. 72008 (2015)
10.1103/PhysRevD.92.072008

Lima, J.; Basilakos, S.; **Solà, J.**
Nonsingular decaying vacuum cosmology and entropy production. Article.
General Relativity and Gravitation, Vol. 47, Num. 4, p. 40 (2015)
10.1007/s10714-015-1888-2

Lizzi, F.
Spectral geometry for quantum spacetime. Article.
Nuovo Cimento della Societa Italiana di Fisica C, Vol. 38, Num. 165 (2015)
10.1393/ncc/i2015-15165-3

Lizzi, F.; Rivera, M.; Vitale, P.
Green's functions for translation invariant star products. Article.
Modern Physics Letters A, Vol. 30, Num. 1550194 (2015)
10.1142/S0217732315501941

Lombriser, L.; **Simpson, F.**; Mead, A.
Unscreening modified gravity in the matter power spectrum. Article.
Physical Review Letters, Vol. 114, Num. 251101 (2015)
10.1103/PhysRevLett.114.251101

López, R.; Acosta-Pulido, J.A.; **Estalella, R.**; Gómez, G.; García-Lorenzo, B.
3D kinematics of the near-IR HH 223 outflow in L723. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 447, Num. 3, p. 2588 - 2602 (2015)
10.1093/mnras/stu2607

Mangano, G.; **Lizzi, F.**; Porzio, A.
Inconstant Planck's constant. Article.
International Journal of Modern Physics A, Vol. 30, Num. 1550209 (2015)
10.1142/S0217751X15502097

Manrique, A.; **Salvador-Solé, E.**
An improved treatment of cosmological intergalactic medium evolution. Article.
Astrophysical Journal, Vol. 803, Num. 2, p. 103 - 109 (2015)
10.1088/0004-637X/803/2/103

Manrique, A.; **Salvador-Solé, E.**; **Juan, E.**; Hatziminaoglou, E.; Rozas, J.M.; Sagristà, A.; Casteels, K.; Bruzual, G.; Magris, G.
Leaving the Dark ages with AMIGA. Article.
Astrophysical Journal Supplement Series, Vol. 216, Num. 1, p. 1 - 17 (2015)
10.1088/0067-0049/216/1/13

Marcote, B.; et al. (ICCUB: **Marcote, B.**, **Ribó, M.**; **Paredes, J.M.**)
Orbital and superorbital variability of LS I +61 303 at low radio frequencies with GMRT and LOFAR. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 456, Num. 2, p. 1791 - 1802 (2015)
10.1093/mnras/stv2771

Marcote, B.; **Ribó, M.**; **Paredes, J.M.**; Ishwara-Chandra, C.H.
Physical properties of the gamma-ray binary LS 5039 through low- and high-frequency radio observations. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 451, Num. 1, p. 4578 - 4592 (2015)
10.1093/mnras/stv940

Martí, J.; Luque-Escamilla, P.L.; Casares, J.; **Marcote, B.**; **Paredes-Fortuny, X.**; **Ribó, M**; **Paredes, J.M.**; **Núñez, J.**
In quest of non-thermal signatures in early-type stars. Article.
Astrophysics and Space Science, Vol. 355, Num. 2, p. 277 - 284 (2015)
10.1007/s10509-014-2157-7

Martín, J.; **Molina, A.**; Ruiz, E.
Perturbation theory and harmonic gauge propagation in general relativity, a particular example. Article.
General Relativity and Gravitation (2015)
10.1007/s10714-015-1947-8

Martínez Torres, A.; Oset, E.; Prelovsek, S.; **Ramos, A.**
Reanalysis of lattice QCD spectra leading to the D^*_{s0} (2317) D^*_{s1} (2460). Article.
Journal of High Energy Physics, Vol. 1505, p. 153-1 - 153-21 (2015)
10.1007/JHEP05(2015)153

Masqué, J.M.; Rodríguez, L.F.; Araudo, A.; **Estalella, R.**; Carrasco-González, C.; Anglada, G.; Girart, J.M.; Osorio, M.
Proper Motions of the Outer Knots of the HH 80/81/80N Radio-jet. Article.
Astrophysical Journal, Vol. 814, p. 44 (2015)
10.1088/0004-637X/814/1/44

Mateu, C.; Antoja, T.; Aguilar, L.; **Figueras, F.**; Brown, A.; **Antiche, E.**; Hernandez-Perez, F.; Valenzuela, O.; Aparicio, A.; Hidalgo, S.; Velazquez, H.
Detectability of ultra faint dwarf galaxies with gaia. Book in Series.
EAS Publications Series, Vol. 67-68, p. 385 - 385 (2015)
10.1051/eas/1567079

Mathe, Z.; **Casajus Ramo, A.**; **Lazovsky, N.**; Stagni, F.
The DIRAC web portal 2.0. Conference Paper.
Journal of Physics: Conference Series, Vol. 664, Num. 62039 (2015)
10.1088/1742-6596/664/6/062039

Mathe, Z.; **Casajus Ramo, A.**; Stagni, F.; Tomassetti, L.
Evaluation of NoSQL databases for Dirac monitoring and beyond. Conference Paper.
Journal of Physics: Conference Series, Vol. 664, Num. 42036 (2015)
10.1088/1742-6596/664/4/042036

Mecca, A.; Lovato, A.; Benhar, O.; **Polls, A.**
Effective-interaction approach to the Fermi hard-sphere system. Article.
Physical Review C, Vol. 91, Num. 3, p. 34325 (2015)
10.113/PhysRevC.91.034325

Mehtar-Tani, Y.; **Tywoniuk, K.**
Jet (de)coherence in Pb-Pb collisions at the LHC. Article.
Physics Letters; Section B: Nuclear; Elementary Particle and High-Energy Physics, Vol. 744, p. 284 - 287 (2015)
10.1016/j.physletb.2015.03.041

Monguió, M.; Grosbol, P.; **Figueras, F.**
First detection of the field star overdensity in the Perseus Arm. Article.
Astronomy and Astrophysics (2015)
10.1051/0004-6361/201424896

Monguio, M.; Grosbol, P.; **Figueras, F.**
The perseus arm stellar overdensity at 1.6 kpc. Book in Series.
EAS Publications Series, Vol. 67-68, p. 205 - 209 (2015)
10.1051/eas/1567037

Mor, R.; Robin, A. C.; Lemasle, B.; **Figueras, F.**
Impact of the uncertainties of the ISM when studying the IMF at intermediate masses. Article.
Memorie della Societa Astronomica Italiana, Vol. 86, p. 562 - 562 (2015)

Mor, R.; Robin, AC.; **Figueras, F.**; Lemasle, B.
The IMF at intermediate masses from galactic cepheids. Book in Series.
EAS Publications Series, Vol. 67-68, p. 387 - 387 (2015)
10.1051/eas/1567080

Muessel, W.; Strobel, H.; Linnemann, D.; Zibold, T.; **Julia-Diaz, B.**; Oberthaler, M.K.
Twist-and-turn spin squeezing in Bose-Einstein condensates. Article.
Physical Review A, Vol. 92, p. 23603 (2015)
10.1103/PhysRevA.92.023603

Muraveva, T.; **Palmer, M.**; Clementini, G.; **Luri, X.**; Cioni, M.-R.L.; Moretti, M. I.; Marconi, M.; Ripepi, V.
New near-infrared period-luminosity-metallicity relations for RR Lyrae stars and the outlook for Gaia. Article.
Astrophysical Journal, Vol. 807, Num. 2 (2015)
10.1088/0004-637X/807/2/127

Nersisyan, H.B.; **Fernández-Varea, J.M.**; Arista, N.R.
Dynamic screening of an ion in a degenerate electron gas within the second-order Born approximation. Article.
Nuclear Instruments & Methods in Physics Research Section B-Beam Interactions With Materials and Atoms, Vol. 354, p. 167 - 171 (2015)
10.1016/j.nimb.2014.11.089

Notari, A.; Quartin, M.
On the proper kinetic quadrupole CMB removal and the quadrupole anomalies. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 047 (2015)
10.1088/1475-7516/2015/06/047

Notari, A.
Higgs mass and gravity waves in standard model false vacuum inflation. Article.
Physical Review D, Vol. 91, Num. 063527 (2015)
10.1103/PhysRevD.91.063527

Núñez, J.; Núñez, A., Montojo, F.J.; Condominas, M.
Improving Space Debris Detection in GEO ring Using Image Deconvolution. Article.
Advances in Space Research, Vol. 55, p. 1 - 10 (2015)
10.1016/j.asr.2015.04.006

Olikara, Z.P.; **Gómez, G.**; Masdemont, J.J.
Dynamic mechanisms for spacecraft disposal from Sun Earth libration points. Article.
Journal of Guidance Control and Dynamics, Vol. 38, Num. 10, p. 1976 - 1989 (2015)
10.2514/1.G000581

Orginos, K.; **Parreno, A.**; Savage, M.J.; Tiburzi, B.C.; Beane, S.R.; Chang, E.; Detmold, W.
Two nucleon systems at $m\pi \sim 450$ MeV from lattice QCD. Article.
Physical Review D, Vol. 92, Num. 11, p. 114512 (2015)
10.1103/PhysRevD.92.114512

Oriola, D.; Roth, S.; Dogterom, M.; **Casademunt, J.**
Formation of helical membrane tubes around microtubules by single-headed kinesin KIF1A. Article.
Nature Communications, Vol. 6, p. 8025-1 - 8025-8 (2015)
10.1038/ncomms9025

Oset, E.; et al (ICCUB: **Feijoo, A.**; **Magas, V.K.**; **Ramos, A.**)
Weak decays of heavy hadrons into dynamically generated resonances. Review.
International Journal of Modern Physics E-Nuclear Physics, Vol. 25, p. 1630001-1 - 1630001-10 (2015)
10.1142/S0218301316300010

Palau, A.; Ballesteros-Paredes, J.; Vázquez-Semadeni, E.; Sánchez-Monge, Á.; **Estalella, R.**; Fall, S.M.; Zapata, L.A.; Camacho, V.; Gómez, L.; Naranjo-Romero, R.; Busquet, G.
Gravity or turbulence? -III. Evidence of pure thermal Jeans fragmentation at ~ 0.1 pc scale. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 453, p. 3785 - 3797 (2015)
10.1093/mnras/stv1834

Palmer, M.; **Luri, X.**; Arenou, F.; **Masana, E.**
Astrostatistics for luminosity calibration in the gaia era. Book in Series.
EAS Publications Series, Vol. 67-68, p. 271 - 274 (2015)
10.1051/eas/1567048

Pan, L.; **Padoan, P.**
Turbulence-induced Relative Velocity of Dust Particles V. Testing Previous Models. Article.
Astrophysical Journal, Vol. 812, Num. 10 (2015)
10.1088/0004-637X/812/1/10

Paredes-Fortuny, X.; **Bosch-Ramon, V.**; Perucho, M.; **Ribó, M.**
Simulations of an inhomogeneous stellar wind interacting with a pulsar wind in a binary system. Article.
Astronomy and Astrophysics, Vol. 574, Num. A77 (2015)
10.1051/0004-6361/201424672

Paredes-Fortuny, X.; **Ribó, M.**; **Bosch-Ramon, V.**; Casares, J.; Fors, O.; **Núñez, J.**
Evidence of coupling between the thermal and nonthermal emission in the gamma-ray binary LS I +61 303. Article.
Astronomy and Astrophysics, Vol. 575, Num. L6, p. 1 - 4 (2015)
10.1051/0004-6361/201425361

Peracaula, M.; Torrent, A.; Masias, M.; Lladó, X.; Freixenet, J.; Martí, J.; Sánchez-Sutil, J.R.; Muñoz-Arjonilla, A.J.; **Paredes, J.M.**
Exploring three faint source detections methods for aperture synthesis radio images. Article.
New Astronomy Reviews, Vol. 36, p. 86 - 99 (2015)
10.1016/j.newast.2014.10.006

Pérez-Palau, D.; Masdemont, J.J.; **Gómez, G.**
Tools to detect structures in dynamical systems using Jet Transport. Article.
Celestial Mechanics and Dynamical Astronomy, Vol. 123, Num. 3, p. 239 - 262 (2015)
10.1007/s10569-015-9634-3

Pérez-Ràfols, I.; **Miralda-Escudé, J.**; Lundgren, B.; Ge, J.; Petitjean, P.; Schneider, D.P.; York, D.G.; Weaver, B.A.
The cross-correlation of Mg II absorption and galaxies in BOSS. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 447, p. 2784 - 2802 (2015)
10.1093/mnras/stu2645

Picatoste, E.; Bigbeder-Beau, C.; Duarte, O.; **Garrido, L.**; **Gascon, D.**; **Grauges, E.**; Lefrançois, J.; Machefert, F.; **Mauricio, J.**; Vilasis, X.
Low noise 4-channel front end ASIC with on-chip DLL for the upgrade of the LHCb Calorimeter. Article.
Journal Of Instrumentation, Vol. 10, Num. C04017 (2015)
10.1088/1748-0221/10/04/C04017

Pomoell, J.; **Aran, A.**; Jacobs, C.; Rodríguez-Gasén; Poedts S.; **Sanahuja, B.**
Modelling large solar proton events with the shock-and-particle model: Extraction of the characteristics of the MHD shock front at the cobpoint. Article.
Journal Of Space Weather And Space Climate, Vol. 5, Num. A12, p. 1 - 20 (2015)
10.1051/swsc/201505

Portell, J.; **Fabricius, C.**; **Torra, J.**; **Garralda, N.**; **González, J.**; **Castañeda, J.**
Daily processing of Gaia data. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain. A. J. Cenarro, F. Figueras, C. Hernández-Monteagudo, J. Trujillo Bueno, and L. Valdivielso (eds.), p. 822 - 827 (2015)

Prieto, J.; **Jimenez, R.**; Haiman, Z.; González, R. E.
The origin of spin in galaxies: clues from simulations of atomic cooling haloes. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 452, p. 784 - 802 (2015)
10.1093/mnras/stv1234

Privon, G. C.; et al (ICCUB: **Iwasawa, K.**)
Excitation Mechanisms for HCN (1-0) and HCO^+ (1-0) in Galaxies from the Great Observatories All-sky LIRG Survey. Article.
Astrophysical Journal, Vol. 814, Num. 39 (2015)
10.1088/0004-637X/814/1/39

Qiu, H.; Zambrini, R.; **Polls, A.**; Martorell, J.; **Julia-Diaz, B.**
Hybrid synchronization in coupled ultracold atomic gases. Article.
Physical Review A, Vol. 92, p. 43619 (2015)
10.1103/PhysRevA.92.043619

Quartin, M. ; **Notari, A.**
On the significance of power asymmetries in Planck CMB data at all scales. Article.

Journal Of Cosmology And Astroparticle Physics, Vol. 2015, Num. 008 (2015)
10.1088/1475-7516/2015/01/008

Quartin, M.; **Notari A.**
Improving Planck calibration by including frequency-dependent relativistic corrections. Article.
Journal Of Cosmology And Astroparticle Physics(826), Vol. 2015, Num. 050 (2015)
10.1088/1475-7516/2015/09/050

Ribó J.M.; Hochberg D.
Competitive exclusion principle in ecology and absolute asymmetric synthesis in chemistry. Article.
Chirality, Vol. 27, Num. 10, p. 722 - 727 (2015)
10.1002/chir.22490

Riesgo, A.; Taboada, S.; Sánchez-Vila, L.; **Solà, J.**; Bertran, A.; Avila, C.
Some like it fat: Comparative ultrastructure o f the embryo in two demosponges of the genus mycale (order poecilosclerida) from Antarctica and the Caribbean. Article.
PLoS ONE, Vol. 10 (2015)
10.1371/journal.pone.0118805

Roca-Fabrega, S.; **Figueras, F.**; Valenzuela, O.; **Romero-Gomez, M.**; Antoja, T.; Colin, P.; Pichardo, B.; Velazquez, H.
Novel kinematic methods to trace spiral arms nature using gaza data. Book in Series.
EAS Publications Series, Vol. 67-68, p. 393 - 394 (2015)
10.1051/eas/1567083

Roca-Maza, X.; **Viñas, X.**; **Centelles, M.**; Agrawal, B.K.; Colò, G.; Paar, N.; Piekarewicz, J.; Vretenar, D.
The neutron skin thickness from the measured electric dipole polarizability iNi , ^{120} Sn, an d ^{208} Pb. Article.
Physical Review C, Vol. 92, Num. 6, p. 064304-1 - 064304-11 (2015)
10.1103/PhysRevC.92.064304

Romero-Gómez, M.; **Figueras, F.**; Antoja, T.; **Abedi, H.**; Aguilar, L.
A view of the Galactic bar in the Gaia space of observables . Article.
EAS Publications Series, Vol. 67-68, p. 87 - 90 (2015)
10.1051/eas/1567015

Romero-Gómez, M.; **Figueras, F.**; Antoja, T.; **Abedi, H.**; Aguilar, L.
The analysis of realistic Stellar Gaia mock catalogues. I. Red Clump stars as tracers of the central bar. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 447, Num. 1, p. 218 - 233 (2015)
10.1093/mnras/stu2457

Russell, T.D.; et al. (ICCUB: **Migliari, S.)**
Radio monitoring of the hard state jets in the 2011 outburst of MAXI J1836-194. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 450, p. 1745 - 1759 (2015)
10.1093/mnras/stv723

Russo, J.G.
Large N_c from Seiberg-Witten Curve and Localization. Article.
Physics Letters B, Vol. 748, p. 19 (2015)
10.1016/j.physletb.2015.06.051

Russo, J.G.; Silva, G.A.
Exact partition function in U(2) x U(2) ABJM theory deformed by mass and Fayet-Iliopoulos terms. Article.
Journal of High Energy Physics, Vol. 12, p. 92 (2015)
10.1007/JHEP12(2015)092

Russo, J.G.; Silva, G.A.; Tierz, M.
Supersymmetric U(N) Chern-Simons-Matter Theory and Phase transitions. Article.
Communications in Mathematical Physics, Vol. 338, Num. 3, p. 1411 (2015)
10.1007/s00220-015-2399-4

Salazar, F.J.T.; Winter, O.C.; Macau, E.E.; Masdemont, J.J.; **Gómez, G.**
Natural formations at the Earth Moon triangular point in perturbed restricted problems. Article.
Advances in Space Research, Vol. 56, p. 144 - 162 (2015)
10.1016/j.asr.2015.03.028

Salazar, F.J.T.; Winter, O.C.; Macau, E.E.; Masdemont, J.J.; **Gómez, G.**
Zero drift regions and control strategies to keep satellite in formation around triangular libration point in the restricted Sun Earth Moon scenario. Article.
Advances in Space Research, Vol. 56, p. 1502 - 1518 (2015)
10.1016/j.asr.2015.07.001

Salvat, F.
The PENELOPE code system. Specific features and recent improvements. Article.
Annals of Nuclear Energy, Vol. 82, p. 98 - 109 (2015)
10.1016/j.anucene.2014.08.007

Sawicki, I.; **Bellini, E.**
Limits of quasistatic approximation in modified-gravity cosmologies. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 92, Num. 84061 (2015)
10.1103/PhysRevD.92.084061

Sharma, B.K.; **Centelles, M.**; **Viñas, X.**; Baldo, M.; Burgio, G.M.
Unified equation of state for neutron stars on a microscopic basis. Article.
Astronomy and Astrophysics, Vol. 584, Num. 9, p. A103-1 - A103-22 (2015)
10.1051/0004-6361/201526642

Sirvent, J.L.; Dehning, B.; Emery, J.; **Dieguez, A.**
Secondary particle acquisition system for the CERN beam wire scanners upgrade. Article.
Journal Of Instrumentation, Vol. 10, Num. 4, p. C04021 - C04031 (2015)
10.1088/1748-0221/10/04/C04021

Solà, J.
Preface of Fundamental Constants in Physics and their Time Variation. Review.
Modern Physics Letters A, Vol. A30, Num. 22, p. 1502004 (2015)
10.1142/S0217732315020046

Solà, J.
The Cosmological Constant and Entropy problems: mysteries of the present with profound roots in the past. Article.
International Journal of Modern Physics D, Vol. D24, Num. 12, p. 1544027 (2015)
10.1142/S0218271815440277

Solà, J.; **Gómez-Valent, A.**
The ΛCDM model: From inflation to dark energy through running Λ. Article.
International Journal of Modern Physics D, Vol. 24, Num. 4, p. 1541003 - 1541039 (2015)
10.1142/S0218271815410035

Solà, J.; *Gómez-Valent, A.*; **Cruz, J.**
Hints of dynamical vacuum energy in the expanding universe. Article.
Astrophysical Journal, Vol. 811, Num. 1, p. L14 (2015)
10.1088/2041-8205/811/1/L14

Su, N.; **Tywoniuk, K.**
Massless mode and positivity violation in hot QCD. Article.
Physical Review Letters, Vol. 114, Num. 161601 (2015)
10.1103/PhysRevLett.114.161601

Taboada, S.; Riesgo, A.; Blasco, G.; **Solà, J.**; Xavier, J.R.; López-Legentil, S.
Development of 10 microsatellite markers for the Atlanto-Mediterranean sponge Petrosia ficiformis. Letter.
Conservation Genetics Resources, Vol. 7, p. 895 - 897 (2015)
10.1007/s12686-015-0496-5

Tarasov, V.N.; Gridnev, K.A.; Greiner, W.; Kuprikov, V.I.; Gridnev, D.K.; Tarasov, D.V.; **Viñas, X.**; Godbey, K.S.
Investigating the properties of nuclei with extreme neutron excess and $2 \leq Z \leq 8$. Article.
Bulletin of the Russian Academy of Sciences: Physics, Vol. 79, p. 819 - 822 (2015)
10.3103/S1062873815070242

Tarasov, V.N.; Gridnev, K.A.; Kuprikov, V.I.; Gridnev, D.K.; Tarasov, D.V.; Godbey, K.S.; **Viñas, X.**; Greiner, W.
Ligth exotic nuclei with extreme neutron excess and $2 \leq Z \leq 8$. Article.
International Journal of Modern Physics E-Nuclear Physics, Vol. 24, p. 1550057-1 - 1550057-14 (2015)
10.1142/S0218301315500573

Tetarenko, A.J.; Sivakoff, G.R.; Miller-Jones, J.C.A.; Curran, P.A.; Russell, T.D.; Coulson, I.M.; Heinz, S.; Maitra, D.; Markoff, S.B.; **Migliari, S.**; Petitpas, G.R.; Rupen, M.P.; Rushton, A.P.; Russell, D.M.; Sarazin, C.L.
SUB-mm JET PROPERTIES OF THE X-RAY BINARY SWIFT J1745-26. Article.
Astrophysical Journal, Vol. 805, Num. 30 (2015)
10.1088/0004-637X/805/1/30

Trenti, M.; **Padoan, P.**; **Jimenez, R.**
The Relative and Absolute Ages of Old Globular Clusters in the LCDM Framework. Article.
Astrophysical Journal Letters, Vol. 808, Num. L35 (2015)
10.1088/2041-8205/808/2/L35

Trenti, M.; Perna, R.; **Jimenez, R.**
The Luminosity and Stellar Mass Functions of GRB Host Galaxies: Insight Into the Metallicity Bias. Article.
Astrophysical Journal, Vol. 802, Num. 103 (2015)
10.1088/0004-637X/802/2/103

Tywoniuk, K.
Decoding the quark-gluon plasma with jet quenching observables. Conference Paper.
Journal of Physics: Conference Series, Vol. 612, Num. 12010 (2015)
10.1088/1742-6596/612/1/012010

Verde, L.
Neutrino properties from cosmology. Article.
Journal of Physics: Conference Series, Vol. 598, Num. 12010 (2015)
10.1088/1742-6596/598/1/012010

Verde, L.
Precision cosmology; Accuracy cosmology and Statistical cosmology. Article.
Proceedings of the International Astronomical Union, Vol. 10, p. 223 - 234 (2015)
10.1017/S1743921314013593

Vidaña, I.; Providência, C.; **Polls, A.**
Effect of tensor correlations on the density dependence of the nuclear symmetry energy. Article.
Symmetry, Vol. 7, p. 15 - 31 (2015)
10.3390/sym7010015

Vignali, C.; et al (ICCUB: **Iwasawa, K.**)
The XMM deep survey in the CDF-S. IX. An X-ray outflow in a luminous obscured quasar at z ~ 1.6. Article.
Astronomy and Astrophysics, Vol. 583, Num. A141 (2015)
10.1051/0004-6361/201525852

Vincent, A. C.; Vecchio, A.; **Miralda-Escudé, J.**; Peña Garay, C.
The Galactic 511 keV Line and the Intergalactic Positron Density. Article.
Physics Procedia, Vol. 61, p. 796 - 801 (2015)
10.1016/j.phpro.2014.12.102

Viñas, X.; Bhagwat, A.; Castillo, R.; **Centelles, M.**; Schuck, P.
Gogny-force inspired mass formula within the Wigner-Kirkwood averaging scheme. Proceeding.
Acta Physica Polonica B Proceedings Supplement, Vol. 8, Num. 3, p. 699 - 706 (2015)
10.5506/APhysPolBSupp.8.699

Viñas, X.; Bhagwat, A.; **Centelles, M.**; Schuck, P.; Wyss, R.
Applications to nuclear properties of the microscopic-macroscopic model based on the semiclassical Wigner-Kirkwood method. Article.
Physica Scripta, Vol. 90, Num. 11, p. 114001-1 - 114001-10 (2015)
10.1088/0031-8949/90/11/114001

Viñas, X.; Roca-Maza, X.; **Centelles, M.**
The determination of the bulk symmetry incompressibility from the isoscalar giant monopole resonance revisited. Proceeding.
Acta Physica Polonica B Proceedings Supplement, Vol. 8, Num. 3, p. 707 - 714 (2015)
10.5506/APhysPolBSupp.8.707

Walton, N.A.; Bailer-Jones, C.A.L.; Brown, A.G.A.; Clementini, G.; Eyer, L.; Feltzing, S.; **Figueras, F.**; Grebel, E.K.; Michalowski, T.; De Ridder, J.; Santos, N.; Smith, M.C.; Soubiran, C.
Great-itn and gaia: preparing for science. Book in Series.
EAS Publications Series, Vol. 67-68, p. 7 - 14 (2015)
10.1051/eas/1567002

Yun, J. L.; Elia, Davide; Djupvik, A. A.; **Torrelles, J.M.**; Molinari, S.
Not a galaxy: IRAS 04186+5143, a new young stellar cluster in the outer Galaxy. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 452, Num. 2, p. 1523 - 1534 (2015)
10.1093/mnras/stv1431

Zhang, J.; Blanco-Pillado, J.J.; **Garriga, J.**; Vilenkin, A.
Topological defects from the multiverse. Article.
Journal Of Cosmology And Astroparticle Physics, Vol. 59, Num. 5, p. 1 - 34 (2015)
10.1088/1475-7516/2015/05/059

Zilhão, M.; Noble, S.C.; Campanelli, M.; Zlochower, Y.
Resolving the relative influence of strong field spacetime dynamics and MHD on circumbinary disk physics. Article.
Physical Review D - Particles; Fields; Gravitation and Cosmology, Vol. 91, Num. 24034 (2015)
10.1103/PhysRevD.91.024034

Zilhão, M.; Witek, H.; Cardoso, V.
Nonlinear interactions between black holes and Proca fields. Article.
Classical and Quantum Gravity, Vol. 32, Num. 234003 (2015)
10.1088/0264-9381/32/23/234003

NON-SCI PUBLICATIONS

Abedi, H.; **Figueras, F.**; Aguilar, L.; Mateu, C.; **Romero-Gómez, M.**; López-Corredoira, M.; Garzón, F.
Galactic warp kinematics: model vs. Observations. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 423 - 428 (2015)

Acosta-Pulido, J.A.; et al. (ICCUB: **Estalella, R.**; **Migliari, S.**; **Moldón, J.**; **Paredes, J.M.**; **Ribó, M.**; **Solanes, J.M.**)
The Spanish Square Kilometre Array White Book. Book Published by Sociedad Española de Astronomía, p. 1 - 289 (2015)

Agueda, N.; Klein, K.L. (SEPServer Consortium)
Proportionality law between the flare SXR intensity and the number of released solar near-relativistic electrons. Proceeding.
Highlights of Spanish Astrophysics VIII; Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12; 2014; in Teruel; Spain, p. 659 - 666 (2015) (2015)

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Marcote, B.**; Munar-Adrover, P.; **Paredes, J. M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
VizieR Online Data Catalog: The 2009 multiwavelength campaign on Mrk421. Multimedia Publication (2015)

Aleksić, J.; et al (MAGIC Collaboration. ICCUB: **Galindo, D.**; **Marcote, B.**; Munar-Adrover, P.; **Paredes, J. M.**; **Paredes-Fortuny, X.**; **Ribo, M.**; **Zanin, R.**)
VizieR Online Data Catalog: Mrk421 in March 2010. Multimedia Publication (2015)

Antiche, E.; **Voss, H.**; **Luri, X.**; Anglada Escudé, G.; **Jordi, C.**
From Gaia to NEAT (Theia): synergies and results of the NEAT double-blind experiment. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 631 - 636 (2015)

Antoja, T.; **Figueras, F.**; Helmi, A.; Monari, G.
Dynamical processes in the disk of the Milky Way and Gaia perspective. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 384 - 389 (2015)

Balaguer-Núñez, L.; **Casamiquela, L.**; **Jordi, C.**; Muiños, J. L.; Galadí-Enríquez, D.; **Masana, E.**
Studying the Outskirts of NGC 2682 (M 67) open cluster. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 593 - 593 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (1). Article.
Minor Planets and Comets Circulars (MPC), p. 90966 - 90966 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (2). Article.
Minor Planets and Comets Circulars (MPC), p. 91854 - 91854 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (3). Article.
Minor Planets and Comets Circulars (MPC), p. 92474 - 92474 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (4). Article.
Minor Planets and Comets Circulars (MPC), p. 93114 - 93114 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (5). Article.
Minor Planets and Comets Circulars (MPC), p. 93768 - 93768 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (6). Article.
Minor Planets and Comets Circulars (MPC), p. 94438 -

94438 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (7). Article.
Minor Planets and Comets Circulars (MPC), p. 94830 - 94830 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (8). Article.
Minor Planets and Comets Circulars (MPC), p. 95374 - 95374 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (9). Article.
Minor Planets and Comets Circulars (MPC), p. 95855 - 95855 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (10). Article.
Minor Planets and Comets Circulars (MPC), p. 96415 - 96415 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Minor Planet Observations [006 Fabra Observatory] of year 2015 (11). Article.
Minor Planets and Comets Circulars (MPC), p. 97001 - 97001 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Observations and orbits of comets: [006 Fabra Observatory] comet C/2014 Q2 (Lovejoy). Article.
Minor Planets and Comets Circulars (MPC), p. 90859 - 90859 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Observations and orbits of comets: [006 Fabra Observatory] comet C/2014 Q2 (Lovejoy) (2). Article.
Minor Planets and Comets Circulars (MPC), p. 91799 - 91799 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Observations and orbits of comets: [006 Fabra Observatory] comet C/2014 Q2 (Lovejoy) (3). Article.
Minor Planets and Comets Circulars (MPC), p. 92701 - 92701 (2015)

Bernal, A.; Codina, J.M.; **Núñez, J.**; Torras, N.
Observations and orbits of comets: [006 Fabra Observatory] (4). Article.
Minor Planets and Comets Circulars (MPC), p. 94397 - 94397 (2015)

Bravina, L.; **Magas, V.K.**; **Feijoo, A.**
Proceedings of the International Workshop on Collectivity in Relativistic Heavy Ion Collisions (IWoC 2014), Kolymbari, Crete, Greece, September 14-20, 2015.
Editors.

Bulgarelli, A.; et al. (ICCUB: **Graciani, R.**; **Ribo, M.**)
The On-Site Analysis of the Cherenkov Telescope Array.
Article.
Proceedings of Science, Vol. ICRC2015, Num. 763 (2015)

Carmona, M.; **Gómez, J.M.**; Bosch, J.; López, M.; Ruiz, O.
Elmer - Guide to FEM simulations - v02. Book.
Universitat de Barcelona (2015)

Carmona, M.; **Gomez, J.M.**; Bosch, J.; Lopez, M.; Ruiz, O.
GMSH-Guide for mesh generation v03. Book.
Universitat de Barcelona (2015)

Carrasco, J. M.; **Voss, H.**; **Jordi, C.**; **Fabricius, C.**; Pancino, E.; Altavilla, G.
Selection of stars to calibrate Gaia. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain.p. 786 - 791 (2015)

Carrera, R.; **Casamiquela, L.**; **Balaguer-Núñez, L.**; **Jordi, C.**; Pancino, E.; Allende-Prieto, C.; Blanco-Cuaresma, S.; Martínez-Vázquez, C. E.; Murabito, S.; del Pino, A.; et al.
The Open Cluster Chemical Abundances from Spanish Observatories survey (OCCASO). Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 453 - 453 (2015)

Casamiquela, L.; **Balaguer-Núñez, L.**; **Jordi, C.**; **Masana, E.**
Physical parameters of NGC6705 (M11) open cluster using Strömgren photometry. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 596 - 596 (2015)

Castañeda, J.; **Fabricius, C.**; **Torra, J.**; **Clotet, M.**; **González, J.**; **Garralda, N.**; **Portell, J.**
First cycle processing of Gaia data. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 792 - 797 (2015)

Cenarro, A. J.; **Figueras, F.**; Hernández-Monteagudo, C.; Trujillo Bueno, J.; Valdivielso, L.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, . Book Editors.
(2015)

Cuesta, A. J.; **Verde, L.**; Riess, A.; **Jimenez, R.**
The cosmic inverse distance ladder: baryon acoustic oscillations and type-Ia supernovae. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 172 - 176 (2015)

De Oña Wilhelmi, E.; **Zanin, R.**; Carreto Fidalgo, D.; **Galindo, D.**; Garrido Terrats, D.; Idec, W.; Schweizer, T.; Bednarek, W.; Torres, D.F.; Saito, T.; Gaug, M.; López, M.; Hirotani, K.
Tera-electron-Volt pulsed emission from the Crab detected by MAGIC. Conference Paper.
Proceedings of Science, Vol. 30-July-2015, Num. 746 (2015)

Dieguez, A.; Alonso, O.; Vilella, E.; Vila, A.
A Verilog-A model for the design of devices for fluorescence life-time measurement with CMOS SPADs. Conference Paper.
2015 International Conference on Synthesis; Modeling; Analysis and Simulation Methods and Applications to Circuit Design; SMACD 2015, Num. 7301680 (2015)
10.1109/SMACD.2015.7301680

Dobado, A.; Delgado, R.L.; Llanes-Estrada, F.J.; **Espriu, D.**
Comparing mesons and W_L W_L TeV-resonances. Conference Paper.
Bled Workshops Phys. 16 (2015)

Fernández-Barral, A.; Blanch, O.; De Oña Wilhelmi, E.; Torres, D.F.; Fruck, C.; Hadasch, D.; López-Oramas, A.; Munar-Adrover, P.; **Zanin, R.**
VHE gamma-ray observations of transient and variable stellar objects with the MAGIC Telescopes. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 732 (2015)

Figueras, F.; **Jordi, C.**
The Gaia scientific exploitation networks. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 600 - 600 (2015)

Fioretti, V.; Bulgarelli, A.; Zoli, A.; Markoff, S.; **Ribo, M.**; Inoue, S.; Grandi, P.; De Cesare, G.
Real-Time Analysis sensitivity evaluation of the Cherenkov Telescope Array. Article.
Proceedings of Science, Vol. ICRC2015, Num. 699 (2015)

Fontich, E.; Simó, C.; **Vieiro, A.**
The Discrete Hamiltonian-Hopf Bifurcation for 4D Symplectic Maps. Book Chapter.
Trends in Mathematics, Vol. 4 , p. 77 - 81 (2015)
10.1007/978-3-319-22129-8_14

González Hernández, J.I.; **Ruiz-Lapuente, P.**; Tabernero, H.M.; Montes, D.; **Canal, R.**; Méndez, J.; Bedin, L.R.
Origin of Galactic Type-Ia Supernovae: SN 1572 and SN 1006. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 488 - 493 (2015)

Glicenstein, J.F.; et al (ICCUB: **Gascon, D.**; **Sanuy, A.**)
NectarCAM: A camera for the medium size telescopes of the cherenkov telescope array. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 937 (2015)

Gridnev, K.A.; Tarasov, V.N.; Gridnev, D.K.; **Viñas, X.**; Greiner, W.
Stability peninsulas in the neutron-rich sector. Book Chapter.
Nuclear Physics: Present and Future, p. 99 - 105 (2015)
10.1007/978-3-319-10199-6_10

Herrero, E.; Ribas, I.; **Jordi, C.**; Morales, J. C.; Perger, M.; Rosich, A.
Modelling the photosphere of active stars for planet detection and characterization. Proceeding.
p. 494 - 494 (2015)
10.1051/0004-6361/201425369

Iwasawa, K.
X-ray view of luminous infrared galaxies in nearby

Universe and beyond. Proceeding.
IAU General Assembly, Vol. 22, Num. 2255161 (2015)

Jordi, C.
Gaia is now a reality. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 390 - 401 (2015)

Klein, K.-L.; **Agueda, N.**; Bütikofer, R.
On the origin of relativistic solar particle events: Interplanetary transport modelling and radio emission. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 121 (2015)

Luque-Escamilla, P. L.; Martí, J.; Ramírez-Valenzuela, E.; Muñoz-Arjonilla, A. J.; **Paredes, J. M.**
Variability of the counterpart to the γ-ray blazar GT0106+613. Proceeding.
Proceedings of the International Astronomical Union, Vol. 313, p. 87 - 88 (2015)
10.1017/S1743921315001957

Malandraki, O.E.; Klein, K.-L.; Vainio, R.; **Agueda, N.**; Núñez, M.; Heber, B.; Buetikofer, R.; Sarlanis, C.; Crosby, N.; Share, G.; Tylka, A.J.; Murphy, R.; Bazilevskaya, G.; Bindi, V.
High energy solar particle events forecasting and analysis: The HESPERIA project. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 215 (2015)

Maneu, J.; **Parreño, A.**
Hypernuclear decay of strangeness -2 hypernuclei: Weak and strong baryon-baryon-meson vertices. Proceeding.
Proceedings of Science, Vol. 29-June-2015, Num. 129 (2015)

Masana, E.
Gaia outreach activities in spain. Proceeding.
Highlights of Spanish Astrophysics VIII; Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12; 2014; in Teruel; Spain, p 918 - 922 (2015)

Masana, E.; **Luri, X.**; **Julbe, F.**; **Borrachero, R.**; **Antiche, E.**
Gaia data simulations. Proceeding.
Highlights of Spanish Astrophysics VIII; Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12; 2014; in Teruel; Spain, p. 870 - 870 (2015)

Monguió, M.; Grosbøl, P.; **Figueras, F.**
The Perseus arm in the anticenter direction. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 506 - 511 (2015)

Mor, R.; **Figueras, F.**; Robin, A. C.; Lemasle, B.
Constraining the intermediate-mass range of the Initial Mass Function using Galactic Cepheids. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 607 - 607 (2015)

Niro, V.
IceCube and high-energy neutrinos. Journal.
Nuovo Cimento C-Colloquia And Communications In Physics, Vol. 38 (2015)
10.1393/ncc/i2015-15039-8

Núñez, J.
Les Escombraries Espacials. Article.
Butlletí del Consell Assessor del Parlament sobre Ciència i Tecnologia (CAPCIT), p. 1 - 4 (2015)

Núñez, J.; Muiños, J.L.; Montojo, R.; **Merino, M.T.**; Lóopez-Morcillo, R.; **del Ser, D.**; Rosich. A.; Canals, Ll.
Minor Planet Observations [G27 Fabra Observatory, Montsec] of year 2014 (1). Article.
Minor Planets and Comets Circulars (MPC), p. 91341 - 91341 (2015)

Núñez, J.; Muiños, J.L.; Montojo, R.; **Merino, M.T.**; Lóopez-Morcillo, R.; **del Ser, D.**; Rosich. A.; Canals, Ll.
Minor Planet Observations [G27 Fabra Observatory, Montsec] of year 2014 (2). Article.
Minor Planets and Comets Circulars (MPC), p. 92144 - 92144 (2015)

Núñez, J.; Muiños, J.L.; Montojo, R.; **Merino, M.T.**; Lóopez-Morcillo, R.; **del Ser, D.**; Rosich. A.; Canals, Ll.
Minor Planet Observations [G27 Fabra Observatory, Montsec] of year 2015 (3). Article.
Minor Planets and Comets Circulars (MPC), p. 95086 - 95086 (2015)

Núñez, J.; Muiños, J.L.; Montojo, R.; **Merino, M.T.**; Lóopez-Morcillo, R.; **del Ser, D.**; Rosich. A.; Canals, Ll.

Observations and orbits of comets [G27 Fabra Observatory, Montsec] comet C/2014 Q2 (Lovejoy). Article.
Minor Planets and Comets Circulars (MPC), p. 91800 - 91800 (2015)

Ollé, M.; Barrabés, E.; **Gómez, G.**; Mondelo, J.M.
Transport Dynamics: From the Bicircular to the Real Solar System Problem. Book Chapter.
Trends in Mathematics, Vol. 4, p. 45 - 48 (2015)
10.1007/978-3-319-22129-8_8

Palmer, M.; **Luri, X.**; Arenou, F.; **Masana, E.**; Clementini, G.
Luminosity calibration in the LMC with Gaia. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 610 - 610 (2015)

Palmer, M.; **Luri, X.**; Arenou, F.; **Masana, E.**; de Bruijne, J.; **Antiche, E.**; Babusiaux, C.; **Borrachero, R.**; Sartoretti, P.; **Julbe, F.**, et al.
Overview and stellar statistics of the expected Gaia Catalogue using the Gaia Object Generator. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 530 - 530 (2015)

Paredes-Fortuny, X.; **Ribó, M.**; **Bosch-Ramon, V.**; Casares, J.; **del Ser-Badía, D.**; Fors, O.; *Núñez, J.*
γ-ray binaries and recent results on LS I +61 303. Book Chapter.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 1 - 5 (2015)

Paredes-Fortuny, X.; **Ribo, M.**; **Bosch-Ramon, V.**; Casares, J.; Fors, O.; **Nunez, J.**
VizieR Online Data Catalog: Optical observations of LS I +61 303. Multimedia Publication.
VizieR Online Data Catalog, Vol. 357 (2015)

Portell, J.; Artigues, G.; Iudica, R.; García-Berro, E.
FAPEC-based lossless and lossy hyperspectral data compression. Conference Paper.
Proceedings of SPIE - The International Society for Optical Engineering, Vol. 9646, Num. 96460D (2015)
10.1117/12.2195033

Ribas, S. J.; Paricio, S.; Canal-Domingo, R.; Gustems, L.; Calvo, C. O.
Monitoring Light Pollution on the Starlight Reserve of Montsec. Proceeding.
Highlights of Spanish Astrophysics VIII; Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12; 2014; in Teruel; Spain, (2015)

Rio, J.; **Jordi, C.**; **Antiche, E.**; **Luri, X.**; Roelens, M.; **Palmer, M.**
Age determination of open clusters with Gaia. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 547 - 552 (2015)

Roca-Fàbrega, S.; Antoja, T.; **Figueras, F.**; Valenzuela, O.; **Romero-Gómez, M.**; Pichardo, B.
Vertex deviation maps to bracked the Milky Way resonant radius. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 612 - 612 (2015)

Roca-Fàbrega, S.; Valenzuela, O.; **Figueras, F.**
Towards a new cosmological Milky Way like galaxy simulation in the Gaia Era. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 310 - 315 (2015)

Roiser, S.; et al (ICCUB: **Casajus, A.**)
The LHCb Distributed computing model and operations during LHC runs 1; 2 and 3. Proceeding.
Proceedings of Science, Vol. 15-20-March-2015, Num. 5 (2015)

Roma, D.; Bosch, J.; Carmona, M.; **Casas, A.**; Herms, A.; **Gomez, J.M.**; Lopez, M.; **Sabater, J.**; Baumgartner, J.; Maue, T.; Nakai, E.; Schmidt, W.; Volkmer, R.
A space grade camera for image correlation. Article.
IEEE Explore. New Circuits and Systems Conference (NEWCAS), 2015 IEEE 13th International (2015)
10.1109/NEWCAS.2015.7182110

Romero-Gomez, M.; **Figueras, F.**; Antoja, T.; **Abedi, H.**; Aguilar, L.
Gaia capabilities to determine the length and angle of the

Galactic bar. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain p. 564 - 569 (2015)

Sabater, J.; **Gómez, J. M.**; **Torra, J.**; López, M.; Raines, S. N.; Eikenberry, S. S.
Kinematics and trajectory generation for MIRADAS arms. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 871 - 871 (2015)

Sadoun, R.; Zheng, Z.; **Miralda-Escude, J.**; Trac, H.; Cen, R.
Effect of reionization on the observational properties of Lyman-α emitters. Proceeding.
IAU General Assembly , Num. 2255636 (2015)

Salvador-Solé, E.
Constraints on reionization from the observed properties of the high-z universe. Book Chapter.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, Vol. 8, p. 50 - 61 (2015)

Sezima, T.; Galadí-Enríquez, D.; Paunzen, E.; **Jordi, C.**; **Balaguer-Núñez, L.**; Jilkova, L.
Membership determination of Open Clusters with CLUSTERIX. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain, p. 594 - 594 (2015)

Solanes, J.M.; Darriba, L.
Modeling the Production of Intergalactic Light in the Pre-Collapse Phase of Galaxy Groups. Proceeding.
Recent Developments in Theoretical and Experimental General Relativity, Astrophysics and Relativistic Field Theories, p. 1727 - 1729 (2015)

Solanes, J.M.; Perea, J.
The dependency of merger time-scales on the properties and orbital parameters of merging galaxies. Implications for AGN pairs. Proceeding.
Highlights of Spanish Astrophysics VIII, Proceedings of the XI Scientific Meeting of the Spanish Astronomical Society held on September 8-12, 2014, in Teruel, Spain.p. 337 - 342 (2015)

Soto, J.
α_s from hadronic quarkonia decays. Proceeding.
Proceedings of the Workshop on high-precision α_s measurements: from LHC to FCC-ee, CERN, Geneva, p. 41 - 44 (2015)

Topchiev, N.P.; et al (ICCUB: **Gascon, D.**; **Paredes, J.M.**)
GAMMA-400 gamma-ray observatory. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 1026 (2015)

Verdes-Montenegro, L.; et al (ICCUB: **Solanes, J.M.)**
HI and galaxy evolution. Book Chapter.
Spanish SKA White Book, p. 47 - 62 (2015)

Vignali, C.; **Iwasawa, K.**; Comastri, A.; Gilli, R.; Lanzuisi, G.; Ranalli, P.; Cappelluti, N.; Mainieri, V.; Georgantopoulos, I.; Carrera, F. J.; Fritz, J.; Brusa, M.; Brandt, W. N.; Bauer, F. E.; Fiore, F.; Tombesi, F.
An X-ray outflow in a luminous obscured quasar at z=1.6 in the Chandra Deep-Field South. Proceeding.
Demographics and Environment of AGN from Multi-Wavelength Surveys, Proceedings of a conference held 21-24 September, 2015 on Crete Island, Chania, Num. 108 (2015)

Weinstein, A.; **Aliu, E.**; Casanova, S.; Di Girolamo, T.; Dyrda, M.; Hahn, J.; Majumdar, P.; Rodriguez, J.; Tibaldo, L.
Creating a high-resolution picture of cygnus with the cherenkov telescope array. Proceeding.
Proceedings of Science, Vol. 30-July-2015, Num. 824 (2015)

Yun, J. L.; Palmeirim, P.M.
Active star formation at intermediate Galactic latitude: the case of IRAS 06345-3023. Article.
Monthly Notices of the Royal Astronomical Society, Vol. 448, Num. 3, p. 2013 - 2018 (2015)
10.1093/mnras/stv152

Zanin, R.; Blanch, O.; Lopez, R.; Torres, D.F.; Horns, D.; Mayer, M.; Martin, J.
The most precise measurements of the Crab nebula inverse Compton spectral component. Proceeding.
Proceedings of Science, Vol. 30, Num. 831 (2015)

TECHNICAL DOCUMENTS AND REPORTS

Antiche, E.; Gutierrez, R.
Validation Tests Specification (WP940)
GAIA-C9-TN-UB-ELA-017 (jan-15)

Antiche, E.; **Romero-Gomez, M.**; **Figueras, F.**; **Mor, R.**
TGAS astrometric parameter distribution validation within WP942
GAIA-C9-TN-UB-ELA-021 (nov-15)

Arenou, F.; Babusiaux, C.; Boudreault, S.; Di Matteo, P.; Eyer, L.; **Fabricius, C.**; Findeisen, K.; **Antiche, E.**; Helmi A.; Mignard, F.; Manteiga-Outeiro, M.; Reyle, C.; Robin, A.; Tanga, P., Vallenari, V., Ziaeepour, H.
Validation Tests Specification (WP940)
GAIA-C9-SP-OPM-FA-061 (sep-15)

Arenou, F.; **Fabricius, C.**
Prototype of internal validation tools
GAIA-CO-TN-OPM-FA-062 (nov-15)

Babusiaux, C.; Sartoretti, P.; Leclerc, N.; Chereau, F.; **Weiler, M.**
The Gaia instrument and basic image simulator (GIBIS) - User Guide
GAIA-C2-SP-OPM-CB-003-15 (jan-15)

Brown, A.G.A.; Vallenari, A.; **Luri, X.**, et al.
DPAC and Gaia Collaboration Publication Plan
GAIA-CD-PL-LEI-AB-041-020 (aug-15)

Carrasco, J. M.; **Jordi, C.**; **Fabricius, C.**; **Voss, H.**
XP and G passbands sensitivity calibration Functions
GAIA-C5-TN-UB-JMC-012 (jan-15)

Carrasco, J. M.; **Weiler, M.**; **Jordi, C.**
Deriving synthetic photometry from XP spectra
GAIA-C5-TN-UB-JMC-020 (sep-15)

Carrasco, J.M.; **Voss, H.**; **Jordi, C.**; **Fabricius, C.**
Tilt effects on XP spectra
GAIA-C5-TN-UB-JMC-019 (feb-15)

Carrasco, J.M.; **Weiler, M.**; **Jordi, C.**
Deriving synthetic photometry from XP spectra (2015)

Casas, A.; Carmona, M.; Redondo, M.; Llamas, X.; **Gómez Cama, J.M.**
Ttc Qm Test Plan 1_1_1 (2015)

Casas, A.; Llamas, X.; Carmona, M.; **Gómez Cama, J.M.**
Ttc Qm Abcl 3_1_0 (2015)

Casas, A.; Llamas, X.; Carmona, M.; **Gómez Cama, J.M.**
Ttc Qm Abcl 3_1_1 (2015)

Casas, A.; Llamas, X.; Carmona, M.; **Gómez Cama, J.M.**
Ttc Qm End Item Data Package 1_1_0 (2015)

Castañeda, J.; **Clotet, M.**
Data Segment 0 CrossMatch Reprocessing Report
GAIA-DB-PR-UB-JC-074-1 (oct-15)

Clotet, M.
DPCB Cycle 0 Briefings
GAIA-DB-TN-UB-MCL-032 (sep-15)

Clotet, M.
DPCB Cycle 1 Briefings
GAIA-DB-TN-UB-MCL-036 (sep-15)

Clotet, M.
DPCB QA Report – 2013/05
GAIA-DB-TN-UB-MCL-013 (oct-15)

Clotet, M.; **Gonzalez, J.**
DPCB Data Manager
GAIA-DB-TN-UB-MCL-002 (jul-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 17.1 Software Release Note
GAIA-DB-SP-UB-MCL-027 (apr-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 17.2 Software Release Note
GAIA-DB-SP-UB-MCL-031 (may-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 17.3 Software Release Note
GAIA-DB-SP-UB-MCL-034 (jul-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 18.0 Software Release Note
GAIA-DB-TR-UB-JG-036 (jun-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 18.1 Software Release Note
GAIA-DB-TR-UB-MCL-035 (nov-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools 19.0 Software Release Note
GAIA-DB-SP-UB-MCL-037 (nov-15)

Clotet, M.; **Gonzalez, J.**
DpcbTools-Data Manager (DDM) Software User Manual
GAIA-DB-UG-UB-MCL-003-04 (apr-15)

Clotet, M.; **Gonzalez, J.**; **Castaneda, J.**, et al.
IDU Cross-Match Resolution - algorithm and implementation approach
GAIA-C3-TN-UB-MCL-033-01 (dec-15)

Clotet, M.; **Gonzalez, J.**; **Portell, J.**
DPCB Procedures Handbook
GAIA-DB-PL-UB-MCL-008-05 (oct-15)

Gracia, G; **Luri, X.**; **Castañeda, J.**; **Masana, E.**, **Clotet, M**; **Portell, J.**
CU, DPC, and Workgroup leadersDPAC Management Report #9
GAIA-PO-PR-ESAC-SE-059 (may-15)

Els, S.; Siddiqui, H.; **Gracia, G.**, et al.
Minutes of the CU1/DPC/PO meeting
GAIA-C1-MN-ESAC-SE-060-1 (apr-15)

Fabricius, C.
Initial In-Orbit Characterisation of radiation damage in SM (IIOC-152)
GAIA-CH-TN-UB-CF-031-01 (feb-15)

Fabricius, C.; de Bruijne, J.; Biermann, M.; **Portell, J.**; **Castañeda, J.**; **Torra, J.**
On ground reconstruction of windows
GAIA-C3-TN-UB-CF-011-07 (mar-15)

Gómez Cama, J.M.; Carmona, M.; **Roma, D.**; Volkmer, R.
Iss Control Firmware Interface Control Document 3_1_1 (2015)

Gonzalez, J.; **Clotet, M.**
DpcbTools 17.2 Software Test Report
GAIA-DB-TR-UB-JG-033 (may-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**
DpcbTools 17.3 Software Test Report
GAIA-DB-TR-UB-JG-037 (jul-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**
DpcbTools 18.0 Software Test Report
GAIA-DB-TR-UB-JG-035 (jun-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**
DpcbTools 18.1 Software Test Report
GAIA-DB-TR-UB-JG-038 (sep-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**
DpcbTools 19.0 Software Test Report
GAIA-DB-TR-UB-JG-040 (oct-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**
IDU Cross-Match - Test Results
GAIA-C3-TN-UB-JG-040-1 (dec-15)

Gonzalez, J.; **Clotet, M.**; **Castaneda, J.**, et al.
DPCB test specification
GAIA-DB-SP-UB-NBM-004-06 (apr-15)

Gonzalez, J.; **Clotet, M.**; **Portell, J.**; **Castaneda, J.**
DPCB Test Report for Cycle 17
GAIA-DB-TR-UB-JG-030 (apr-15)

Gracia, G.
Gaia Archive Publication Rehearsal #1 Plan
GAIA-C9-PL-ESAC-GGA-043 (dec-15)

Gracia, G.
Gaia Data Delivery Note GAIA-GDR0-01_SIM
GAIA-C9-SP-ESAC-GGA-039 (apr-15)

Gracia, G.
Procedure to include data in Gaia Archive Data Releases
GAIA-C9-PL-ESAC-GGA-042 (jul-15)

Gracia, G.
Gaia Archive HW Environment
GAIA-C9-TN-ESAC-GGA-040 (mar-15)

Gracia, G.
Minutes of the 1st CU9/WP974 Cross match meeting
GAIA-C9-MN-ESAC-GGA-041 (apr-15)

Gracia, G.; **Luri, X.**; Arenou, F.; de Bruijne, J.; Hernandez, J.; Jordan, S.; Marrese, P.M.; Moitinho, A.; Salgado, J.; Walton, N.
CU9 roadmap for publication of TGAS results
GAIA-C9-PL-ESAC-GGA-038 (mar-15)

Gurpide, A.; **Clotet, M.**; **Gonzalez, J.**
Clustering Analysis for IDU XM
GAIA-C3-TN-UB-AGL-001-01 (nov-15)

Gurpide, A.; **Clotet, M.**; **Gonzalez, J.**
IDU XM clustering calibration
GAIA-C3-TN-UB-AGL-002-01 (dec-15)

Julbe, F.
Gaia Data Analytics Framework (GDAF) description
GAIA-C9-SP-UB-FJL-001 (nov-15)

Aaij, R.; et al (LHCb collaboration. ICCUB: Baladov, A.; **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Measurements of long-range near-side angular correlations in $\sqrt{s_{NN}}$ *= 5 TeV proton-lead collisions in the forward region*
LHCb-PAPER-2015-040, CERN-PH-EP-2015-308 (2015)

Aaij, R.; et al (LHCb collaboration. ICCUB: Baladov, A.; **Garrido, L.**; **Gascon, D.**; **Graciani Diaz, R.**; **Graugés, E.**; **Marin Benito, C.**; **Picatoste Olloqui, E.**; **Rives Molina, V.**)
Production of associated Y and open charm hadrons in pp collisions at $\sqrt{s}$ *= 7 and 8TeV via double parton scattering*
CERN-PH-EP-2015-279, CERN-LHCB-PAPER-2015-046, LHCB-PAPER-2015-046 (oct-15)

Llamas, X.; **Casas, A.**; Carmona, M.; **Gómez Cama, J.M.**
Ttc Qm Abdcl 1_1_0
(2015)

Llamas, X.; **Roma, D.**; Bosch, J.; Carmona, M.; López, M.; Nakai, E.; **Gómez Cama, J.M.**; Volkmer, R.
Ctc Schematics 1_2_0
(2015)

Llamas, X.; **Roma, D.**; Carmona, M.; Bosch, J.; **Gómez Cama, J.M.**
Ctc Fpga Pcb Procurement Specification 2_2_0
(2015)

Luri, X.; **Gracia, G.**; Hambly, N.; Marrese, P.M.; Salgado, J.; **Gonzalez, J.**
CU9 roadmap for cross-match with external catalogues
(2015)

Luri, X.; **Gracia, G.**; Salgado, J., et al.
Gaia Archive Interface Control Document with External Data Centres
GAIA-C9-SP-UB-XL-034 (apr-15)

Mor, R.; Robin, A.C.; **Figueras, F.**; **Antiche, E.**; **Fabricius, C.**; **Romero-Gomez, M.**
Besançon Galaxy Model Simulation for CU9-WP943 (BGMBTG 2.0) I. First steps comparing Tycho-2 and Model
GAIA-C9-TN-UB-RMC-001-1 (may-15)

Portell, J.; **Castaneda, J.**; **Clotet, M.**
DPCB interface control document
GAIA-DB-SP-UB-JP-041-07 (dec-15)

Portell, J.; **Clotet, M.**
DPCB Backup Policy
GAIA-DB-TN-UB-JP-039 (jul-15)

Portell, J.; **Clotet, M.**
DPCB-MareNostrum deployment procedure
GAIA-DB-TN-UB-JP-037 (jul-15)

Redondo, M.; **Casas, A.**; Carmona, M.; López, M.; Heidecke, F.; Nakai, E.; Llamas, X.; **Gómez Cama, J.M.**; Volkmer, R.
Ttc Schematics 1_2_0
(2015)

Redondo, M.; **Casas, A.**; Heidecke, F.; Llamas, X.; **Gómez, J.M.**; Volkmer, R.
TTC Worst Case Circuit Analysis (WCA)
(2015)

Roma, D.; Bosch, J.; Carmona, M.; López, M.; Diez, L.; Llamas, X.; **Gómez Cama, J.M.**; Volkmer, R.
Ctc Test Plan 1_1_0
(2015)

Roma, D.; Carmona, M.; Maroto, O.; **Gómez Cama, J.M.**
Ctc Fpga Pet Test Report 1_1_0
(2015)

Roma, D.; **Casas, A.**; Carmona, M.; Llamas, X.; **Gómez Cama, J.M.**; Volkmer, R.
Iss Egse User Manual 1_1_0
(2015)

Roma, D.; **Casas, A.**; Carmona, M.; Llamas, X.; **Gómez Cama, J.M.**; Volkmer, R.
Iss Egse User Manual 1_1_1
(2015)

Vallenari, A.; Alfaro, E.; Bragaglia, A.; Frasca, A.; **Jordi, C.**; Carrera, R.; **Balaguer, L.**; Lanzafame, A.; Allende-

Prieto, C.; Randich, S.; Magrini, L.; Sordo, R.; Spagna, Aparicio, A.; **Casamiquela, L.**; Gallart, C.; Monelli, M.
WEAVE Science Case: Preliminary Selection of targets for the Open Cluster Survey
(2015)

Vallenari, A.; Bragaglia, A.; Spagna, A.; Arenou, F.; **Jordi, C.**; **Balaguer, L.**; Soubiran, C.; Marrese, P.
GWP-947 Cluster selection requirements for the first Gaia data release
(2015)

Vallenari, A.; **Gracia, G.**; Els, S.; **Luri, X.**, et al.
OR# 5 Planning
GAIA-CD-PL-OAPD-AV-011 (jan-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.03
GAIA-C5-PR-IOA-FVL-138 (apr-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.04
GAIA-C5-PR-IOA-FVL-138 (may-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.05
GAIA-C5-PR-IOA-FVL-138 (jun-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.06
GAIA-C5-PR-IOA-FVL-138 (jul-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.07
GAIA-C5-PR-IOA-FVL-138 (sep-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.08
GAIA-C5-PR-IOA-FVL-138 (oct-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.09
GAIA-C5-PR-IOA-FVL-138 (nov-15)

van Leeuwen, F.; Brown, A.; Bunclark, P.; Cacciari, C.; De Angeli, F.; Evans, D.W.; Hambly, N.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P.J.
CU5 internal report 2015.10
GAIA-C5-PR-IOA-FVL-138 (dec-15)

van Leeuwen, F.; Busso, G.; Cacciari, C.; Cowell, S.; De Angeli, F.; Evans, D. W.; Fyfe, D.; Harrison, D. L.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P. J.; Riello M.
CU5 Internal Report 2015.01
GAIA-C5-PR-IOA-FVL-137 (feb-15)

van Leeuwen, F.; Busso, G.; Cacciari, C.; Cowell, S.; De Angeli, F.; Evans, D. W.; Fyfe, D.; Harrison, D. L.; Hodgkin, S.; **Jordi, C.**; Pancino, E.; Richards, P. J.; Riello M.
CU5 Internal Report 2015.02
GAIA-C5-PR-IOA-FVL-138 (mar-15)

van Leeuwen, M., Utrilla, E., **Farras Casas, M.**, et al.
Instructions for the preparation of documentation for GDR1
GAIA-C9-TN-IOA-MVL-002 (dec-15)

Voss, H.
Saturation fraction for gated observation after decontamination step 5 and refocus of FoV 1.
GAIA-CH-TN-UB-HV-028 (jan-15)

THESES 6

PHD THESES

Finished Theses

AdS/CFT Correspondence and Superconductivity: Various Approaches and Magnetic Phenomena
Author: Aldo Dector
Director/s: Jorge Russo
Date: 28/10/2015

Astrostatistics for Luminosity Calibration in the Gaia Era
Author: Max Palmer
Director/s: Xavier Luri, Fréderic Arenou (Observatoire de Paris)
Date: 13/01/2015

Correspondences in higher-dimensional gravity
Author: Adriana Di Dato
Director/s: Roberto Emparan
Date: 12/11/2015

COTS Analog Prototype for LHCb's Calorimeter Upgrade
Author: Carlos Abellán
Director/s: David Gascón, Atilà Herms
Date: 30/01/2015

Dynamical transport mechanisms in celestial mechanics and astrodynamics problems
Author: Daniel Pérez
Director/s: Gerard Gómez, Josep Masdemont
Date: 30/11/2015

High performance computing of massive Astrometry and Photometry data from Gaia
Author: Javier Castañeda
Director/s: Claus Fabricius, Jordi Torra
Date: 15/12/2015

Monitor amb control strategies to reduce the impact of process variations in digital circuits
Author: Joan Mauricio Ferré
Director/s: Francesc Moll (UPC)
Date: 14/12/2015

Non-thermal emission from high-energy binaries through interferometric radio observations
Author: Benito Marcote
Director/s: Marc Ribó, Josep Mª Paredes
Date: 27/10/2015

On the characterisation of the galactic warp in the Gaia era
Author: Hoda Abedi
Director/s: Francesca Figueras, Luís Alberto Aguilar (UNAM, Méjico)
Date: 27/02/2015

Physics of the Intergalactic Medium: a Study of the Power Spectrum ot the Lyman Alpha Forest and the Metal Content of Damped Lyman Alpha Systems
Author: Andreu Ariño
Director/s: Jordi Miralda Escudé
Date: 03/02/2015

Probing Gauge Theories: Exact Results and Holographic Computations
Author: Blai Garolera
Director/s: Bartomeu Fiol
Date: 27/02/2015

Some theoretical and experimental aspects of axion physics
Author: Albert Renau
Director/s: Federico Mescia, Domènec Espriu
Date: 11/11/2015

Ongoing PhD theses

Agujeros negros y modelos de gravitación emergente
Author: Luciano Gabbanelli
Director/s: Domènec Espriu
Tentative Date: March 2019

Anàlisi i modelització de l'entorn de radiació de partícules solars energètiques a l'heliosfera interior com a preparació per Solar Orbiter
Author: Daniel Pacheco Mateo

Director/s: Àngels Aran, Neus Àgueda
Tentative Date: 2018

Analysis of Periodic and Quasi-Periodic Orbits in Three Body Problems and Applications
Author: Yu Cheng
Director/s: JianPing Yuan, Gerard Gómez Muntané
Tentative Date: December 2017

Analytic Derivation of Non-Linear Dark Matter Clustering from the Filtering of the Primordial Density Field
Author: Enric Juan Rovira
Director/s: Eduard Salvador Solé
Tentative Date: February 2016

Caracterització de la contaminació lumínica en zones protegides i urbanes
Author: Salvador Ribas Rubio
Director/s: Jordi Torra
Tentative Date: January 2016

Contribution to GNSS inter-satellite link technique: Usage of low earth orbital and sub-orbital measurements
Author: David Roma Dollase
Director/s: J. Maria Gómez Cama, M. Hernandez-Pajares (UPC)
Tentative Date: January 2017

Coupling fluid-dinamics and non-thermal processes to study sources of high energy emission
Author: Víctor Moreno de la Cita
Director/s: Valentí Bosch-Ramon
Tentative Date: October 2016

Disseny d'algoritmes de control per posicionadors multiobjectes per instrumentació de telescopis terrestres
Author: Pablo Oriol Bitaubé
Director/s: J. Maria Gómez Cama, J.Torra

Electronics control and signal processing for the LHCb fast calorimeter detectors
Author: Eduardo Picatoste
Director/s: David Gascón, Atilà Herms
Tentative Date: December 2016

Estudi de la polarització del fotó en desintegracions radiatives d'hadrons B amb el detector LHCb
Author: Carla Marín Benito
Director/s: Lluís Garrido , Albert Puig (U. Zurich)
Tentative Date: September 2017

Estudi de l'halo de matèria fosca de la Via Làctia amb rieres de marea dels seus satèl·lits
Author: Carles García Palau
Director/s: Jordi Miralda
Tentative Date: 2019

Estudi Espectròscopic de cúmuls oberts per a l'anàlisi de la metal·licitat de la galàxia
Author: Laia Casamiquela
Director/s: Carme Jordi, Dolores Balaguer, Ricardo Carrera (IAC)
Tentative Date: 2017

Explotació científica del survey TFRM-PSES: detecció d'exoplanetes per trànsit, caracterització de la variabilitat d'estrelles Ks i Ms, i optimització de l'anàlisi de dades TFRM-PSES
Author: Daniel del Ser Badia
Director/s: Jorge Núñez de Murga, Octavi Fors

Far-from-equilibrium holography and heavy ion collisions
Author: Miquel Triana i Iglesias
Director/s: David Mateos
Tentative Date: 2017

Gauge/string duality for strongly coupled theories with finite baryon density
Author: David Pravos Fernández
Director/s: David Mateos
Tentative Date: May 2019

Gravitació anàloga amb metamaterials
Author: Isabel Fernández Núñez
Director/s: Enric Verdaguer, Oleg Bulashenko
Tentative Date: 2018

Jets as Probes of Strongly Coupled Quark-Gluon Plasma
Author: Daniel Pablos Alfonso
Director/s: Jorge Casalderrey Solana
Tentative Date: 22/06/2016

Lyman-alpha autocorrelation, small scales structures and fluctuations of the radiation background
Author: Satya Gontcho A Gontcho
Director/s: Jordi Miralda Escudé
Tentative Date: Fall 2017

Meson-Baryon interactions in free space and in the medium from effective Chiral Lagrangians
Author: Albert Feijoo Aliau
Director/s: Volodymyr Magas, Àngels Ramos
Tentative Date: July 2016

Multipartite quantum entanglement
Author: Daniel Alsina Leal
Director/s: José Ignacio Latorre
Tentative Date: June 2017

New insights into holography from supersymmetric localization
Author: Genís Torrents Verdaguer
Director/s: Tomeu Fiol Núñez
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Observational and theoretical study of the interaction of relativistic winds from young pulsars with inhomogeneous stellar winds
Author: Xavier Paredes-Fortuny
Director/s: Marc Ribó Gomis, Valentí Bosch-Ramon
Tentative Date: October 2016

Physical Cosmology in the epoch of large surveys
Author: José Luís Bernal Mera
Director/s: Licia Verde, Antonio J. Cuesta
Tentative Date: Fall 2019

Physical mechanisms in high-energy pulsars and their environments
Author: Daniel Galindo
Director/s: Roberta Zanin, Josep M. Paredes
Tentative Date: May 2019

Probing the correlation between IR and X-ray emission in Luminous Infrared Galaxies
Author: Núria Torres Albà
Director/s: Josep M. Paredes, Kazushi Iwasawa
Tentative Date: October 2019

Rotation, magnetic fields and fragmentation in the earliest stages of star formation: High angular resolution observations in the ALMA and JVLA era
Author: Carmen Juárez Rodríguez
Director/s: J. Miquel Girart Medina
Tentative Date: September 2016

Studies of Black Hole Horizons
Author: Marina Martínez Montero
Director/s: Roberto Emparan García de Salazar
Tentative Date: October 2016

Study of b-hadron decays into two hadrons and a photon at LHCb and first observation of b-baryon radiative decays
Author: Vicente Rives Molina
Director/s: Albert Puig
Tentative Date: October 2016

Super-resolución de imágenes astronómicas y teledetección
Author: Mª Teresa Merino Espasa
Director/s: Jorge C. Núñez

Teories Gauge i Principi de Máxima Entropia
Author: Alba Cervera Lierta
Director/s: José Ignacio Latorre
Tentative Date: June 2019

The cross-correlation among tracers of the underlying large-scale mass distribution in the universe
Author: Ignasi Pérez Ràfols
Director/s: Jordi Miralda Escudé
Tentative Date: October 2016

The Milky Way stellar population in the Gaia era: Archive validation tasks and First Science
Author: Roger Mor
Director/s: Francesca Figueras, Annie Robin (Observatori de Besançon, França)
Tentative Date: 2018

Vacuum energy in Quantum Field Theory and Cosmology
Author: Adrià Gómez Valent
Director/s: Joan Solà Peracaula
Tentative Date: Spring 2018

Weak decay of doubly strange systems
Author: Jordi Maneu Victoria
Director/s: Assumpta Parreño
Tentative Date: November 2017

Wideband pulse amplifiers for the integrated cameras of the Cherenkov Telescope Array
Author: Andreu Sanuy
Director/s: David Gascón
Tentative Date: February 2016

MASTER THESES

A consistency test of General Relativity using Expansion History and Growth of Structures
Author: José Luís Bernal
Director/s: Alexander Knebe, Licia Verde, Antonio J. Cuesta
Date: 01/07/2015

Black holes in the limit of very many dimensions
Author: Joan Garcia i Tormo
Director/s: Roberto Emparan
Date: 01/09/2015

Bose-Fermi mixtures in 1D few-sites optical lattices
Author: Gerard Pelegrí
Director/s: Bruno Julià-Díaz, Artur Polls
Date: 01/07/2015

Dynamical Vacuum Energy in the Expanding Universe
Author: Javier de Cruz
Director/s: Joan Solà
Date: 03/07/2015

Effects of Lambda on gravitational waves and possible measurement through pulsar timing arrays
Author: Galo Gallardo
Director/s: Domènec Espriu
Date: 01/09/2015

Effects of the Image Cleaning on Monte Carlo simulations of Moonlight observations with the MAGIC telescopes
Author: Cristina Sans
Director/s: Marc Ribó
Date: 06/07/2015

Emission spectrum and inclusive cross section of strings
Author: Ramon Masachs
Director/s: Jorge G. Russo
Date: 03/07/2015

Majorana Fermions in a Spin ladder
Author: Felipe LeVot
Director/s: Sofyan Iblisdir
Date: 03/07/2015

Neutron and spin-orbit effects in nuclear charge density and electron-nucleus scattering
Author: Clàudia González
Director/s: Mario Centelles, Xavier Viñas
Date: 01/09/2015

Non-Unitary Neutrino Oscillations
Author: Alba Cervera
Director/s: M. Concepción González García
Date: 03/07/2015

Oblique corrections in the Dine-Fischler-Srednicki-Zhitnitsky axion model
Author: Alisa Katanaeva
Director/s: Domènec Espriu
Date: 01/09/2015

Photometric Follow-up of Gaia Alerts From Montsec Observatory
Author: Umut Burgaz
Director/s: Josep Manel Carrasco
Date: 06/07/2015

Probing the correlation between IR and X-ray emission in luminous infrared galaxies
Author: Núria Torres
Director/s: Kazushi Iwasawa, Valentí Bosch Ramon
Date: 06/07/2015

Quantum properties of two-component two-sites Bose-Hubbard model
Author: Pere Mujal
Director/s: Bruno Julià-Díaz, Artur Polls
Date: 01/07/2015

Sensitivity to the photon polarization in B^0_s -> \phi \ gamma at the LHCb experiment
Author: Miguel Larrañaga
Director/s: Lluís Garrido
Date: 03/07/2015

Shorcut to adiabaticity in spin fragmented condensates
Author: David López
Director/s: Bruno Julià-Díaz, Artur Polls
Date: 01/07/2015

Static Fields via Probe Computations using AdS/CFT duality
Author: Susana Vásquez
Director/s: Bartomeu Fiol
Date: 03/07/2015

Study of radiative B decays in LHCb: B_s->KK\gamma
Author: Joan Arnau
Director/s: Ricardo Graciani
Date: 03/07/2015

Study of the efficiency of the LHCb Trigger for the K^0_S --> eepipi decay
Author: Carles García
Director/s: Lluís Garrido
Date: 03/07/2015

Unitarisation of WW scattering and interpretation of a 2 TeV resonance in a strongly interacting electroweak symmetry-breaking sector
Author: Pere Arnan
Director/s: Domènec Espriu
Date: 01/09/2015

ACTIVITIES

7

ICCUB COLLOQUIA

The ICCUB Colloquia are talks given by world renowned speakers about cutting-edge topics in cosmology, astrophysics and particle and nuclear physics. The talks are directed to a diverse audience, including not only ICCUB members but also external researchers and graduate and master students.

Colloquia Comission

- Bartomeu Fiol
- Bruno Julià
- Federico Mescia
- Paolo Padoan

2015 ICCUB Colloquia

Ignacio Cirac (Max-Planck Institut für Quantenoptik)
Quantum simulations
26/02/2015

Gordon Baym (Deptt of Physics, Univ. of Illinois at Urbana-Champaign)
Quarks and cold atoms: Many particle physics from the hottest to the densest to the coldest places in the Universe
09/04/2015

Roberto Emparan (ICREA / ICCUB)
Heart of darkness: General Relativity @ 100
07/05/2015

Bernard J. T. Jones (Kapteyn Astronomical Institute, Univ. of Groningen)
Discovering the environment in which galaxies form - The Cosmic Web
04/06/2015

Rashid Sunyaev (Max-Planck Institute for Astrophysics, Garching)
Two important milestones in the history of the Universe: The surface of the last scattering and the black body photosphere. The inevitable spectral distortions of the Cosmic Microwave Background Radiation
02/07/2015

Ruprecht Machleidt (Department of Physics, University of Idaho)
Understanding the Atomic Nucleus: Recent Dramatic Advances and Remaining Challenges
15/10/2015

Laura Baudis (Physik Institut, Univ. of Zurich)
Direct detection of dark matter in the Milky Way
26/11/2015

ICCUB COLLOQUIA 2016 SPEAKERS

From left to right: I. Cirac , G. Baym, B.J.T. Jones, R. Emparan, L. Baudis, R. Machleidt

SEMINARS

Seminars are more specialized talks given by either ICCUB members or visitors.

We distinguish those seminars organized directly by the institute (ICCUB Seminars), and group seminars organized in collaboration with UB departments:

- High Energy Physics Group (HEP),
- Atomic, Molecular and Nuclear Physics Group (FAN),
- Department of Astronomy and Meteorology (DAM).

2015 Seminars

Sugumi Kanno (UPV)
Quantum Entanglement in the Multiverse
HEP Seminar
08/01/2015

Petra H. Huentemeyer (MTU, Michigan Tech University, USA)
First Preliminary Results and Science Potential of the HAWC Observatory
DAM Seminar
09/01/2015

Javier Virto (Siegen University)
Flavoured Conspiracies and Conundrums
HEP Seminar
15/01/2015

Marina Martínez (ICCUB)
Bumpy Black Holes
HEP Seminar
16/01/2015

Nathalie Palanque-Delabrouille (CEA, Paris)
Constraints on massive neutrinos using Lyman-alpha forest
DAM Seminar
22/01/2015

Matteo Viel (Trieste Observatory)
Modelling the effect of neutrinos on cosmological structure formation
DAM Seminar
22/01/2015

Vladimir Mitev (Humboldt University)
The Exact Effective Couplings of 4D N=2 Gauge Theories
HEP Seminar
22/01/2015

Suresh Nampuri (University of the Witwatersrand)
Hot Attractors
HEP Seminar
23/01/2015

Geraldine Servant (IFAE & ICREA)
Higgs (and axion) implications for baryogenesis
HEP Seminar
29/01/2015

Matteo Viel (Trieste Observatory)
Modelling the effect of neutrinos on cosmological structure formation
ICCUB Seminar
03/02/2015

Nathalie Palanque-Delabrouille (CEA, Paris)
Constraints on massive neutrinos using Lyman-alpha forest
ICCUB Seminar
04/02/2015

Antonio Cuesta (ICCUB)
Robustness of cosmic neutrino background detection in the Cosmic Microwave Background
ICCUB Seminar
04/02/2015

Gloria Koenigsberger (I. de Ciencias Físicas, UNAM)
On the Evolution of Asynchronous Binary Systems
DAM Seminar
05/02/2015

Eric Bergshoeff (Groningen University)
Supergravity, T-duality and Double Field Theory
HEP Seminar
05/02/2015

Miguel Zilhão (ICCUB)
Simulating Accreting Binary Black Holes
HEP Seminar
06/02/2015

Hugo F. Arellano (U. Chile)
Di-nucleons in infinite nuclear matter
FAN Seminar
10/02/2015

Dimitrios Giataganas (Oxford University)
Gauge/Gravity duality with Anisotropies
HEP Seminar
12/02/2015

Stephan Stetina (Vienna University of Technology)
From Field Theory to the Hydrodynamics of Relativistic Superfluids
HEP Seminar
13/02/2015

José Oñorbe (MPIA, Heidelberg, Germany)
Characterisation of the Intergalactic Medium Jeans Scale and its measurement using quasar pairs
ICCUB Seminar
18/02/2015

Drazen Glavan (Utrecht University)
Late-time quantum backreaction in cosmology
HEP Seminar
19/02/2015

Jakub Zakrzewski (Jagiellonian University)
Beyond the standard Bose-Hubbard model description of cold bosons in optical lattice potential
FAN Seminar
19/02/2015

Jorge Russo (ICREA / ICCUB)
Large N Phase Transitions in massive N = 2 Gauge Theories
HEP Seminar
19/02/2015

Yuko Urakawa (Nagoya University)
Infrared radiative stability and the initial state of the inflationary universe
HEP Seminar
20/02/2015

Justin Read (University of Surrey)
Astrophysical probes of dark matter
DAM Seminar
26/02/2015

Lorenzo Calibbi (ULB, Brussels)
Phenomenology of low-energy flavour models: rare processes and dark matter
HEP Seminar
26/02/2015

Cristiano Germani (Munich University)
Firewalls as artefacts of inconsistent truncations of quantum geometries
HEP Seminar
27/02/2015

Marina Martínez-García & **Xavier Xirgu** (H2020 Programme Officer / Chieff Innovation Officer at Univ. Girona)
H2020: how to increase the impact and success
ICCUB Seminar
03/03/2015

Giorgio Arcadi (LPT, Orsay)
Phenomenology of minimal decaying dark matter scenarios
HEP Seminar
05/03/2015

Mallory Roberts (New York University Abu Dhabi (NYYAD) and Eureka Scientific)
Heating Before Eating: X-Ray observations of Redback Millisecond Pulsar Systems in the ablation state
DAM Seminar
06/03/2015

Yuichiro Kiyo (Juntendo University)
Update of quarkonium phenomenology at NNNLO
HEP Seminar
06/03/2015

Lorenzo Vitale (ITPP Lausanne)
Hamiltonian Truncation methods for strongly-coupled QFTs
HEP Seminar
09/03/2015

Clifford Burgess (McMaster University & Perimeter Institute & CERN)
Self-Tuning Under the Microscope: Last Chance to be Wrong About What the LHC Will Find?
HEP Seminar
12/03/2015

Daniel Pablos (ICCUB)
A Hybrid Strong/Weak Coupling Approach to Jet Quenching
HEP Seminar
13/03/2015

Pau Figueras (DAMPT)
New frontiers in numerical general relativity
ICCUB Seminar
16/03/2015

Vitor Cardoso (IST Lisbon)
Black holes as particle detectors
HEP Seminar
19/03/2015

Konrad Tywoniuk (ICCUB)
Long-range correlations in the quark-gluon plasma from QCD confinement
HEP Seminar
20/03/2015

Sophie Renner (DAMTP, Cambridge)
Composite leptoquarks and anomalies in B decays
HEP Seminar
26/03/2015

Nicola Serra (Zurich University)
*Angular analysis of the decay B0->K*mm at LHCb*
HEP Seminar
27/03/2015

Luca Merlo (IFT UAM/CSIC)
Masses and Mixings from Continuous Symmetries
HEP Seminar
09/04/2015

Adrià Gómez (ICCUB)
Dynamical vacuum energy in the expanding Universe
HEP Seminar
10/04/2015

Daniel Ceverino (CAB-CSIC-INTA)
Simulations of Galaxy Formation: Feedback and the early formation of compact spheroids by disc instabilities
DAM Seminar
16/04/2015

Rafel Escribano (IFAE)
Combined analysis of the decays $\tau \to KS\pi\nu\tau$ and $\tau \to K\eta\nu\tau$
HEP Seminar
16/04/2015

Andrés Goya (Universidad de Buenos Aires)
Three Dimensional Bigravity as a New Model for AdS/CFT Extensions
HEP Seminar
23/04/2015

Federico Mescia (ICCUB)
Axion - Higgs interplay in the 2 Higgs Doublet model
HEP Seminar
24/04/2015

Alessio Celi (ICFO)
Synthetic Lattices & applications
HEP Seminar
30/04/2015

Luca di Luzio (Genova University)
Invisible axion and massive neutrinos: are they related?
HEP Seminar
30/04/2015

Carles Badenes (University of Pittsburgh)
The Persistence of Memory: What Supernova Remnants Can Tell Us about Type Ia Supernova Progenitors
ICCUB Seminar
04/05/2015

Roberto Terlevich (Institute of Astonomy, Cambridge Univ.)
Towards precision Cosmology with HII galaxies
ICCUB Seminar
06/05/2015

David Martínez-Delgado (Universidad de Heildelberg)
Stellar Tidal Streams in nearby spiral galaxies
DAM Seminar
07/05/2015

David Berenstein (UCSB)
Extremal chiral ring states in AdS/CFT are described by free fermions
HEP Seminar
07/05/2015

Konrad Tywoniuk (ICCUB)
Transport in a plasma of confining gluons
HEP Seminar
08/05/2015

Christian Duval (Aix-Marseille University & CPT)
On the Schrödinger-Newton equation and its symmetries
HEP Seminar
14/05/2015

Joaquim Gomis (ICCUB)
The symmetries of Carroll (super) particle
HEP Seminar
15/05/2015

Thomas Zojer (Groningen University)
Non-relativistic (super)gravity
HEP Seminar
21/05/2015

Juan M. Torres-Rincon (Subatech, U. Nantes)
Flavor dependence of hadron melting temperatures
FAN Seminar
21/05/2015

Adriana di Dato (ICCUB)
Hydrodynamics from charged dilaton black brane
HEP Seminar
22/05/2015

Giulio Bonelli (SISSA)
The N=2, D=4 gauge theory partition function on compact toric manifolds
HEP Seminar
28/05/2015

Daniel Alsina (ICCUB)
Absolutely maximally entangled states
HEP Seminar
29/05/2015

Andreas Reisenegger (Pontificia Universidad Catolica de Chile, Santiago, Chile)
Internal heating and ultraviolet emission of old neutron stars
ICCUB Seminar
03/06/2015

Jordi Miralda & Àngels Egea (ICCUB / UB)
Una nova aportació a la terminologia catalana per l'astrofísica
ICCUB Seminar
03/06/2015

Juliana Saponara (Facultad de Ciencias Astronómicas y Geofísicas (FCAG-UNLP). La Plata)
Cinemática de galaxias enanas locales y CHIPÁ
DAM Seminar
04/06/2015

Ignatios Antoniadis (AEC Bern & LPTHE Paris)
Aspects of string phenomenology in particle physics and cosmology
HEP Seminar
04/06/2015

Benjo Fraser (Athens University)
Higher Rank Wilson Loops in the N=2 SU(N)xSU(N) Conformal Quiver
HEP Seminar
05/06/2015

Genís Torrents (ICCUB)
Holographic D3 probes
HEP Seminar
11/06/2015

Alexander Andrianov (St. Petersburg State University)
PT symmetric classical and quantum cosmology: PToms against phantoms
HEP Seminar
12/06/2015

Hideo Kodama (KEK, Tsukuba)
Non-linear evolution of the black hole-axion system and its observation by gravitational waves
HEP Seminar
15/06/2015

Álvaro Sánchez Monge (Physikalisches Institut, Universität zu Köln)
High-mass star formation: from dense cores to HII regions
DAM Seminar
16/06/2015

Álvaro Sánchez Monge (Physikalisches Institut, Universität zu Köln)
XCLASS: Introduction and tutorial
DAM Seminar
18/06/2015

Leonardo Giusti (Milano-Bicocca University)
Implications of Poincare symmetry for thermal field theories
HEP Seminar
18/06/2015

Alexei Bazavov (UC Riverside & University of Iowa)
The equation of state and deconfinement in QCD
HEP Seminar
19/06/2015

Diego Correa (Universidad Nacional de La Plata)
Cusped Wilson lines and ladders
HEP Seminar
25/06/2015

Alexander Andrianov (St. Petersburg State University)
PT symmetric classical and quantum cosmology: PToms against phantoms
HEP Seminar
26/06/2015

Francisco Moldón Vara (ASTRON, Netherlands Institute for Radio Astronomy)
Exploring the low-frequency universe. High resolution images of extragalactic jets with the international LOFAR
DAM Seminar
08/07/2015

Michal Heller (Perimeter Institute)
Hydrodynamics Beyond the Gradient Expansion: Resurgence and Resummation
HEP Seminar
09/07/2015

Chunshan Lin (YITP, Kyoto University)
Resonant Primordial Gravitational Waves amplification
HEP Seminar
10/07/2015

Joseph Hennawi (Max-Planck-Institut für Astronomie, Heidelberg)
The First Characterization of the Jeans Filtering Scale of the Intergalactic Medium
ICCUB Seminar
13/07/2015

Enrique Ruiz Arriola (Univ. Granada)
Coarse graining of Interactions and the Renormalization Group in Nuclear and Molecular Physics
FAN Seminar
21/07/2015

Albert Gallemí (UB)
Coherent Quantum Phase Slip in Bosonic Atomtronic Circuits
FAN Seminar
28/07/2015

Christian Reichardt (University of Melbourne)
Measuring CMB Polarization with POLARBEAR and the Simons Array: Towards New Constraints on Neutrino Masses and Inflation
ICCUB Seminar
07/09/2015

Douglas Scott (University of British Columbia)
New Results from the Planck satellite
ICCUB Seminar
10/09/2015

Giacomo Cacciapaglia (Université de Lyon)
Recent developments in composite dynamics: Dark Matter and the di-boson excess
HEP Seminar
17/09/2015

Takahiro Tanaka (Kyoto University)
An attempt to derive bigravity from braneworld
HEP Seminar
21/09/2015

Miguel Quartin (Univ. Federal de Rio de Janeiro)
Measuring cosmic structure with supernovae
HEP Seminar
23/09/2015

Daniele Musso (ICTP)
A goldstino at the end of the cascade
HEP Seminar
25/09/2015

Markus Froeb (ITP Leipzig)
Constructing interacting QFTs via OPEs
HEP Seminar
01/10/2015

Daniel Elander (Purdue University)
Walking dynamics from gauge-gravity duality
HEP Seminar
02/10/2015

Xavier Bekaert (LMPT Tours)
Torsional Newton-Cartan geometry and its relativistic origin
HEP Seminar
08/10/2015

Sayantani Bhattacharyya (Indian Inst. Tech., Kanpur)
A Charged Membrane Paradigm at Large Dimension
HEP Seminar
19/10/2015

Nico Wintergerst (Stockholm University)
Approaches to black hole information
HEP Seminar
20/10/2015

Rainer Stiele (Goethe University & Institute of Space Sciences)
Thermodynamics and Phase Structure of Strongly-Interacting Matter in a Quark-Meson Model
FAN Seminar
20/10/2015

Tamas Csorgo (Wigner Research Centre for Physics, Budapest, & KRF, Gyongyos, Hungary)

Multiple diffraction theory of elastic pp scattering, with applications to pp data at LHC energies
FAN Seminar
21/10/2015

Sera Markoff (API/GRAPPA, University of Amsterdam)
Understanding the dominant state of active black holes in the Universe
DAM Seminar
26/10/2015

Ishwara Chandra CH. (National Center for Radio Astrophysics, TIFR, Pune, India)
GMRT Survey of Cygnus OB2 region at 610 and 325 MHz
DAM Seminar
28/10/2015

Juan Rojo (Oxford University)
Discovery through precision: perturbative QCD at the dawn of Run II
HEP Seminar
29/10/2015

Martin Heinze (DESY)
Orbit method quantization of AdS superparticles
HEP Seminar
30/10/2015

Johannes Schmude (Universidad de Oviedo)
Supersymmetric Field Theories on Five-Manifolds
HEP Seminar
05/11/2015

Óscar Dias (Southampton University)
Black resonators: evading black hole theorems and the path to cosmic censorship violation
HEP Seminar
09/11/2015

Eulogio Oset (U. Valencia /IFIC)
Testing the nature of resonances in B, D, Lambda_b weak decays
FAN Seminar
12/11/2015

Jorge Rocha (ICCUB)
Rotating thin shells, gravitational collapse and cosmic censorship
HEP Seminar
13/11/2015

Vitor Cardoso (Universidade de Lisboa)
The gravity of fundamental fields
HEP Seminar
19/11/2015

Andreas Crivellin (CERN)
New Physics in B decays and correlations with LHC searches
HEP Seminar
20/11/2015

Alfonso Ramallo (Universidad de Santiago de Compostela)
Cold holographic matter in a top-down model
HEP Seminar
26/11/2015

Jon Marcaide (Astronomy and Astrophysics Department. University of Valencia)
Supernova 1987A seen with ALMA's eyes
DAM Seminar
27/11/2015

Keisuke Izumi (ICCUB)
Causal Structures in Gauss-Bonnet gravity
HEP Seminar
27/11/2015

Miguel Vázquez-Mozo (Universidad de Salamanca)
On the double copy structure of soft gravitons
HEP Seminar
03/12/2015

Guillem Domènech (Kyoto University)
Metric transformations: cosmological disformal invariance and frame "dependence" of inflation
HEP Seminar
04/12/2015

Lars Hofer (ICCUB)
Flavour and high-energy physics in the precision era of the LHC
HEP Seminar
11/12/2015

Enric Verdaguer (ICCUB)
Riemann correlator in de Sitter including loop corrections from matter fields
HEP Seminar
18/12/2015

EVENT ORGANIZATION

Jordi, C.
Chair of organizing committee
Gaia CU5 Plenary meeting #14
Facultat de Física, UB
27/01/15-29/01/15

Calvo, M.; **Garrido, L.**; Lefèvre, R.; Oyanguren, A.; Puig. A.
Organizing committee
2nd Workshop on Radiative decays at LHCb
Physics Faculty, UB
04/05/15-06/05/15

Bertolín, A.; **Espriu, D.**; García-Berro, E.; Martín, V.; Nofrarias, M.; Sopuerta, C.F.
Organizing committee
IGMW2015 "5th Iberian Gravitational Wave Meeting"
Universitat Politècnica de Catalunya (UPC)
12/05/15-14/05/15

Gómez, G.
Organizing committee
AstroNet-II International Final Conference
Gran Hotel Reymar, Tossa de Mar (Spain)
15/06/15-19/06/15

Gascón, D.; Paredes, J.M.; Ward, J.
Organizing committee
GAMMA-400 Workshop
Physics Faculty, UB
29/06/15-30/06/15

Figueras, F.; Romero-Gómez, M.; Balaguer-Núñez, L.; Mor, R.; Molina, D.
Organizing committee
Third Gaia Challenge Workshop
Physics Faculty, UB
31/08/15-04/09/15

Espriu, D.; Pons, J.M.; Russo, J.; Simon, J.
Organizing committee
Symmetries in Particles and Strings - A Conference to celebrate the 70th birthday of Quim Gomis
Physics Faculty, UB
04/09/2015

Luri, X.; Jordi, C.; Balaguer-Núñez, L.; Molina, D.
Organizing committee
Gaia Catalogue Access CU9 plenary Meeting
Physics Faculty, UB
21/09/15-23/09/15

Àgueda, N.; **Aran, A.**; **Sanahuja, B.**
Organizing committee
HESPERIA Progress Meeting
Physics Faculty, UB
19/10/15-20/10/15

Germani, C.
Organizing committee
Vacuum2015 "The vacuum of the Universe: from cosmology to particle physics"
Physics Faculty, UB
21/10/15-23/10/15

Aran, A.; Bosch-Ramon, V.; Garrido, L.; Mescia, F.; Miralda-Escudé, J.; Parreño, A.
Organizing committee
ICCUB Christmas Meeting
Physics Faculty, UB
21/12/15-22/12/15

Verde. L.
Chair of organizer committee
Beyond LCDM
Scandic Holmenkollen Park Hotel
13/01/2015-17/01/2015

Figueras, F.
Coordinator
Gaia Perseus Weave
University of Alicante
16/02/2015

Jordi, C.
Secretary of organizer committee
Un matí d'astrobiologia a la UB
University of Barcelona
06/03/2015

Verde. L.
Participation in organizing committee
Aspen Winter Conference, Closing in on the cosmological model
Aspen Center for Physics
08/03/2015-14/03/2015

Bosch-Ramon, V.
Main Organizer
Variable Galactic Gamma-Ray Sources III
Max-Planck-Haus
04/05/15-06/05/15

Iwasawa, K.
Science Organizing Committee
The 2nd Spanish X-ray Astronomy 2015
Instituto de Física de Cantabria (IFCA)
03/06/15-05/06/15

Verdaguer-Oms, E.
Secretary of organizer committee
Peyresq Physics-20 International Workshop: 'Micro and Macro Structure of Spacetime'
Peyresq, Alpes de Haute-Provence
13/06/15-19/06/15

Solà, J.
Participation in the organizingcommittee
12th International Symposium on Radiative Corrections (RADCOR 2015)
University of California
15/06/15-19/06/15

Jordi, C.
Participation in organizing committee
First Science From Gaia - the Gaia Research for European Astronomy Training (GREAT) Network Science Symposium, EWASS 2015
University of La Laguna
23/06/15-24/06/15

Parreño, A.
Participation in organizing committee
The 8th International Workshop on Chiral Dynamics 2015
Pisa University
29/06/15-03/07/15

Emparan, R.
Chair of organizing committee
Gravity: New perspectives from strings and higher dimensions
Centro de Ciencias de Benasque Pedro Pascual
12/07/15-24/07/15

Verde. L.
Participation in organizer committee
MPA/ESO/MPE/EXC Joint Conference on the large-scale structure
Headquarters of the European Southern Observatory (ESO)
20/07/2015-24/07/2015

Russo, J.
Co-Organizer
Gauge theories, supergravity and superstrings, 2015
Centro de Ciencias de Benasque Pedro Pascual
02/08/15-15/08/15

Verde. L.
Chair of organizing committee
From Inflation to Galaxies: a workshop in honour of Sabino Matarrese
Castello Pasquini
31/08/2015-03/09/2015

Bosch-Ramon, V.; Paredes, J.M.
Participation in organizing committee
High Energy Phenomena in Relativistic Outflows V (HEPRO V)
Honorable House of Senators of the Province of Buenos Aires
05/10/15-08/10/15

PUBLIC OUTREACH

Courses and Workshops

Masterclass on Particle Physics 2015

The Masterclass on Particle Physics is an activity addressed to high school students in their final year, as part of the International activity Hands on Particle Physics. The workshop has taken place at the UB since 2005 and lasts one day, during which students attend talks about Physics and study real data from LHC. The students also visit the laboratories and attend a presentation about the courses of the Faculty of Physics.

In 2015 two sessions were held on the 2nd and 9th of March at the Faculty of Physics. 157 students from 114 high schools all around Catalonia attended.

Web: http://www.lhc.cat/taller.php

MASTERCLAS ON PARTICLE PHYSICS 2015

Attendees talk to each other and ICCUB members during the coffee break.

Exhibitions

The ICCUB owns four travelling exhibitions. These exhibitions have different printed versions in the Institute that are annually exposed in different external centers, like high schools, libraries or community centers. Two of these exhibitions have been entirely produced by the ICCUB.

A thousand million eyes for a thousand million stars

This is an exhibition about the Gaia mission produced by the ICCUB in 2013. It consists of 14 informative plaques edited and printed in Catalan and Spanish.

http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/ExpoGaia

Itinerary 2015 of the Catalan version:

- IES Menéndez y Pelayo, Barcelona
  From November 9th to December 4th

Itinerary 2015 of the Spanish version:

- Parque Metropolitano Marismas de los Toruños y Pinar de la Algaida de El Puerto de Santa María, Cádiz
  From the 3rd to the 12th of July, in the context of the *IIIrd Astrophysical Conference*.
- Physics Faculty, Univ. of Santiago de Compostela (Galicia)
  From the 10th to the 31st of July

Amb A d'AstrònomA

This is the Catalan version of the exhibition *Con A de AstrónomA*, dedicated to all woman astronomers from different eras and countries, whose contribution to Astronomy have been relevant in a worldwide scale. The ICCUB translated and edited it in 2010.

http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/AmbA

Itinerary 2015:

- Centre d'Observació de l'Univers, Àger, Lleida
  From August 1st to November 15th
- Escola Virolai, Barcelona
  From the 12th to the 26th of November
- Centre Cívic Espai Gatassa, Mataró.
  From November 27th to January 15th

The cosmic distances

This exhibition, which was fully created by ICCUB members in 2012, shows the methods scientists use to calculate the distances to celestial objects, and how these methods have progressively evolved throughout the years

depending on how far away observed objects were.

Nowadays, the ICCUB is responsible for the informative plaques and manages the itinerary of the exhibition.

http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/DistanciesCosmiques

Itinerary in 2015:

- Escola Octavio Paz, Barcelona,
  From February 23rd to March 1st
- Institut Castellbisbal, Castellbisbal
  From September 17th to November 4th
- Escola Jesuïtes Sarrià Sant Ignasi, Barcelona
  From the 9th to the 23rd of November

From the Earth to the Universe

This is the Catalan version of the exhibition *From the Earth to the Universe*, which was translated and edited by the ICCUB in 2009. It shows the Universe through astronomical images of great importance for science dissemination, taken from different observatories around the world, as well as from Spain.

http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/DistanciesCosmiques

- Escola Pia de Mataró. Mataró
  From September 21st to November 5th
- Museu Darder de Banyoles, Banyoles
  From November 20th to January 11th

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EXHIBITIONS

1 and 2: "From the Earth to the Universe" on exhibition at Darder Museum of Banyoles

3, 4: The cosmic distances", on exhibition at Castellbisbal High Schooll

Web Sites

ServiAstro

ServiAstro is the website for public outreach on astronomy of the Institut of Cosmos Sciences (ICCUB) and the Quantum Physics and Astrophysics Department (FQA) of the University of Barcelona.

ServiAstro offers all kind of outreach material, as well as information about past and future astronomical events which are visible from Catalonia. Furthermore, on ServiAstro visitors will find a compilation of astronomical ephemerides, tools for astronomical calculations, news, answers to frequently asked questions and links to lots of other websites about astronomy.

http://serviastro.am.ub.edu/

Descobrint la Física de Partícules amb l'LHCb

This a web site maintained by the Experimental Particle Physics Group at the ICCUB. It contains didactic material produced by the group and information about their outreach activities, such has the *Masterclass on Particle Physics*.

http://www.lhc.cat/

The ICCUB in the social networks

Institutional Channels

@ICC_UB ICCUB

Group and members channels:

Gaia Group: channels devoted to the outreach of the Gaia Mission, particularly those activities in which ICCUB members participate.

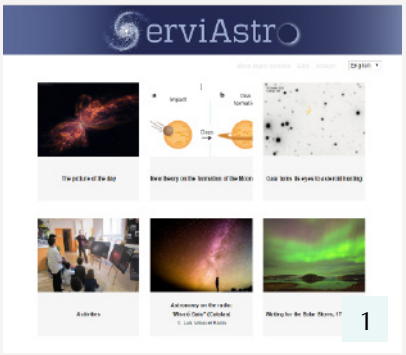
- https://www.facebook.com/gaiaub,
- https://twitter.com/GaiaUB
- https://www.youtube.com/channel/UCAdmF8h-oCIlZZjMFxWPm5A

Cataquark: piulades de recerca i divulgació: Twitter account managed by ICCUB researcher J. Guasch which offers interesting outreach news about particle physics.

- https://twitter.com/cataquark

Thermalization: Twitter account managed by ICCUB researcher M. Attems about Theoretical particler Physics.

- https://twitter.com/thermalization



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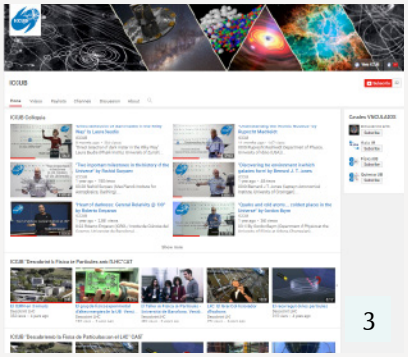

3

THE ICCUB ON THE INTERNET

1. ServiAstro front wepage

2. lch.cat webpage about the Masterclass on Particle Physics

3. ICCUB Youtube channel

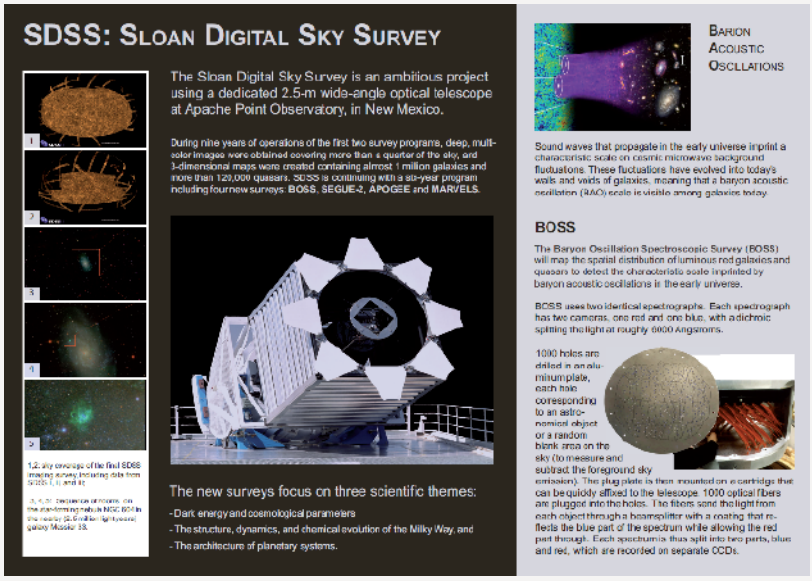
Outreach Material

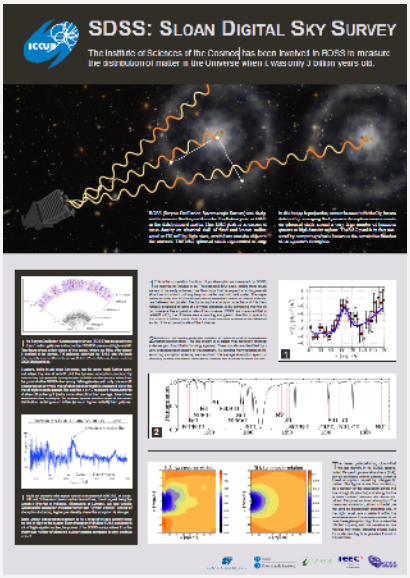
Posters SDSS-BOSS

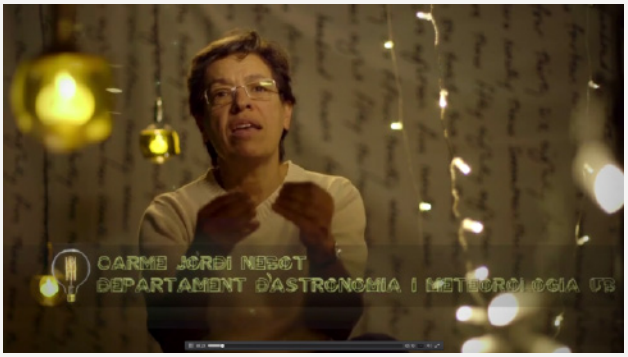
Two posters were created to describe the operation of BOSS as well as the activity of the Institute in this exploration, to accompany an original plate from the telescope that has been hung in the office area ICCUB-IEEC.

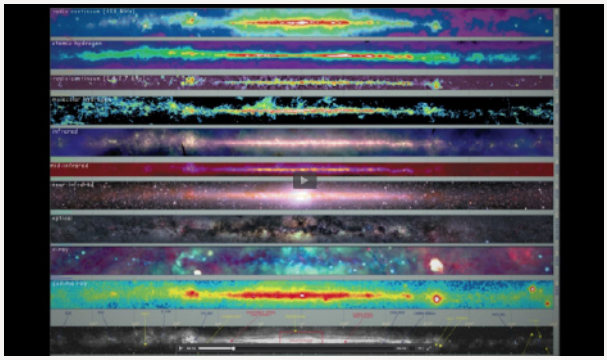
Short documentary *La ciència de la llum - La llum que ens parla des dels estels*

Short documentary produced by the University of Barcelona in the occasion of the *Setmana de la Ciència* in which the ICCUB researcher Carme Jordi talks about how astronomers use light coming from the stars to determine their characteristics and study different phenomena.









SOME OUTREACH MATERIAL PRODUCED AT ICCUB IN 2016

1 and 2: Posters about SDSS and BOSS for the ICCUB-IEEC area

3 and 4: Screenshots from the video "La ciència de la llum - La llum que ens parla des dels estels"

Publications

Carrasco, J.M., Masana, E., *Cómo ver el eclipse solar en Barcelona*, Newspaper *La Vanguardia*, 18/03/2015
http://www.lavanguardia.com/ciencia/20150318/54429070811/eclipse-solar-2015-barcelona.html

DAM-ICCUB, *La UB retransmitirá por internet el eclipse de sol del próximo viernes*, Newspaper *La Vanguardia*, 18/03/2015
http://www.lavanguardia.com/vida/20150318/54429071187/la-ub-retransmitira-por-internet-el-eclipse-de-sol-del-proximo-viernes.html

Carrasco, J.M., *Una casualitat còsmica i meteorològica permet veure durant uns instants l'eclipsi a Barcelona*, Newspaper *ARA*, 20/03/2015
http://www.ara.cat/societat/Expectacio-maxima-davant-eclipsi-repetira_0_1324067681.html

Carrasco, J.M., *Por fin, hemos llegado a Plutón,* Newspaper *La Vanguardia* 15/07/2015
http://www.lavanguardia.com/ciencia/fisica-espacio/20150715/54433927101/new-horizons-pluton-paso-enorme-historia-exploracion-espacial.html

Carrasco, J.M., *Plutón tiene montañas de agua helada de 3.500 metros*, 16/07/2015
http://www.lavanguardia.com/ciencia/fisica-espacio/20150716/54433435515/pluton-montana-agua-helada.html

Emparan, R. *¿Fuego en el horizonte?*, Magazine *Investigación y Ciencia*, Vol. 463, p. 21 - 21 (2015)
http://www.investigacionyciencia.es/revistas/investigacion-y-ciencia/numero/463/fuego-en-el-horizonte-13018

Luri, X., *La meitat de les estrelles de l'Univers van per lliure,* Newspaper *ARA,* 04/01/2015
http://www.ara.cat/suplements/diumenge/meitat-estrelles-lunivers-lliure_0_1279072091.html

Gaia Team, *La imatge més inusual de la Via Làctia*, 06/07/2015
http://www.ara.cat/societat/lactia-via-fotografia-agencia-espacial-europea_0_1388861346.html

Gaia Team, *Contando estrellas con Gaia*, 06/07/2015
http://www.lavanguardia.com/ciencia/fisica-espacio/20150706/54433741802/satelite-gaia-esa-via-lactea-nubes-de-magallanes.html

Jordi, C., *Diez desafíos caídos del cielo*, 12/08/2015
http://www.elcultural.com/noticias/ciencia/Diez-desafios-caidos-del-cielo/8154

Jordi, C. *Gaia ya ha cubierto el cielo completo,* 23/06/2015
http://www.iac.es/divulgacion.php?op1=16&op2=402&id=78

Jordi, C. *Gaia alerta de asteroides desconocidos*, 30/06/2015, Newspaper *El País*
http://elpais.com/elpais/2015/06/25/ciencia/1435230474_414196.html

Luri, X., *La piràmide i altres misteris marcians.* Newspaper *ARA*, 30/082015
http://www.ara.cat/premium/piramide-altres-misteris-marcians_0_1421258108.html

Luri, X., *La Terra i la Lluna, un naixement d'impacte.* Newspaper *ARA*, 31/05/2015
http://www.ara.cat/suplements/diumenge/Terra-lluna-naixement-dimpacte_0_1367263271.html

Luri, X., *Mart, el primer pas per a somiar en la colonització d'un altre planeta.* Newspaper *ARA*, 29/09/2015
http://www.ara.cat/societat/pas-somiar-colonitzacio-dun-planeta_0_1440455971.html

Luri, X., *Tot a punt per a l'eclipsi solar* Newspaper *ARA*, 18/03/2015
http://www.ara.cat/societat/punt-leclipsi-solar_0_1323467669.html

Luri, X.
Astronomía: la ciencia más mágica. Article. Newspaper *El Mundo* 11/09/2015
http://www.elmundo.es/ciencia/2015/09/11/55eff8d122601d995a8b4572.html

Press releases

The Faculty of Physics hosts the exhibition "Physics in our lives"
13 January 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/01/010.html

The UB participates in second period of operation of the Large Hadron Collider
25 February 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/02/042.html

The UB live-streams the solar eclipse that happens on 20 March and provides two telescopes to observe it
19 March 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/03/060.html

ICCUB and BGSMath receive the distinction María de Maeztu and IBEC obtains Severo Ochoa Excellence Award
23 April 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/04/038.html

CMS and LHCb experiments reveal new rare particle decay
14 May 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/05/034.html

UB participates in the new season of the large hadron collider
8 June 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/fotonoticies/06/001.html

Award-winning UB centres receive the distinctions María de Maeztu and Severo Ochoa
17 July 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/07/035.html

Gaiaverse, the new portal of the Gaia mission
28 July 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/07/050.html

Gaia's first year of scientific observations
28 August 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/08/017.html

Foundation stone laying ceremony of a new generation of high-energy telescopes
13 October 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/fotonoticies/10/002.html

Jordi Torra, new director of the Institute of Space Studies of Catalonia (IECC)
26 October 2015
http://www.ub.edu/web/ub/en/menu_eines/noticies/2015/10/058.html

Public Talks

Jordi, C., *La lluna mentidera*, January 2015

Ribó, M., *Què fa un astrònom?*, Escola del Parc, El Prat de Llobregat, 27/01/15

Bosch-Ramon, V., *Coupling non-thermal radiation to hydrodynamics*, Univ. Valencia, Burjassot, 01/03/15

Jordi, C., *La professió d'astrònoma. Què faig a la Universitat?*, Escola Octacio Paz, 09/03/15

Jordi, C., *Projecció i comentari d'imatges del cel profund*, in the context of the *International Year of Light*, Associació Astronòmica Sant Cugat-Valldoreix. Sant Cugat del Vallès, 14/03/15

Jordi, C., *La detecció dels exoplanetes: tècniques i limitacions*, in the context of the event *Un matí d'astrobiologia a la UB*, Aula Magna Faculty of Biology, 06/03/2015

Salvador, E., *La primera llum de l'Univers i els seus efectes*, Agrupació Astronòmica de Sabadell, 18/03/15

Miralda, J., *Present i futur de la vida a l'Univers*, Centre Cultural de Terrassa, organized by Agrupació Astronòmica de Terrassa i Òmnium Cultural Terrassa, 20/03/15

Torra, J., *Gaia, els primers resultats i nous reptes*, Central office of the Agrupació Astronòmica de Sabadell, 25/03/15

Romero, M., *Gaia, la màquina dels descobriments*, Col·legi La Farga, St Cugat del Vallès, 26/03/15

Roca, S., *Gaia, la màquina dels descobriments*, Col·legi La Farga, St Cugat del Vallès, 27/03/15

Jordi, C., *El color de les estrelles*, Associació astronòmica Sant Cugat-Valldoreix. Sant Cugat del Vallès, 16/04/15

Carrasco, J.M.; Luri, X., *Misión Gaia: imágenes para crear el mapa de la galaxia*, Saló del Còmic, 18/04/15

Carrasco, J.M.; Luri, X., *Mapas del cielo: la misión Gaia y la Vía Láctea en 3D*, Festival *Pint of Science*. Michael Collins, 18/05/15

Ribó, M., *Forats negres silenciosos amagats en sistemes binaris*, central office of ASTER, Barcelona, 11/06/15

Luri, X., *El mètode científic aplicat a la vida diària*, Museu Darder de Banyoles, 13/06/15

Ribó, M., *Els forats negres... i altres objectes compactes*, Faculty of Physics, 19/06/15

Carrasco, J.M.; Luri, X., *Gaia: El primer pas per a la fundació d'un imperi galàctic*, Biblioteca Zona Nord, 07/07/15

Jordi, C., *La revolució observacional de Gaia*, Inaugural talk for the II congress PRO-AM (collaboration between professional and amateur astronomers) Alcalá de la Real, 12-14/07/2015

Solanes, J. M., *Cosmogènesi: l'origen de l'univers*, Àger, 10/10/15

Jordi, C., Round table, *La ciència de la llum. Mostra de vídeo científic*, Aula Magna from the Faculty of Physics (UB), 02/11/15

Casamiquela, L.; Romero, M., *Gaia*, Escola Minguella. Badalona, 12/11/15

Romero, M., *Amb A d'astrònoma: Dones i astronomia*, Les Avellanes Town Hall (Lleida), 21/11/15

Paredes, J.M., *Radiació gamma: L'univers de raigs gamma*, Sala Gabriel Oliver, aulari Josep Carner, Faculty of Philology (Barcelona), 21/11/15

Participation in Workshops and Schools

Ribó, M., Eclipse tracing 23/03/15, CEIP Escola de la Concepció, 20/03/15

Carrasco, J.M., *Rockets workshop*, Escola Font de Fargàs de Barcelona, 26/05/15

Casamiquela, L.; Moreno, V.; Pérez, I.; Roca-Fàbrega, S., *Astronomy project*, MónNatura Pirineus, 22/06/15-04/07/15

Broadcast of Astronomical Events

Comet Lovejoy (22/01/2015)

To mark the occasion of comet Lovejoy passing near Earth, news and a special section concerning the comet were added to Serviastro's website.

http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/LovejoyC2014Q2

Total solar eclipse, 20 March 2015

Several material and activities were prepared on occasion of this ephemeris:

- Informative page in the website ServiAstro: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/InfEc200315
- Online broadcast of the event: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/EcSol200315BCN
- Public observation from Espai Ciència Saló de l'Ensenyament, Barcelona. Plaça Universitat, Barcelona.
- Material distribution to schools: Observation Pack (8 protection glasses + 15 informative brochure, http://serviastro.am.ub.edu/twiki/pub/ServiAstro/InfEc200315/Diptic_eclipsi_200315_baixa2.pdf)

The following schools participated:

- CEIP Lluis Vives, Barcelona
- Escola El Calderí, Caldes de Monbui
- Escola Nostra Senyora de Montserrat, Rubí
- CEIP Fort Pienc. Barcelona
- CEIP Sant Lluis Gonçaga, La Garriga
- CEIP Abat Oliba, Cornellà de Llobregat
- Escola Costa Llobera, Barcelona
- INS EUROPA, Hospitalet de Llobregat
- Escola de la Concepció. Barcelona

The eclipse in the press: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/MitjansComunicacio200315

Perseids 2015

The Perseids meteor shower was noted on Serviastro's website and in a radio program.

- Special web page in Serviastro: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/PerSeides2015
- Radio program where J.M Carrasco talks about Perseids 2015: *El matí a Radio 4*, 12-08-2015

Total Lunar eclipse the 28 September 2015

Serviastro broadcasted this event online and created an informative webpage and poster.

- Informative web page: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/InfEc280915
- Online broadcast of the eclipse: http://serviastro.am.ub.edu/twiki/bin/view/ServiAstro/EcLu280915
- A poster: http://serviastro.am.ub.edu/twiki/pub/ServiAstro/InfEc280915/Poster_-_final_baixa_EclipsiLluna280915.pdf

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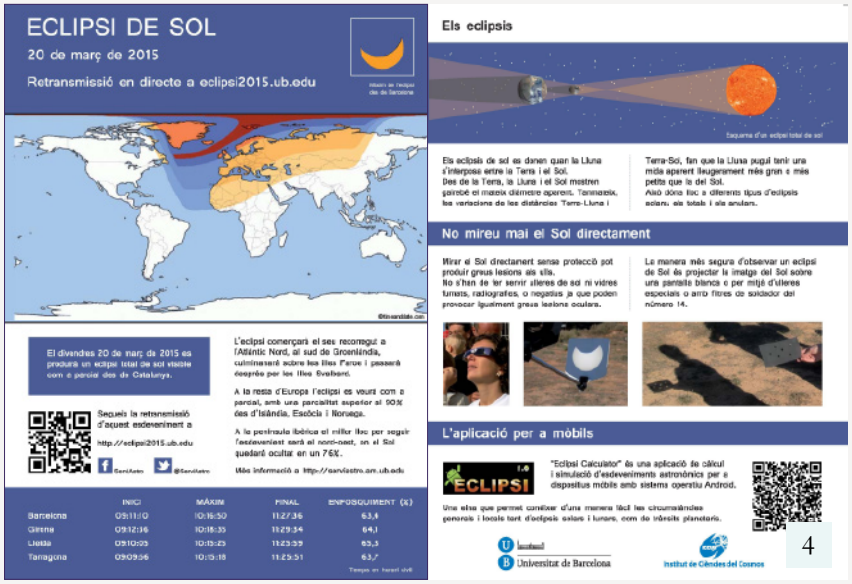

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BROADCASTS OF ASTRONOMICAL EVENTS

1: Comet Lovejoy on January 10th 2015 from Camporells (France). Image: Eduard Masana (ICCUB); 2, 3: Broadcast of the Total Lunar eclipse on September 28th 2015; 4: Leaflet of the eclipse 20/03/15; 5: Poster of the Total Lunar eclipse on September 28th 2015

Astronomy Sessions

Masana, E.; **Balaguer, L.**; **Molina, D.**, *Sun eclipse observation*, Faculty of Physics, UB, Barcelona, 20/03/15

Mor, R.; **Pérez-Ràfols, I.**; **Olarte, S.**, *Sun eclipse observation*, Edifici Històric UB, Barcelona 21/03/15

Roca, S.; **Luri, X.**, *Sun eclipse observation*, Saló ensenyament. 22/03/15

Moreno de la Cita, V.; **Pérez-Ràfols, I.**; **Roca, S.**; **Olarte, S.**, *Sun observation*, Edifici Històric UB, 22/05/15

Pérez-Ràfols, I.; **Roca, S.**, *Public star observation*, Castellterçol. 28/08/15

Masana, E.; **Jordi, C.**, *Public observation at the refuge of Rebost*, Refugi del Rebost, Parc Natural del Cadí-Moixeró (Bagà, Berguedà), 17/10/15

ASTRONOMY SESSIONS

1: Poster of the observation in Castellterçol; 2: Solar eclipse observation, Edifici Històric UB. 21/03/15

Participation in TV and radio shows

Carrasco, J.M., Interview *Eclipsi solar* in the program *Lletra lligada*, Ràdio 4, 16/03/15

Luri, X., Interview *Eclipsi solar* in the program *Què vols ser?,* Ràdio Barcelona, 19/03/15

Carrasco, J.M., Interview *Eclipsi solar* in the program *El Matí*, Cope Catalunya, 19/03/15

Carrasco, J.M., Interview *Eclipsi solar* in the program *Hora Barcelona*, Ràdio Barcelona, 19/03/15

Carrasco, J.M., Interview *Eclipsi solar* in the program *Amics i coneguts*, Ràdio 4, 21/03/15

Carrasco, J.M., Interview *Eclipsi solar* in the program *Connexió Barcelona*, BTV, 20/03/15

Carrasco, J.M., *La New Horizons arriba a Plutó*, in the News program at 10 o'clock of BTV, 14/07/15

Carrasco, J.M., *Gaia*, in the program *La república santboiana*, Science Section, Ràdio Sant Boi, 18/11/15

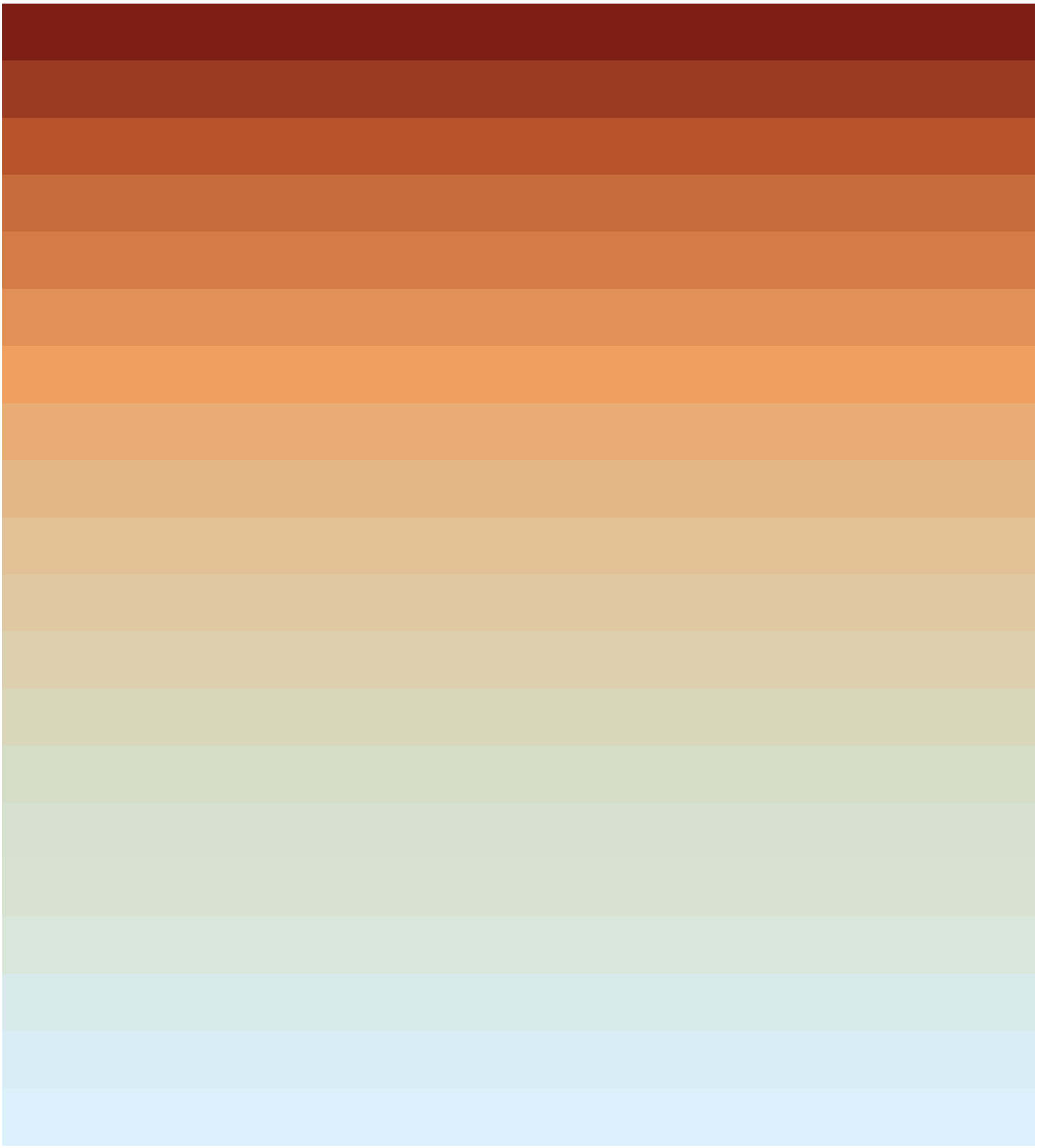
Carrasco, J.M., *L'estel de Nadal*, in the program *La república santboiana*, Science Section, Ràdio Sant Boi, 16/12/15

Carrasco, J.M., *Sopa d'estrelles: Secció d'Astronomia a "El matí de Catalunya Ràdio"*, Fridays from 24 July to 28 August 2015, http://www.am.ub.edu/twiki/bin/view/ServiAstro/WebRadio

- Program 1, 24/07/15: *Arribada de New Horizons. Cançó Pluto Mars. El cel de l'estiu*
- Program 2, 31/07/15: *Planetas extrasolars. Blue Moon. Astronomia* Program 3, 07/08/15: *Astronomia i cinema*
- Program 4, 14/08/15: *Els sons de l'astronomia*
- Program 5, 21/08/15: *Gaia*

PARTICIPATION IN TV AND RADIO SHOWS

1: Carrasco, J.M., "La New Horizons arriba a Plutó", in the News program at 10 o'clock of BTV, 14/07/15
2: Carrasco, J.M., "La república santboiana, secció de ciència", (Ràdio Sant Boi), "L'estel de Nadal", 16/12/15



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